

Cofinal strata and ordinal absorption

Finitary powersets of finite lexicographic sums

An exploratory research article

Prepared for Vladimir Reshetnikov with ChatGPT

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Abstract

We investigate a concrete instance of the published problem of enlarging classes of well-quasi-orders whose finitary powersets have compositionally computable ordinal invariants. For a finite poset Q , we stratify the finitely generated downsets of a lexicographic sum $\sum_{q \in Q}^{\text{lex}} P_q$ by the inclusion-maximal antichains of Q . Under explicit hypotheses on the fibers, each stratum has a pure maximal order type ω^{γ_A} . A cofinal-absorption theorem shows that its contribution survives exactly when no higher stratum has a strictly larger exponent.

For uniform ordinal fibers $\alpha = \omega^\rho$, $\rho > 0$, this yields

$$\text{o}(\mathcal{K}(S_\alpha(Q))) = \sum_{k=\text{w}(Q)}^1 \omega^{\rho \otimes k} c_k(Q), \quad \text{h}(\mathcal{K}(S_\alpha(Q))) = \alpha \cdot \text{h}(\mathcal{M}(Q)).$$

Here $c_k(Q)$ counts maximal antichains of size k not dominated by a larger one, and $\mathcal{M}(Q)$ is their domination poset. The first formula extends to nonuniform WPO fibers of prescribed pure finitary-powerset type. We also obtain a leading-coefficient counting reduction, an explicit maximal-chain construction, and a Cantor-normal-form truncation rule for finite nonuniform lexicographic sums. Complete proofs accompany exact-arithmetic Python code and finite structural checks. This is a partial answer to the broader research direction, not a solution of its general width or arbitrary-iteration problems; priority of the derived formulas is not independently established.

Addition of 28 September 2026 (batch 36). Parts II and III, merged from four later manuscripts, determine the height for nonuniform fibres: a coordinate is charged, by ordinary ordinal addition, only when its vertex leaves the frontier.

Addition of 29 September 2026 (batch 42). Part IV shows, via uniquely restricted matchings, that the height problem is NP-complete already for height-two skeletons with every fibre ω .

Scope of the claims. The mathematical results below are proved from stated standard WPO facts. The accompanying computations test finite identities and the implementation; they are not a formal verification of infinite ordinals. The general extension question remains broader than the class treated here.

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1 The selected open direction and the precise target

Abriola, Halfon, Lopez, Schmitz, Schnoebelen and Vialard ask, in the conclusion of *Measuring well-quasi-ordered finitary powersets*, which extensions of their elementary family retain computable ordinal invariants [1, Section 6, p. 23]. Their grammar uses disjoint unions, Cartesian products, words, multisets and finitary powersets. Arbitrary finite-poset lexicographic gluing is not one of its constructors. We take this missing constructor as a specific target.

Problem 1.1 (Finite-gluing instance). Let Q be a finite poset and let P_q , $q \in Q$, be WPOs. Determine the maximal order type and height of the finitary powerset of

$$S = \sum_{q \in Q}^{\text{lex}} P_q,$$

from finite information about Q and appropriate invariants of the fibers. In particular, obtain exact formulas when every fiber is the ordinal ω^ρ , for a fixed $\rho > 0$.

This formulation is our concrete subproblem, not a quotation of a separately numbered conjecture. The broad published question does not prescribe this particular class. The results here give an additional compositional rule with explicit hypotheses, rather than a claim that the whole elementary calculus has been closed under a new operation.

1.1 What is established here

The general maximal-order-type result is Theorem 7.3. Its hypotheses require that the fibers have no finite cofinal subsets and that $\text{o}(\mathcal{K}(P_q)) = \omega^{\beta_q}$, with $\beta_q > 0$. The exponents may vary with q , and the fibers need not be chains. The two uniform formulas are Corollary 8.1 and Theorem 9.1. The latter includes an actual chain attaining the computed height.

The main proof mechanism is independent of finitary powersets: a finite stratification with pure stratum types and upward cofinality admits an exact survivor formula. The finitary-powerset construction supplies a canonical such stratification. The distinction between a finite stratification and a lexicographic decomposition is essential: comparable strata need not have all their points comparable.

We do not determine ordinal width in general. Nor do we claim a formula for arbitrary repeated applications of $\mathcal{K}$, arbitrary impure fiber types, or infinite indexing posets. A concrete failure of a tempting nonuniform height formula is proved in Section 13. [Added 28 September 2026, batch 36: the nonuniform height itself, which the report determined only for uniform fibers at its pinned state e21766d04, is now proved in Parts II and III; see Section 15.]

1.2 Status and relation to existing results

The source for the open direction is the July 2024 version of [1]; its arXiv record and the final-page question were checked on 19 September 2026. A targeted literature search did not locate the frontier formulas proved below. That search is not a proof of novelty, and the results should be checked against specialist literature before priority is claimed.

A nearby problem must not be confused with our target: maximal order types of uniform lexicographic products have already been treated in Vialard's published 2025 paper [4]. We use that publication, rather than an older preprint's status statement, in assessing the literature. Appendix A derives a finite nonuniform formula as a supporting result and recovers the known finite uniform case. It is not presented as a resolution of an open lexicographic-product problem.

The background facts about maximal linearizations and natural operations are classical [2, 3]. The contribution attempted here is the cofinal-stratum reduction and its consequences for the chosen finite-gluing instance.

2 Definitions and ordinal tools

2.1 WPOs, rank and maximal order type

A *well-partial-order* (WPO) is a partial order in which every infinite sequence $x_0, x_1, \dots$ has $i < j$ with $x_i \leq x_j$. All posets considered below are sets, and we work in ordinary set theory with choice. A well-quasi-order can be replaced by its quotient under mutual comparison.

A *linearization* is a total order extending the partial order on the same underlying set. Every linearization of a WPO is a well-order. Its largest possible order type is denoted $o(P)$. The de Jongh–Parikh maximal-linearization theorem guarantees that this maximum is attained [2].

For a well-founded poset, define

$$r_P(x) = \sup_{y < x} (r_P(y) + 1), \quad h(P) = \sup_{x \in P} (r_P(x) + 1).$$

We use $h(\emptyset) = 0$. This is the height defined by the rank of the tree of finite strictly decreasing sequences. In particular, an ordinal chain α has height α . An order-preserving rank bound will mean a map into ordinals which is *strictly* increasing on strict comparisons; such a map bounds r_P pointwise by well-founded induction.

The ordinal width $w(P)$ is the rank of the tree of finite sequences of pairwise incomparable elements. When P has finite maximum antichain cardinality d , this ordinal width equals the integer d . We use width only in this finite sense for index posets and the input examples; no general output-width formula is asserted.

2.2 Finite downsets and the Hoare quotient

For a poset P , set

$$\mathcal{K}(P) = \{\downarrow F : F \subseteq P \text{ finite}\}, \quad \downarrow F = \{x \in P : \exists f \in F (x \leq f)\},$$

ordered by inclusion. The empty set belongs to $\mathcal{K}(P)$. Finite subsets have the Hoare relation

$$F \leq_H G \iff (\forall f \in F)(\exists g \in G) f \leq g.$$

This is equivalent to $\downarrow F \subseteq \downarrow G$, so $\mathcal{K}(P)$ is exactly its antisymmetric quotient. These are finitely generated *downsets*; they are not required to be directed ideals. Equivalently, they are the compact elements of the lattice of all downsets of P .

For a WPO P , $\mathcal{K}(P)$ is a WPO. One standard proof applies the finite-word WPO theorem to arbitrary enumerations of finite subsets: a word embedding implies Hoare domination. We write $\mathcal{K}(P)^+ = \mathcal{K}(P) \setminus \{\emptyset\}$.

A subset $C \subseteq P$ is *cofinal* if every $p \in P$ satisfies $p \leq c$ for some $c \in C$. Thus

$$P \text{ has no finite cofinal subset} \iff P \notin \mathcal{K}(P). \quad (1)$$

This condition ensures that a whole fiber cannot be generated internally by finitely many points.

2.3 Lexicographic sums

For pairwise disjoint nonempty posets P_q indexed by a finite poset Q , write $S = \sum_{q \in Q}^{\text{lex}} P_q$. Within a fiber we use its original order; for different fibers all points of P_q lie below all points of P_r exactly when $q < r$. Fibers over incomparable indices remain incomparable. The resulting poset is a WPO whenever every fiber is a WPO: an infinite sequence has an infinite subsequence in a single fiber.

For uniform ordinal fibers $P_q = \alpha$, denote this poset by $S_\alpha(Q)$. This notation avoids choosing between competing conventions for the order of factors in a lexicographic product.

2.4 Natural arithmetic and standard monotonicity

Unmarked $+$ and $\cdot$ are ordinary ordinal addition and multiplication. We use $\#$ for Hessenberg natural sum and $\otimes$ for natural product. In Cantor normal form, natural sum adds coefficients at equal exponents; natural product multiplies the corresponding formal polynomials, using natural sum for exponents. Consequently

$$\omega^\beta \otimes \omega^\gamma = \omega^{\beta \# \gamma}, \quad \underbrace{\rho \# \cdots \# \rho}_{k \text{ times}} = \rho \otimes k. \quad (2)$$

Natural sum is strictly increasing in each argument. For any β ,

$$\mu_1, \dots, \mu_t < \omega^\beta \implies \mu_1 \# \cdots \# \mu_t < \omega^\beta \quad (3)$$

when t is finite. For $\beta = 0$, all the μ_i are zero. For $\beta > 0$, this follows immediately from the finite Cantor normal forms: every appearing exponent is below β .

We use the following standard WPO facts [2, 3]:

$$o(P \sqcup R) = o(P) \# o(R), \quad o(P \times R) = o(P) \otimes o(R). \quad (4)$$

An induced subposet cannot have larger maximal order type. Adding comparisons on the same set cannot increase maximal order type, because it only removes possible linearizations. It follows that, for a finite partition $P = \bigsqcup_i P_i$ into induced subposets,

$$o(P) \leq \#_i o(P_i). \quad (5)$$

These background results are the nontrivial external inputs to our proofs. The new reductions below do not assume a general formula for the finitary-powerset operation.

3 Cofinal absorption

The first lemma explains why certain low-exponent strata can disappear entirely from a maximal order type.

Lemma 3.1 (Cofinal absorption). *Let X be a WPO with a disjoint partition $X = U \sqcup V$. Suppose that U is cofinal in X and*

$$o(U) = \kappa = \omega^\beta, \quad o(V) < \kappa.$$

Then $o(X) = \kappa$.

Proof. The lower bound follows because U is an induced subposet of X . For the upper bound, let L be any linearization of X . Suppose its order type is greater than κ . Let I be its initial segment consisting of the first κ positions.

The restriction of L to $U \cap I$ has order type at most κ . If it had type $\mu < \kappa$, the restriction to $V \cap I$ would have some type $\nu \leq o(V) < \kappa$. The order I is a linearization of the disjoint union of these two restricted well-orders. Hence

$$\kappa = \text{type}(I) \leq \mu \# \nu < \kappa,$$

by (4) and (3), a contradiction. Therefore $L \upharpoonright (U \cap I)$ has type exactly κ .

Choose x at a position after I . Cofinality gives $u \in U$ with $x \leq u$. Since L extends X , u also occurs after I . Thus $L \upharpoonright U$ contains its initial suborder of type κ and another point after it, so its type is at least $\kappa + 1$. This contradicts $o(U) = \kappa$. No linearization can exceed κ , proving the result. $\square$

Cofinality alone is insufficient when the cofinal part has an impure maximal type. For example, let U have type $\omega + 1$, let v be incomparable with the initial ω points of U , and put v below its last point. Then U is cofinal but $o(U \cup \{v\}) = \omega + 2$. The pure-power assumption in Lemma 3.1 is doing real work.

4 A finite cofinal-stratification theorem

Definition 4.1. A *finite cofinal stratification* of a WPO X consists of a finite poset I and a partition $X = \bigsqcup_{i \in I} X_i$ into nonempty induced subposets such that

$$x \in X_i, y \in X_j, x \leq y \implies i \leq j, \quad (6)$$

$$i \leq j \implies (\forall x \in X_i)(\exists y \in X_j) x \leq y. \quad (7)$$

The second condition is upward cofinality, not universal comparability.

Theorem 4.2 (Surviving strata). *Suppose $X = \bigsqcup_{i \in I} X_i$ has a finite cofinal stratification and $o(X_i) = \omega^{\beta_i}$. Define*

$$R = \{i \in I : \text{there is no } j > i \text{ with } \beta_j > \beta_i\}.$$

Then

$$\boxed{o(X) = \#_{i \in R} \omega^{\beta_i}.} \quad (8)$$

Moreover, some maximal linearization places each entire stratum in a single block.

Proof. Upper bound. For each $i \notin R$, choose $r(i) \geq i$ for which $\beta_{r(i)}$ is maximal among the exponents of successors of i , including i itself. Since i is not a survivor, this maximum is strictly larger than β_i . Furthermore, $r(i) \in R$: a successor of $r(i)$ with still larger exponent would contradict its choice.

For each $r \in R$, form the induced subposet

$$G_r = X_r \cup \bigcup_{\{i \notin R : r(i)=r\}} X_i.$$

These groups partition X . By upward cofinality, X_r is cofinal in G_r . By the partition bound and finite natural-sum closure,

$$o(G_r \setminus X_r) \leq \#_{\{i : r(i)=r\}} \omega^{\beta_i} < \omega^{\beta_r}.$$

The inequality also holds when the group has no other strata, since the left side is zero. Lemma 3.1 yields $o(G_r) = \omega^{\beta_r}$. Applying the partition bound once more proves $o(X) \leq \#_{r \in R} \omega^{\beta_r}$.

Lower bound. If $i, j \in R$ and $i < j$, then $\beta_i \geq \beta_j$; otherwise i would not survive. Order the elements of R by nonincreasing exponent, breaking ties by any linear extension of their induced order. This total order is compatible with the order inherited from I .

Any linear extension of an induced subposet of a finite poset extends to a linear extension of the whole poset. To see this directly, add the chosen comparisons between elements of the subposet. A directed cycle would give a chain of newly added comparisons, linked by old directed paths, whose endpoints lie in the subposet. Each old path already induces a comparison in its inherited order. Such a cycle would contradict compatibility of the chosen total order. The enlarged acyclic relation therefore has a linear extension.

Apply this observation to $R \subseteq I$. Concatenate maximal linearizations of all the X_i in the resulting order of I . Condition (6) ensures that this concatenation is a linearization of X . Its restriction to survivor blocks is an ordinary sum of the monomials ω^{β_i} in nonincreasing exponent order. That ordinary sum equals their natural sum. The entire linearization has at least this order type, giving the lower bound. The upper bound already proved shows that the concatenation is maximal. $\square$

Remark 4.3. The theorem permits repeated exponents. A higher stratum of *equal* exponent does not absorb a lower one; both contributions survive unless a strictly larger exponent occurs still higher. This strict inequality is responsible for the finite coefficients in later formulas.

The proof also explains why an algorithm considering only block orders is not by itself enough. Such an algorithm supplies a lower bound. The cofinal-absorption argument controls all possible interleavings inside linearizations and supplies the matching upper bound.

5 The frontier poset of a finite order

Definition 5.1. For a nonempty finite poset Q , let $\mathcal{M}(Q)$ be its set of *inclusion-maximal* antichains. For $A \in \mathcal{M}(Q)$ put

$$D(A) = \{q \in Q : \exists a \in A (q < a)\}.$$

Order maximal antichains by domination:

$$A \leq B \iff (\forall a \in A)(\exists b \in B) a \leq b. \quad (9)$$

We call $(\mathcal{M}(Q), \leq)$ the *frontier poset*.

Maximality here is by inclusion, not cardinality. A maximal antichain can have fewer elements than the width of Q . This is exactly what happens in the two orientations of a three-point fork.

Lemma 5.2. For $A, B \in \mathcal{M}(Q)$,

$$A = \min(Q \setminus D(A)), \quad A \leq B \iff D(A) \subseteq D(B). \quad (10)$$

In particular, $A \mapsto D(A)$ is injective and $\leq$ is a partial order. The least and greatest frontiers are $\min Q$ and $\max Q$.

Proof. Every predecessor of an element of A lies in $D(A)$, while A itself is disjoint from $D(A)$. Conversely, suppose $x \in \min(Q \setminus D(A))$. By maximality, x is comparable with an element $a \in A$. It cannot lie strictly below a , since then $x \in D(A)$. It cannot lie strictly above a , since $a \notin D(A)$ would contradict minimality of x . Thus $x = a$.

If $A \leq B$, every strict predecessor of an $a \in A$ is a strict predecessor of some $b \in B$, proving $D(A) \subseteq D(B)$. Conversely, assume this inclusion and take $a \in A$. If $a \in D(B)$, it is

strictly below some $b \in B$. Otherwise every strict predecessor of a lies in $D(A) \subseteq D(B)$, so $a \in \min(Q \setminus D(B)) = B$. Either way a is dominated by B .

Injectivity follows from the first identity. Every element of a finite poset lies above a minimal element and below a maximal one, giving the last assertion. $\square$

For the N -shaped poset with comparisons $a < c$, $b < c$, $b < d$, the frontiers form the three-element chain

$$\{a, b\} < \{a, d\} < \{c, d\}. \quad (11)$$

The middle frontier is not a level of the original height-two poset. This already suggests why its finitary powerset can have three ordinal height stages rather than two.

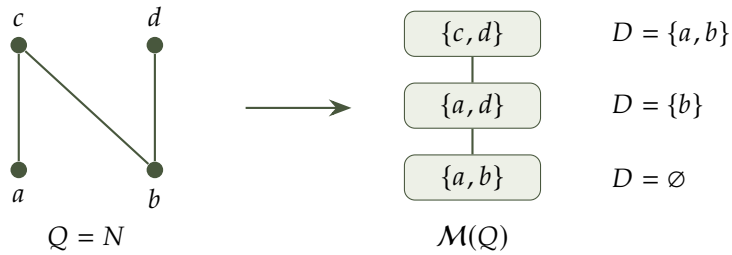


Figure 1: A height-two skeleton can have a height-three frontier poset. The right-hand labels are the completed-fiber sets.

6 Canonical strata of the finitary powerset

Let $S = \sum_{q \in Q}^{\text{lex}} P_q$, where Q is finite and nonempty, and assume that every nonempty fiber P_q has no finite cofinal subset. For $K \in \mathcal{K}(S)$ define its set of *completed fibers* by

$$D_K = \{q \in Q : P_q \subseteq K\}.$$

Lemma 6.1 (Canonical frontier). *Every $K \in \mathcal{K}(S)$ has a unique frontier $A \in \mathcal{M}(Q)$ with $D_K = D(A)$. For this frontier, K contains all of each fiber indexed by $D(A)$, intersects fibers indexed by A in finitely generated downsets, and is empty on all other fibers.*

Proof. Write $K = \downarrow F$ with F finite, let B be the set of indices meeting F , and let $A_0 = \max B$. For $F = \emptyset$, take $B = A_0 = \emptyset$.

If $q < a$ for some $a \in A_0$, a generator in P_a lies above every point of P_q , so $P_q \subseteq K$. If no generator has index strictly above q , then $K \cap P_q$ is generated internally by the finite set $F \cap P_q$. By (1), it cannot be all of P_q . Therefore

$$D_K = D(A_0) = \bigcup_{a \in A_0} \{q : q < a\}. \quad (12)$$

Write this downset as D and put $A = \min(Q \setminus D)$. Every member of A_0 belongs to A : its predecessors lie in D , and it does not lie in D itself. Conversely, every strict predecessor of an element of A lies in D . Thus $D(A) = D$, since $D = D(A_0) \subseteq D(A)$ as well.

The set A is a maximal antichain. An element outside D lies above a minimal element of $Q \setminus D$, while an element of D lies below some member of $A_0 \subseteq A$. Thus every element of Q is comparable with a member of A .

A nonempty intersection $K \cap P_q$ with $q \notin D$ requires a generator at index q itself, with no generator strictly above it. Hence $q \in A_0 \subseteq A$. These intersections are generated by $F \cap P_q$. The uniqueness of A follows from Lemma 5.2. $\square$

Let X_A be the set of all $K \in \mathcal{K}(S)$ with completed-fiber set $D(A)$. For $a \in A$, write $K_a = K \cap P_a$. The lemma gives a precise product representation, with one support condition:

$$X_A \cong \left\{ (K_a)_{a \in A} \in \prod_{a \in A} \mathcal{K}(P_a) : D(A) = \bigcup_{\{a: K_a \neq \emptyset\}} \{q : q < a\} \right\}. \quad (13)$$

The order is coordinatewise inclusion. To verify sufficiency, choose finite generators for the nonempty K_a and take their union inside S . The support condition makes its completed-fiber set exactly $D(A)$. Necessity follows from (12).

In particular, as induced subposets,

$$\prod_{a \in A} \mathcal{K}(P_a)^+ \subseteq X_A \subseteq \prod_{a \in A} \mathcal{K}(P_a). \quad (14)$$

Every X_A is nonempty, since all active coordinates can be chosen nonempty. The empty downset belongs to the stratum indexed by $\min Q$; there is no extra empty stratum.

Lemma 6.2 (Cofinal frontier stratification). *The partition*

$$\mathcal{K}(S) = \bigsqcup_{A \in \mathcal{M}(Q)} X_A$$

is a finite cofinal stratification indexed by $\mathcal{M}(Q)$.

Proof. If $K \subseteq L$, every fiber completed by K is completed by L . Thus $D_K \subseteq D_L$, and Lemma 5.2 gives the required comparison of frontier indices.

For upward cofinality, let $A \leq B$ and $K \in X_A$. Choose nonempty finitely generated downsets $L_b \in \mathcal{K}(P_b)^+$ for every $b \in B$. When $b \in A \cap B$, choose L_b to contain K_b : adjoin one point to its finite generating set if necessary. Use these coordinates and completed fibers $D(B)$ to form $L \in X_B$.

We have $D(A) \subseteq D(B)$. Moreover, each $a \in A \setminus B$ lies strictly below some $b \in B$, so its entire fiber is completed in L . The remaining active coordinates, indexed by $A \cap B$, were explicitly enlarged. Hence $K \subseteq L$. $\square$

The last argument chooses L in response to K . It does not say that *every* member of X_B dominates K . For example, for the skeleton consisting of an isolated point x and a chain $y < z$, the frontiers $\{x, y\} < \{x, z\}$ share an x -coordinate. A downset with a large x -coordinate in the first stratum is not below one with a small x -coordinate in the second. This is why replacing the strata by fully ordered blocks would give only a bound without Theorem 4.2.

7 The general maximal-order-type formula

Lemma 7.1 (Removing the least downset). *If $\text{o}(\mathcal{K}(P))$ is infinite, then $\text{o}(\mathcal{K}(P)^+) = \text{o}(\mathcal{K}(P))$.*

Proof. The empty downset is below everything else, so every linearization is a single least point followed by a linearization of $\mathcal{K}(P)^+$. Attainment of maximal linearizations gives $\text{o}(\mathcal{K}(P)) = 1 + \text{o}(\mathcal{K}(P)^+)$. The latter type must be infinite; for every infinite ordinal η , $1 + \eta = \eta$. $\square$

Remark 7.2 (Purity already excludes finite cofinality). In the next theorem the no-finite-cofinality assumption is stated to make the structural argument visible, but it follows from the positive

pure type assumption. Indeed, a finite cofinal subset of P would make P itself the greatest member of $\mathcal{K}(P)$. Then

$$o(\mathcal{K}(P)) = o(\mathcal{K}(P) \setminus \{P\}) + 1$$

would be a successor ordinal, whereas ω^β is a limit for $\beta > 0$. Thus the numerical hypothesis alone implies this boundary condition (and also implies that P is nonempty).

Theorem 7.3 (Finite lexicographic gluing). *Let Q be a finite nonempty poset. For each $q \in Q$, let P_q be a nonempty WPO having no finite cofinal subset and satisfying*

$$o(\mathcal{K}(P_q)) = \omega^{\beta_q}, \quad \beta_q > 0.$$

For $A \in \mathcal{M}(Q)$ define

$$\gamma_A = \#_{a \in A} \beta_a, \quad \mathcal{R} = \{A \in \mathcal{M}(Q) : \nexists B \succ A (\gamma_B > \gamma_A)\}.$$

Then

$$o\left(\mathcal{K}\left(\sum_{q \in Q}^{\text{lex}} P_q\right)\right) = \#_{A \in \mathcal{R}} \omega^{\gamma_A}. \quad (15)$$

Proof. For each frontier A , the two outer products in (14) have the same maximal order type, by Lemma 7.1 and the Cartesian-product formula:

$$\bigotimes_{a \in A} \omega^{\beta_a} = \omega^{\#_{a \in A} \beta_a} = \omega^{\gamma_A}.$$

Induced-subposet monotonicity therefore gives $o(X_A) = \omega^{\gamma_A}$. Lemma 6.2 provides a finite cofinal stratification. Apply Theorem 4.2. $\square$

The theorem is compositional in a precise, limited sense. Once the pure values $o(\mathcal{K}(P_q))$ are known and the no-finite-cofinality hypotheses are verified, no additional internal data about the fibers enter the output maximal order type. All remaining work takes place in the finite weighted frontier poset. This does not supply a method for discovering the required fiber values for arbitrary input WPOs.

Example 7.4 (A nonuniform weighted frontier). Take the N -poset from (11) and put

$$(\beta_a, \beta_b, \beta_c, \beta_d) = (3, 1, 1, 2).$$

The successive frontier exponents are 4, 5, 3. The first is absorbed by the second, while the second and third survive. Therefore

$$o(\mathcal{K}(S)) = \omega^5 + \omega^3.$$

This is realized, for example, by the ordinal fibers $(\omega^3, \omega, \omega, \omega^2)$.

8 Uniform ordinal fibers: an ordinal polynomial

Fix $\rho > 0$ and $\alpha = \omega^\rho$. Finitely generated downsets of the chain α are the empty set and principal downsets $\downarrow \xi$ for $\xi < \alpha$. Thus

$$\mathcal{K}(\alpha) \cong 1 + \alpha \cong \alpha. \quad (16)$$

The ordinal α is a nonzero limit and has no finite cofinal subset. All hypotheses of Theorem 7.3 hold with $\beta_q = \rho$.

Define the nonnegative integers

$$c_k(Q) = \#\{A \in \mathcal{M}(Q) : |A| = k \text{ and } (\forall B \geq A) |B| \leq k\}. \quad (17)$$

The notation $\#$ outside ordinal operations in this display denotes ordinary finite cardinality. In words, $c_k(Q)$ counts size- k maximal antichains not dominated by a larger maximal antichain.

Corollary 8.1 (Uniform maximal order type). *For a finite nonempty poset Q of width W and $\alpha = \omega^\rho$, $\rho > 0$,*

$$\boxed{o(\mathcal{K}(S_\alpha(Q))) = \omega^{\rho \otimes W} \cdot c_W(Q) + \omega^{\rho \otimes (W-1)} \cdot c_{W-1}(Q) + \cdots + \omega^\rho \cdot c_1(Q).} \quad (18)$$

Terms with zero coefficients are omitted. The displayed additions and coefficient multiplications are ordinary ordinal operations.

Proof. The exponent of a frontier is $\rho \otimes |A|$, by (2). Since $\rho > 0$, this is strictly increasing with the positive integer $|A|$. Therefore a frontier survives precisely under the condition in (17). Collect equal exponents in Theorem 7.3 and list them in decreasing order. $\square$

For $\rho = 1$, the output is literally a finite Cantor-normal-form polynomial in ω whose coefficients depend on the finite skeleton:

$$o(\mathcal{K}(S_\omega(Q))) = \sum_{k=W}^1 \omega^k c_k(Q).$$

Thus the infinite ordinal calculation separates into a finite combinatorial statistic and an explicit substitution in ordinal arithmetic.

Corollary 8.2 (Leading coefficient). *The coefficient $c_W(Q)$ is the number of antichains of Q having maximum cardinality W .*

Proof. Every maximum-cardinality antichain is inclusion-maximal, and none can be dominated by an antichain of larger cardinality. Conversely, every frontier counted by c_W has cardinality W . $\square$

Remark 8.3 (Natural multiplication cannot be replaced). For $\rho = \omega + 1$ and a two-point antichain Q ,

$$\rho \otimes 2 = \omega \cdot 2 + 2, \quad \rho \cdot 2 = \omega \cdot 2 + 1.$$

Consequently the maximal order type is $\omega^{\omega \cdot 2 + 2}$, not $\omega^{\omega \cdot 2 + 1}$. The code keeps these two multiplication operations separate.

9 Exact height and an attaining chain

Let

$$\ell(Q) = h(\mathcal{M}(Q)),$$

where height of the finite frontier poset means the largest number of elements in one of its chains.

Theorem 9.1 (Uniform height). *If Q is finite and nonempty and $\alpha = \omega^\rho$ with $\rho > 0$, then*

$$\boxed{h(\mathcal{K}(S_\alpha(Q))) = \alpha \cdot \ell(Q).} \quad (19)$$

There is a chain of this exact order type inside $\mathcal{K}(S_\alpha(Q))$.

Proof. Upper bound. Choose an order isomorphism $\theta : \mathcal{K}(\alpha) \rightarrow \alpha$ from (16). For a frontier A , let $d(A)$ be the largest number of strict steps in a chain ending at A in $\mathcal{M}(Q)$. Then $0 \leq d(A) < \ell(Q)$, and $A < B$ implies $d(A) < d(B)$.

For $K \in X_A$, define

$$t_A(K) = \#_{a \in A} \theta(K_a), \quad R(K) = \alpha \cdot d(A) + t_A(K).$$

All coordinates $\theta(K_a)$ are below α , so finite natural-sum closure gives $t_A(K) < \alpha$. Thus $R(K) < \alpha \cdot \ell(Q)$.

If $K \subseteq L$ lie in the same stratum, their active coordinate tuples are strictly ordered componentwise. Strict monotonicity of natural sum gives $t_A(K) < t_A(L)$. If their frontiers are distinct, say $A < B$, then

$$R(K) < \alpha \cdot (d(A) + 1) \leq \alpha \cdot d(B) \leq R(L).$$

The index-respecting property rules out all other cases. Hence R is strictly increasing on strict inclusion, so well-founded induction yields $r_{\mathcal{K}(S)}(K) \leq R(K)$. Since $R(K) < \alpha \cdot \ell(Q)$ for every K , the definition of height gives the required upper bound.

Lower bound and attainment. Choose a longest frontier chain

$$A_0 < A_1 < \cdots < A_{\ell-1}, \quad \ell = \ell(Q).$$

For each $i < \ell - 1$, choose $a_i \in A_i \setminus A_{i+1}$. Such a point exists: if one inclusion-maximal antichain were properly contained in another, it would not be maximal. Since $A_i \leq A_{i+1}$ and $a_i \notin A_{i+1}$, we have $a_i \in D(A_{i+1})$. It belongs to every later completed-fiber set as well. At the last stage choose any $a_{\ell-1} \in A_{\ell-1}$.

For each i and $\xi < \alpha$, form $K_{i,\xi} \in X_{A_i}$ by completing all fibers in $D(A_i)$, setting every active coordinate other than a_i to $\downarrow 0$, and setting the a_i -coordinate to $\downarrow \xi$. Every active coordinate is nonempty, so these are valid members of the stratum. For fixed i , the family $(K_{i,\xi})_{\xi < \alpha}$ is a chain of type α .

If $i < j$, the varying coordinate a_i is completed at stage j . Any other coordinate in A_i is either completed at stage j or remains an active coordinate there, whose downset contains $\downarrow 0$. Already completed fibers stay completed. It follows that $K_{i,\xi} \subseteq K_{j,\eta}$ for all $\xi, \eta < \alpha$. Inclusion is strict because $D(A_i) \subsetneq D(A_j)$: some fiber is fully present at stage j but not at stage i .

The ordered concatenation of these ℓ many chains is therefore a chain of type $\alpha \cdot \ell$. This proves the lower bound and attainment. $\square$

The height and maximal-order-type formulas inspect different aspects of the same finite object. Height records the longest chain of frontiers. Maximal order type records the exponents of frontiers that cannot be absorbed from above. Neither statistic determines the other.

10 Examples and comparisons

10.1 Chains, antichains and oriented forks

If Q is a chain of n elements, its frontiers are its n singleton subsets, ordered in a chain. Every frontier survives, so both output invariants are $\alpha \cdot n$. This agrees with $S_\alpha(Q) \cong \alpha \cdot n$ and $\mathcal{K}(\alpha \cdot n) \cong \alpha \cdot n$.

If Q is an antichain of n elements, it has only one maximal antichain, namely Q itself. The output has maximal order type $\omega^{\rho^{\otimes n}}$ and height α . Directly, it is the Cartesian product of n copies of $\mathcal{K}(\alpha) \cong \alpha$.

For the upward fork $a < b, c$, the frontiers are $\{a\} < \{b, c\}$. The size-one frontier is absorbed, giving

$$o(\mathcal{K}(S_\omega(Q))) = \omega^2, \quad h(\mathcal{K}(S_\omega(Q))) = \omega \cdot 2.$$

For the downward fork $a, b < c$, the frontiers are $\{a, b\} < \{c\}$. Both survive, giving

$$o(\mathcal{K}(S_\omega(Q))) = \omega^2 + \omega, \quad h(\mathcal{K}(S_\omega(Q))) = \omega \cdot 2.$$

Reversing a finite skeleton need not preserve the output maximal order type, even though it preserves its size, ordinary height and width.

10.2 Three indistinguishable inputs with different outputs

Consider the following four-point height-two skeletons, all with lower points a, b and upper points c, d :

$$\begin{array}{l|l} Q_2 & a < c, a < d, b < c, b < d \\ Q_3 & a < c, b < c, b < d \\ Q_4 & a < c, b < d. \end{array}$$

The middle one is the N -poset. Their frontier posets have respectively two, three and four elements; all frontiers have size two. The third frontier poset is a square rather than a four-element chain.

Skeleton	$ \mathcal{M}(Q) $	$\ell(Q)$	$o(\mathcal{K}(S_\omega(Q)))$	$h(\mathcal{K}(S_\omega(Q)))$
Complete two levels Q_2	2	2	$\omega^2 \cdot 2$	$\omega \cdot 2$
N -poset Q_3	3	3	$\omega^2 \cdot 3$	$\omega \cdot 3$
Two disjoint chains Q_4	4	3	$\omega^2 \cdot 4$	$\omega \cdot 3$

Table 1: Equal input invariants do not force equal finitary-powerset invariants. These values are consequences of the proved formulas.

For every one of these inputs,

$$(o(S_\omega(Q_i)), h(S_\omega(Q_i)), w(S_\omega(Q_i))) = (\omega \cdot 4, \omega \cdot 2, 2). \quad (20)$$

Indeed, disjoint-union monotonicity bounds the input maximal type by $\omega \cdot 4$, and concatenating the four fibers in a linear extension attains it. The height is $\omega \cdot 2$, by a two-level rank bound and any chain of two comparable fibers. An antichain has at most one point in each fiber and its indices must be an antichain of Q_i , giving width two.

The general failure of functionality of finitary-powerset invariants was already proved in [1]; we do not claim that observation as new. Table 1 supplies a transparent finite-skeleton instance and explains the difference by explicit frontier statistics.

10.3 Disjoint unions of finite chains

Suppose Q is the disjoint union of k nonempty finite chains of lengths $n_1, \dots, n_k$. A maximal antichain chooses exactly one point from each chain. Hence

$$\mathcal{M}(Q) \cong \prod_{i=1}^k n_i, \quad |\mathcal{M}(Q)| = \prod_{i=1}^k n_i, \quad \ell(Q) = 1 + \sum_{i=1}^k (n_i - 1).$$

All frontiers have size k and survive. Thus

$$\begin{aligned} \mathfrak{o}(\mathcal{K}(S_\alpha(Q))) &= \omega^{\rho \otimes k} \cdot \prod_{i=1}^k n_i, \\ \mathfrak{h}(\mathcal{K}(S_\alpha(Q))) &= \alpha \cdot \left(1 + \sum_{i=1}^k (n_i - 1) \right). \end{aligned}$$

The maximal-type result also follows directly from $\mathcal{K}(S_\alpha(Q)) \cong \prod_i (\alpha \cdot n_i)$ and natural multiplication. The frontier derivation additionally explains the height coefficient.

11 Coefficients, counting and restricted reuse

11.1 Counting finite downsets in a leading coefficient

Proposition 11.1 (An explicit counting reduction). *Let R be a nonempty finite poset with n elements. Construct a height-two poset Q_R on $\{x^- : x \in R\} \cup \{x^+ : x \in R\}$ by*

$$x^- < y^+ \iff x \leq_R y,$$

and no other strict comparisons. Then $\mathfrak{w}(Q_R) = n$, and the coefficient of ω^n in $\mathfrak{o}(\mathcal{K}(S_\omega(Q_R)))$ is the number of order ideals of R , where an order ideal here means a downward-closed subset.

Proof. The n comparisons $x^- < x^+$ partition Q_R into n two-element chains, so every antichain has size at most n . All minus points form an antichain of size n , proving the width assertion.

An antichain of size n must choose exactly one of x^- , x^+ for each $x \in R$. Let I be the set of x for which it chooses x^+ . The chosen set is an antichain precisely when no $x \notin I$, $y \in I$ satisfy $x \leq_R y$. This is precisely the downward-closure condition on I . The construction is reversible: from an order ideal I , choose x^+ for $x \in I$ and x^- otherwise. Hence maximum antichains of Q_R are in bijection with order ideals of R . Apply Corollary 8.2. $\square$

This does not assert a computational-complexity classification without an additional counting-complexity theorem. It does show directly that an exact leading coefficient can encode a substantial finite counting problem. Even when the transfinite part of the answer is a single exponent, the coefficient is not generally a simple function of size, height and width of the skeleton.

11.2 When the output remains a single pure power

Proposition 11.2 (Pure-output criterion). *Under the hypotheses of Theorem 7.3, let $T = \max Q$, the greatest frontier. The output maximal order type is a single pure power of ω , with coefficient one, if and only if*

$$\gamma_T > \gamma_A \quad \text{for every } A \neq T. \tag{21}$$

In that case it equals ω^{γ_T} . For uniform fibers, (21) says that $\max Q$ is the unique maximum-cardinality antichain.

Proof. If (21) holds, T absorbs every other frontier and is the only survivor. Conversely, suppose some $A \neq T$ has $\gamma_A \geq \gamma_T$. If A is not already a survivor, repeatedly move to a strict successor of larger exponent. Finiteness forces this process to end at a survivor B . Its exponent is at least γ_A , and $B \neq T$: if a strict increase occurred its exponent exceeds γ_T , while if none occurred $B = A \neq T$. Since T always survives, there are at least two nonzero monomial contributions.

Their natural sum cannot be a single pure power with coefficient one. The uniform assertion follows from strict increase of $\rho \otimes k$ in k . $\square$

If a finite lexicographic sum S satisfies this criterion, it can itself be used as a fiber in another application of Theorem 7.3. Indeed, $\mathcal{o}(\mathcal{K}(S))$ has the required pure form. Also S has no finite cofinal subset: restrict any proposed finite cofinal set to a maximal index fiber; points outside that fiber cannot dominate its elements, contradicting the hypothesis there.

This is a genuine restricted reuse rule for the gluing construction. It must not be confused with unrestricted iteration of the powerset constructor: using $\mathcal{K}(S)$ as a new fiber would require information about $\mathcal{o}(\mathcal{K}(\mathcal{K}(S)))$, which has not been supplied by the criterion.

12 Algorithm and reproducible verification

12.1 Exact finite procedure

For $n = |Q|$ and $m = |\mathcal{M}(Q)|$, a direct algorithm first computes transitive closure and enumerates maximal antichains. It then computes $D(A)$, builds the inclusion order on these downsets, and assigns each frontier its exponent γ_A . A frontier is discarded exactly when a strict successor has a larger exponent. Equal exponents of retained frontiers are collected in Cantor normal form.

Listing 1: The survivor calculation; ordinary and natural operations stay distinct.

```
frontiers = inclusion_maximal_antichains(Q)
D = {A: strict_predecessors_of(A) for A in frontiers}
gamma = {A: natural_sum(beta[a] for a in A) for A in frontiers}
retained = [A for A in frontiers
             if not any(D[A] < D[B] and gamma[A] < gamma[B]
                        for B in frontiers)]
answer = natural_sum(omega_power(gamma[A]) for A in retained)
```

The routine for uniform fibers additionally computes the longest-chain length in the finite frontier poset. This produces the height coefficient and a standard predecessor-pointer implementation can recover a longest frontier chain for the explicit construction in Theorem 9.1.

Naive enumeration costs $O(n^2 2^n)$ elementary relation operations. Building and comparing all frontier downsets costs $O(m^2 n)$ without bit-packing; transitive closure costs $O(n^3)$. Ordinal arithmetic and comparison have their own cost in the chosen notation representation. With exponents regarded as supplied exact data, a convenient bound on the finite combinatorial part is

$$O(n^3 + n^2 2^n + m^2 n).$$

No polynomial-time claim in n is made. The simple implementation is intended as an auditable reference, not an optimized maximal-antichain enumerator.

The mathematical theorems allow arbitrary ordinal exponents. The supplied program represents hereditary Cantor normal forms below ε_0 , using integers and recursively represented exponents, with no floating-point arithmetic. This restriction is an implementation boundary, not a theorem hypothesis. Ordinals at or above ε_0 require a richer notation system or abstract ordinal operations.

12.2 Independent block-order oracle

A second routine computes the greatest ordinary sum obtainable by ordering pure blocks in a linear extension of a finite index poset. For a subset J of indices it uses the recurrence

$$B(\emptyset) = 0, \quad B(J) = \max_{i \in \max J} (B(J \setminus \{i\}) + \omega^{\beta_i}). \quad (22)$$

The last block must be maximal in the remaining index order, so the recurrence exhausts all block linearizations. It is independent of the survivor rule, though not independent of ordinary ordinal arithmetic. Equality of the two computations checks the finite optimization identity. Only the absorption proof upgrades it to a statement about all linearizations of the infinite WPO.

12.3 What was actually checked

The delivered verification run used Python 3.13.5 and passed all 216,386 assertions. The machine-readable report is `data/verification.json`. It records the following coverage.

Check family	Cases
Naturally labelled posets on one through six vertices	5,231
Nonuniform frontier weights from $\{1, 2, 3\}$, through four vertices	3,450
General pure-stratum exponents from $\{0, 1, 2\}$, through four vertices	3,450
Hereditary-CNF weighted examples	24
Canonical finite profiles, over 50 skeletons	1,222
Pairs of profiles checked against Hoare domination	39,370
Deterministic series-parallel constructions	500
Order-ideal / maximum-antichain counting reductions	407
Nonuniform lexicographic CNF cases against expanded-block DP	3,450
Random exact ordinal-arithmetic triples, with fixed seed	500

Table 2: Coverage of the finite regression suite. These are not counts of isomorphism classes or formal-proof obligations.

The naturally labelled census has counts 1, 2, 7, 40, 357, 4824 for sizes one through six. A relation is naturally labelled when $i <_Q j$ implies $i < j$ as integers. Every finite poset admits such a labelling, but isomorphic copies can occur more than once. The census is therefore complete for this labelled convention, not an isomorphism-reduced enumeration.

The profile checks independently generate downsets from subsets of two sample generator positions per fiber, then compare them with the canonical stratum description. A distinguished symbol denotes an *entire* completed ω -fiber; it is not identified with the largest finite sample value. These checks verify the support condition, the equivalence of Hoare domination and profile inclusion, index monotonicity, and the constructed upward-cofinal witnesses on those samples.

The series-parallel tests compare the frontier formula with recursive ordinary addition for ordered sums and natural multiplication for disjoint unions. The final test family checks Appendix A by expanding finite Cantor normal forms into pure blocks and applying (22). Exact test inputs, fixed seeds and the full finite census are included in the archive.

12.4 Reproduction and file map

The project has no third-party Python dependencies. From its root run

```
python code/verify.py
python code/frontiers.py data/N.json
latexmk -pdf article.tex
```

The default verifier regenerates its JSON report and CSV census. Running `frontiers.py` without an input file prints the named examples. The JSON format uses integers for finite ordinals and lists of `[exponent, coefficient]` pairs for a nonfinite Cantor normal form. For example, $\omega + 1$ is represented by `[[1, 1], [0, 1]]`.

`code/ordinals.py` implements exact arithmetic; `code/frontiers.py` contains finite-poset and invariant routines; `code/verify.py` contains the verification suite. `PROOF_AUDIT.md` lists proof dependencies and boundaries, while `SOURCES.md` records the literature-status checks.

13 Boundaries and failed extensions

13.1 Why uniform height is a real restriction

It is tempting to extend the height theorem by assigning each frontier its stratum height and taking the greatest sum along a frontier chain. That fails, even with three ordinal fibers.

Let Q consist of an isolated point x and a chain $y < z$, with

$$P_x = \omega^2, \quad P_y = P_z = \omega.$$

There are two frontiers $\{x, y\} < \{x, z\}$, and each stratum has height ω^2 . The naive chain-sum rule predicts $\omega^2 \cdot 2$. But the whole input is the disjoint union $\omega^2 \sqcup (\omega \cdot 2)$, so

$$\mathcal{K}(S) \cong \omega^2 \times (\omega \cdot 2).$$

For (ξ, η) in this product, $\xi \# \eta < \omega^2$. This natural sum is strictly increasing under strict componentwise comparison, giving height at most ω^2 . A copy of ω^2 in the first coordinate gives the reverse inequality. Thus

$$h(\mathcal{K}(S)) = \omega^2 \neq \omega^2 \cdot 2. \quad (23)$$

The dominant x -coordinate persists across the frontier change and cannot be reset. In the uniform proof, one could always choose a *disappearing* coordinate with full type α at every transition. Here the disappearing coordinate has only type ω , not ω^2 . This pinpoints the obstruction rather than merely exhibiting a numerical failure. The general maximal-type theorem remains valid and gives $\omega^3 \cdot 2$, agreeing with natural multiplication. [Added in batch 36: the exact nonuniform formula of Theorem 21.1 charges only the retiring y -coordinate here, giving $\omega + \omega^2 = \omega^2$; see Section 23.]

13.2 Finite cofinality and impure types

If a fiber has a finite cofinal subset, it may become fully present without any generator in a strict successor fiber. Equation (12) then fails. For the one-point skeleton with a one-point fiber, $\mathcal{K}(S)$ has two elements, and one of its downsets completes the sole fiber. The only maximal antichain has empty strict downset, so completed fibers no longer index the proposed strata. This example violates the theorem's hypotheses; it is a warning against silently dropping them.

Likewise, replacing pure values ω^{β_q} by arbitrary ordinal values inside the same survivor formula is unjustified. The elementary counterexample after Lemma 3.1 shows that small terms may survive beneath the terminal part of an impure cofinal component. A successful extension will need to track relevant cuts of the components, not merely their leading exponents.

13.3 Finite indexing is also essential to the argument

Finiteness is used in three separate places: maximal antichains provide a finite stratification; absorbed strata can be grouped into finitely many lower-exponent pieces; and survivor block orders extend by a finite ordering argument. For infinitely many lower-exponent pieces, their natural-sum bound need not stay below a given ω^β . The present proof therefore does not justify replacing the finite index poset by an arbitrary WPO, even when individual frontiers are finite.

The empty index poset is harmless but needs a separate convention: $S = \emptyset$, so $\mathcal{K}(S) = \{\emptyset\}$ and both output invariants are 1. The main formulas explicitly assume $Q \neq \emptyset$; the implementation handles the empty case separately.

14 What remains open and what the attempt achieved

Problem 1.1 is answered for maximal order type under the pure-type hypotheses, and for height with uniform additively indecomposable ordinal fibers. The finite-skeleton information required is explicit: the domination order on inclusion-maximal antichains, their weighted exponents, and its longest-chain length.

A natural next mathematical target is a nonuniform height formula that tracks which coordinates disappear on each frontier transition. The counterexample in (23) shows that a scalar stratum height is not sufficient for a naive chain-sum recurrence. The canonical product profiles retain enough information to formulate more refined rank bounds, but no exact general formula is claimed here. [Added 28 September 2026, batch 36: the two preceding sentences describe the report as pinned at e21766d04. That target has since been reached. Parts II and III prove the exact nonuniform height formula, which charges each coordinate when it disappears (Theorems 21.1, 22.2 and 27.1), and they re-scope the open questions of this section in Section 37.]

Ordinal width remains a separate target. The inequalities $w(X) \leq o(X)$ and the known invariant theory provide bounds, but the cofinal-absorption lemma does not automatically apply to antichain-tree rank: adding cross-stratum comparisons destroys antichains in ways not captured solely by successor exponents. Proving a corresponding width formula requires another argument.

A third direction is to replace the pure-type assumption by a finite collection of component cuts with controlled cofinality. The successful finite-CNF argument for actual lexicographic sums in Appendix A suggests what information may matter, but arbitrary WPO strata need not inherit the same cut structure from a chosen maximal linearization.

The present outcome is an exact, proof-backed partial answer to the published extension direction. Its main point is that a transfinite optimization problem reduces to a finite weighted frontier problem, with a separate cofinality proof ruling out larger interleavings. This claim is narrower than solving all finitary-powerset invariants, and stronger than an experimentally guessed formula for a few examples.

Part II. Nonuniform heights: ordinal-chain fibres

Sections 1–14 and the appendices form the original report (Part I), unchanged except for dated pointers to this addition. Parts II and III were added in batch 36 of ProveIt’s incoming-report intake, from four manuscripts written against the pinned state of this report. Section 15 records their provenance and fixes the notation of the addition.

15 Provenance, conventions and the answer

15.1 Four manuscripts, one addition

Section 13 proves that the tempting chain sum of stratum heights fails, and Section 14 names “a nonuniform height formula that tracks which coordinates disappear on each frontier transition” as the natural next target. Four independently prepared manuscripts, all pinned to ProveIt commit e21766d04c2b8a9b2cd43e563024b3bd1b9 (abbreviated e21766d04), prove that formula. At that commit this report’s directory was byte-identical to the one to which this addition was made; two of the manuscripts quote the Git blob 57498d1bdc438f469df33d296095edc2b8a5a707 of the inspected article.tex. The archives arrived in commit e13affd32 and were placed in commit 1a1396d4d, which staged their code, data and audit files under the prefixes below. Their manuscripts and delivery READMEs are not shipped; they survive in the arrival commit.

No.	Manuscript (date)	Archive	Prefix
07	<i>Retirement paths and exact heights of nonuniform finitary powersets</i> (24 Sep. 2026)	ProveIt_Nonuniform_Powerset_Heights	02-retire-
09	<i>Persistent frontiers and exact heights: a pure-height substitution theorem, a limit-ordinal calculus, and finite certificates</i> (Sep. 2026)	ProveIt_Persistent_Frontier_Heights	03-wpo-
11	<i>Persistent frontiers and exact ordinal heights: finitary powersets of nonuniform ordinal lexicographic sums, with exact local ranks</i> (24 Sep. 2026)	persistent_frontier_heights	04-peel-
12	<i>Exact ordinal heights of finitely generated downsets: persistent coordinates, mixed ordinal fibers, and a finite-state rank calculus</i> (24 Sep. 2026)	ProveIt_Ordinal_Heights_Research	05-ranks-

Table 3: The four merged manuscripts. The numbers are batch-36 manuscript numbers; the prefixes continue this report’s local file sequence, the unprefixed original counting as 01-. All four were prepared for Vladimir Reshetnikov with ChatGPT and all four are pinned to e21766d04. Archives 09 and 11 wrap inner directories of the same name but are different packages.

All four prove the same core formula for positive pure chain fibres (Theorem 21.1), with four different proof architectures. No statement of one contradicts a statement of another, and every worked value they share agrees. They differ in scope:

- **12** (the base of this part) proves a finite persistence theorem for abstract state systems with convex coordinate lifetimes, an exact ideal-state dynamic program, the height for *every* finite ordinal lexicographic sum (zero, finite and successor fibres included), exact point ranks, NP membership of the threshold problem, and a sufficient extension to non-chain

coordinates.

- **11** proves the same persistence theorem by a different, leading-exponent peeling induction that yields an upper-bound certificate, an exact all-antichain isomorphism for arbitrary ordinal fibres, and exact point ranks in *any* finite persistent system.
- **07** proves the core formula by a strict normalization onto maximal supports and a standalone height theorem for retirement boxes, with a paired stratum-types example and a frontier-graph program.
- **09** proves the formula for fibres that are arbitrary well-partial-orders whose finitary powersets have positive pure *height*. This hypothesis is incomparable with those of 12: more general in fibre structure, less general in ordinal range. It therefore keeps its own part (Part III, Section 26) with its own hypotheses. It also proves unbounded ambiguity of stratum heights and a height-two scheduling formula.

A result proved by several sources is printed once, with its provers named; a genuinely different proof is kept as a marked second route. The choices the merge had to make are listed in Section 39.

15.2 Scope, priority and relation to the published programme

Abriola, Halfon, Lopez, Schmitz, Schnoebelen and Vialard ask how to enlarge a compositional family while keeping its ordinal invariants computable [1, Section 6]; the author’s publication list, consulted by 12 on 24 September 2026, reports that paper’s acceptance in *Mathematical Structures in Computer Science*, but the version used by every source is v2. The finite-gluing height problem of this part is a concrete subproblem in that direction, not a separately numbered conjecture, and its solution does not settle the published programme for arbitrary wqos, width or iteration. Height, width and maximal order type are different invariants; Parts II and III compute height and individual well-founded ranks only.

Ordinal product-height arithmetic is classical: the Milner–Rado formula, recalled in Lemma 23.3 and recorded by Hilton [5, Theorem 2.20], serves only as a test oracle. Existence results connecting heights and chains also belong to the classical theory [3, 7]. The proofs of Part II do not invoke general chain-attainment theorems (12); Part III does, through Wolk’s theorem [6, 9]. Vialard’s published work on maximal order types of lexicographic products [4] concerns a different invariant, and no formula from it is used; nor is the survey’s subsequently corrected general lexicographic-product assertion, whose erratum is in the arXiv v5 revision of [3].

Each source reports a targeted literature search that did not locate the formula in the form proved; each says that this is not a proof of novelty. The accurate claim, in the sources’ words, is a proved extension relative to the inspected report, “not a certification of global novelty” (07), and “not a certified first solution of a famous named conjecture” (09); “resolves the gap” refers to the repository’s stated problem, “not to a verified claim of first publication” (11); priority “has not been independently established” (12). The uniform formula of Theorem 9.1 and the persistent counterexample (23) are prior material of this report, credited as such by all four and recovered here, not presented as new. All four manuscripts are unrefereed and none was checked by a proof assistant. No theorem of Parts II–III uses an unrefereed repository theorem as a premise: the canonical-frontier facts they need are reproved, with the sources’ proofs printed (Lemmas 21.2 and 31.1), and the survivor formula of Theorem 7.3 is not used.

15.3 The answer

For a finite poset Q and ordinals α_q ($q \in Q$) write

$$S_\alpha(Q) = \sum_{q \in Q}^{\text{lex}} \alpha_q$$

for the lexicographic sum of the ordinal chains α_q ; this extends the notation $S_\alpha(Q)$ of Section 2 to unequal fibres. For positive pure fibres $\alpha_q = \omega^{\rho_q}$, $\rho_q > 0$, and a chain $C = (A_0 < \dots < A_m)$ in the frontier poset $\mathcal{M}(Q)$, put

$$W(C) = \sum_{i < m} \max_{q \in A_i \setminus A_{i+1}} \alpha_q + \max_{q \in A_m} \alpha_q. \quad (24)$$

Every sum here is *ordinary ordinal addition, in the written order*; the differences $A_i \setminus A_{i+1}$ are nonempty. The core theorem (Theorem 21.1) is

$$\boxed{h(\mathcal{K}(S_\alpha(Q))) = \max_C W(C),} \quad (25)$$

over nonempty chains of $\mathcal{M}(Q)$, attained by an explicit chain of downsets. A coordinate is charged when it *retires*, that is, when its vertex leaves the frontier and its fibre becomes completed; a coordinate that persists through a transition must respect its old value and cannot supply a fresh copy of its whole fibre there. For arbitrary ordinal fibres, finite Cantor-normal-form refinement and a dynamic program on finite downsets give the answer (Theorem 22.2); the same program applied to a principal lower interval gives the exact rank of any finitely generated downset (Theorem 22.4). The mathematics allows arbitrary ordinal exponents; the executable notations of 07, 11 and 12 stop below ε_0 (07's code takes positive integers, read as order labels), and 09's code accepts symbolic exponent labels whose order it does not check.

15.4 Notation and conventions of the addition

The existing macros and meanings of Part I win: $\#$ is the Hessenberg natural sum (07, 09 and 11 print it $\oplus$; 11's macro of the same name as ours prints $\oplus$), r_p is well-founded rank, h is height, $\leq$ is Hoare domination on $\mathcal{M}(Q)$, a *frontier* is an inclusion-maximal antichain of Q , and $D(A)$ is the set of strict predecessors of an antichain A . Unmarked $+$ and $\cdot$ are ordinary ordinal operations. Table 4 fixes the new symbols and lists what each source called them.

Four conventions differ between the sources and are fixed as follows.

1. **Empty exits.** 11 and 12 give a transition with no departing coordinate, and a coordinate-free state, the weight 1: “an empty exit still costs one point” (12), “the fallback value is one, not zero” (11). Part II uses this convention throughout. 09 gives an empty retirement set the weight 0 ($\max \emptyset := 0$). For positive pure fibres the final answers coincide, since the terminal weight is infinite and absorbs every earlier finite term; intermediate dynamic-programming values differ (Remark 32.2), so certificates are not interchangeable. The weight-0 convention is wrong for finite fibres, which 09 excludes; it is used only in Part III.
2. **Maximization domain.** 07 and 11 maximize over all nonempty chains of $\mathcal{M}(Q)$, 12 over maximal chains, 09 over saturated chains from $\min Q$ to $\max Q$. These agree by Proposition 21.13.
3. **Capacity range.** 12's abstract theorem allows exponents $\rho_q \geq 0$; 11's requires $\rho_q > 0$ with the fallback weight 1. These are equivalent: a coordinate of length $\omega^0 = 1$ takes only the value 0, so deleting it changes neither the order nor any weight.

Here	Meaning; names in the sources
α_q	ordinal length of the chain fibre over q (07, 11, 12). In Part III, 09's $\alpha_q := h(\mathcal{K}(P_q))$ is renamed h_q : it is a height, <i>not</i> a fibre, and not $h(P_q)$.
$\text{Ant}(Q), \leq_H$	all antichains of Q , including $\emptyset$, under Hoare domination (12's Ant ; 07 writes $\leq_H$).
$\mathcal{J}(Q)$, “ideal”	all downsets of the finite skeleton Q ; an “ideal” of Q is <i>any</i> downset, not a directed one (12, 09). Directed ideals, the Idl of the neighbouring report on intrinsic approximations, do not occur.
support	the index antichain of the maximal generators of a downset (07, 11, 12). 09's “support ideal” I is the downset of Q generated by it.
$\max I$	the active set of an ideal I ; 09 calls it the “active frontier” $F(I)$. It is an antichain, <i>not</i> necessarily a frontier.
$\langle A; x \rangle$	the downset generated by $\{(q, x_q) : q \in A\}$ (07's $D(A, x)$, renamed to avoid Part I's $D(A)$).
T, U, A_t, λ_q, Y	state poset, label set, active sets, coordinate lengths and persistence order (12). 11 writes $I, L, A_i, \alpha_q, X(I, A, \alpha)$.
$e(s, t), t(t), L(C), H(T)$	exit and terminal weights with fallback value 1, chain cost and optimum (11's e, t, V, H ; 12's $w_i, L(C)$).
$w(A, B), t(A), W(C)$	frontier weights and chain cost (07; 09's w, t ; 11's $E(A, B), T(A)$).
$\Lambda, U^\top, d_\Lambda, c_\Lambda, Z$	11's peeling data (its λ, T, d, c, E).
$Q^\#, U^\#$	CNF-expanded skeleton and its infinite blocks (09's $Q^\#$; 07's Q^* ; 11's R, U ; 12's R).
$F(J), c(I, v), \tau_Q$	12's ideal-state program, edge cost and terminal cost (its F, c, τ_R); τ_Q is unrelated to the $\tau(q)$ of Appendix A. $F^M(A)$ is 07's frontier program (its $F(A)$), $p(t)$ is 11's pair program (its p).
$\text{cl}(A), \mathcal{K}^{\max}(S), \nu, \iota$	07's minimal completion (its $c(A)$), maximal-support part (its Y), normalization map (its N) and box embedding (its e).
$G_i, B, n_\Lambda(i), \text{res}_i$	07's retirement groups (with λ_i the largest capacity in G_i), box, large-group count and residual (its $R_i, E, c(i), R_i(z)$); 11's rank term θ is written ϑ .
W^0, T^0, V^0, Φ	Part III only: 09's weights with <i>empty maximum</i> 0, path value and optimum (its W, T, V, H); its programs d, g are d^0, g , its component chains $\mathcal{C}_q$ are C_q , and its height-two sets are renamed in Section 35.

Table 4: Symbols of Parts II–III and their names in the sources. Every source symbol not listed keeps its name.

4. **Labels.** The N -poset is $a < c, b < c, b < d$ everywhere here, as in Part I, 07, 11 and 12; 09 labels it $a < c, a < d, b < d$, and its examples are transcribed by the isomorphism $a \leftrightarrow b, c \leftrightarrow d$. The persistent example of (23) is an isolated x beside $y < z$, as in Part I and 12; 07 and 11 call these a and $b < c$, 09 calls them x and $a < b$. The delivered data files keep their own labels.

16 Height, rank bounds and the two additions

All posets and ordinals are sets and we work in ZFC; no large-cardinal hypothesis and no countability restriction is used. Rank and height are those of Section 2: $r_P(x) = \sup_{y < x} (r_P(y) + 1)$ and $h(P) = \sup_{x \in P} (r_P(x) + 1)$, with empty suprema 0. Thus $h(\emptyset) = 0$, a singleton has height 1, an ordinal chain α has height α , and $\mathcal{K}(\emptyset) = \{\emptyset\}$ has height 1, not 0. These agree with the rank of the tree of finite strictly descending sequences, including its empty root.

Lemma 16.1 (Height tools; 07, 11, 12). *Let X and Y be well-founded posets.*

- (i) *If $R : X \rightarrow \eta$ is strictly increasing on strict comparisons, then $r_X(x) \leq R(x)$ for every x and $h(X) \leq \eta$. (A poset admitting such a map is automatically well-founded.)*
- (ii) *If $f : X \rightarrow Y$ satisfies $x < x' \Rightarrow f(x) < f(x')$, then $h(X) \leq h(Y)$; no reflection of incomparability is required. In particular order embeddings cannot decrease height, and a strictly increasing copy of an ordinal η in X gives $h(X) \geq \eta$.*
- (iii) *For the ordinal sum of posets, $h(X + Y) = h(X) + h(Y)$.*
- (iv) *Natural sum is strictly increasing in each argument. Ordinary addition is nondecreasing in each argument, strictly increasing and continuous in its right argument, and need not be strictly increasing in its left argument.*

Proof. A descending sequence in X would give a descending sequence of ordinals. By well-founded induction, $r_X(x) = \sup_{y < x} (r_X(y) + 1) \leq \sup_{y < x} (R(y) + 1) \leq R(x)$, and $R(x) < \eta$ gives $r_X(x) + 1 \leq \eta$; this is (i). For (ii), induction in X gives $r_X(x) \leq r_Y(f(x))$, since each strict predecessor of x maps to a strict predecessor of $f(x)$; take suprema. Along an embedded chain the point at position ξ has rank at least ξ . In $X + Y$ the rank of a point of X is unchanged and that of $y \in Y$ is $h(X) + r_Y(y)$, by induction and continuity of ordinary addition in its right variable; this proves (iii). Natural addition aligns exponents and adds coefficients in Cantor normal form, which gives its strict monotonicity; the assertions on ordinary addition follow from its recursive definition. $\square$

No assertion about linearizations is involved: height is not maximal order type, and no maximal-linearization theorem enters Part II. Throughout, both an upper rank bound and an attaining chain are supplied; height is never inferred merely by listing some chains, since it is the rank argument that excludes larger, branching contributions (11). The two additions play different roles and are never interchanged:

$$\omega + \omega^2 = \omega^2, \quad \omega^2 + \omega > \omega^2, \quad \omega \# \omega^2 = \omega^2 + \omega.$$

Appending a nonzero ordinal on the right deletes the lower-exponent tail of the left operand below the leading exponent of the right operand, then combines coefficients at that exponent if present. A pure ordinal is ω^ρ , including $\omega^0 = 1$. By (3), finite natural sums of ordinals below a pure ordinal stay below it; moreover, if $\kappa = \omega^\sigma < \lambda = \omega^\rho$ with $\rho > 0$ and $n < \omega$, then $\kappa \cdot n < \lambda$, and every finite n is below every such λ (11). These facts hold for uncountable exponents as well.

Lemma 16.2 (Absorption with a strict margin; 12). *For any ordinal p , pure ordinal $w = \omega^\rho$ and $\delta < w$,*

$$p \# \delta < p + w. \quad (26)$$

Proof. For $\rho = 0$, $\delta = 0$ and the claim is $p < p + 1$. Suppose $\rho > 0$. Split p into its terms of exponent at least ρ , written $p_{\geq \rho}$, and its tail $p_{< \rho} < \omega^\rho$. Natural addition gives $p \# \delta = p_{\geq \rho} + (p_{< \rho} \# \delta) < p_{\geq \rho} + \omega^\rho = p + \omega^\rho$, by finite natural-sum closure below ω^ρ . $\square$

The strict margin controls transitions between states: coordinates that have not yet retired stay in a natural-sum term, while a completed stage contributes to an ordinary-sum prefix. Part III uses a different mechanism, deferred absorption (Lemma 29.2), for the same purpose.

17 A finite persistence theorem

The rank argument is isolated from the representation of finitely generated downsets by first proving it for abstract finite state systems (12, 11).

17.1 State systems with convex coordinate lifetimes

Definition 17.1 (Persistence system; 12, 11). Let T be a finite poset and U a finite set of coordinate labels. For each $t \in T$ choose an active set $A_t \subseteq U$, and require

$$s \leq t \leq u \implies A_s \cap A_u \subseteq A_t. \quad (27)$$

Give each label q a nonzero ordinal length λ_q . The *persistence order* is

$$Y = Y(T, A, \lambda) = \left\{ (t, x) : t \in T, x \in \prod_{q \in A_t} \lambda_q \right\},$$

with the empty product a singleton, ordered by

$$(t, x) \leq (u, y) \iff t \leq u \text{ and } x_q \leq y_q \text{ for every } q \in A_t \cap A_u. \quad (28)$$

Condition (27), 12's convex lifetimes and 11's no-resurrection condition (NR), says that along a chain a label may be absent at first, become active and disappear, but cannot disappear and then reappear: each label's set of active states is order-convex. Empty active sets are allowed. Coordinates shared by two states persist; a coordinate absent at the later state imposes no comparison there.

Lemma 17.2 (12, 11). *Relation (28) is a well-founded partial order.*

Proof. Reflexivity is immediate, and antisymmetry forces equality of states and then of every coordinate. For transitivity, let $(s, x) \leq (t, y) \leq (u, z)$; a label shared by s and u is active at t by (27), so $x_q \leq y_q \leq z_q$. Within one state, $x \mapsto \prod_q x_q$ strictly increases on strict product comparisons, so each state's product is well-founded; a strictly decreasing sequence changes its state only finitely often, and afterwards lies in one of these products. (Alternatively, use well-founded induction on the finite state order.) $\square$

The finite-product and finite-union closure of well-quasi-orders also shows that Y is a WPO; only well-foundedness is needed for the height proof (11).

Example 17.3 (Why (27) is structural; 12, 11). Take states $0 < 1 < 2$ with $A_0 = A_2 = \{q\}$ and $A_1 = \emptyset$. Then $(0, 1) \leq (1, \emptyset) \leq (2, 0)$ by (28), but $(0, 1) \not\leq (2, 0)$. Without (27) the relation need not be transitive, and taking its transitive closure would define a different construction. The condition is used twice below: for transitivity, and to stop a coordinate varied in an earlier block of a chain from reappearing.

17.2 Retirement costs

For the rest of this section the lengths are pure, $\lambda_q = \omega^{\rho_q}$ with $\rho_q \geq 0$. For states $s < t$ put

$$\mathbf{e}(s, t) = \max(\{1\} \cup \{\lambda_q : q \in A_s \setminus A_t\}), \quad \mathbf{t}(t) = \max(\{1\} \cup \{\lambda_q : q \in A_t\}). \quad (29)$$

For a nonempty chain (a *state path*) $C = (t_0 < \dots < t_m)$, not necessarily saturated and not necessarily starting at a minimal state, write $A_i = A_{t_i}$ and define the exit sets and stage costs

$$\begin{aligned} E_i &= A_i \setminus A_{i+1} \ (i < m), & E_m &= A_m, \\ w_i &= \max(\{1\} \cup \{\lambda_q : q \in E_i\}), & L(C) &= w_0 + w_1 + \dots + w_m. \end{aligned} \quad (30)$$

Thus $w_i = \mathbf{e}(t_i, t_{i+1})$ for $i < m$ and $w_m = \mathbf{t}(t_m)$. The fallback value is one, not zero: a coordinate-free state is not absent, and a strict transition with no departing coordinate still contributes a finite step. Finally put $H(T) = \max_C L(C)$ over all nonempty chains, and $H(\emptyset) = 0$.

Theorem 17.4 (Finite persistence theorem; 12 Thm 3.3, 11 Thm 3.3). *For a finite persistence system with pure lengths,*

$$\boxed{h(Y) = \max_{C \text{ a maximal chain of } T} L(C) = H(T).} \quad (31)$$

For every nonempty chain C of T , the configurations over the states of C form an induced suborder Y_C of height exactly $L(C)$ containing a chain of that exact type. There is an explicit strictly increasing ordinal-valued map on Y with all values below $H(T)$.

12 proves the theorem with $\rho_q \geq 0$ by a rank map over a fixed chain and a finite cover of the descending-sequence tree. 11 proves it with $\rho_q > 0$ and the fallback weight by a leading-exponent induction; by convention 3 of Section 15.4 the two statements are equivalent. The statement does not claim that either rank map equals the pointwise rank (11); exact ranks are computed separately in Theorem 19.4.

17.3 First proof (12): a rank map over a fixed chain

Fix $C = (t_0 < \dots < t_m)$, write $A_i = A_{t_i}$, and let $p_i = \sum_{k < i} w_k$ be the completed prefix. At a configuration x in stage i set

$$R_C(i, x) = p_i \# \left(\#_{q \in A_i} x_q \right), \quad (32)$$

with the natural sum over the empty set equal to 0.

Lemma 17.5 (12). *R_C is strictly increasing on strict comparisons of Y_C , and $R_C(i, x) < L(C)$ for every configuration over C .*

Proof. Range: every coordinate active at stage i retires later, either before a later state or at the final stage, and by (27) it cannot return after retiring. Hence its length is at most $W_i = \max_{i \leq k \leq m} w_k$, the natural sum δ_i of the values at stage i is below W_i , and Lemma 16.2 gives $R_C(i, x) = p_i \# \delta_i < p_i + W_i \leq p_i + \sum_{k=i}^m w_k = L(C)$.

Within one stage a componentwise strict increase raises at least one summand and so raises R_C . Let $(i, x) < (j, y)$ with $i < j$, and split the active set at stage i into $P = A_i \cap A_j$ and $D = A_i \setminus A_j$. Every member of D retires at some transition $k \in [i, j)$, since it cannot disappear and return. With $W = \max_{i \leq k < j} w_k$, the natural sum $\delta_D = \#_{q \in D} x_q$ is below W (also when $D = \emptyset$), so $p_i \# \delta_D < p_i + W \leq p_j$. Adding the persistent coordinates by natural sum and using $x_q \leq y_q$ for $q \in P$,

$$R_C(i, x) = (p_i \# \delta_D) \# \#_{q \in P} x_q < p_j \# \#_{q \in P} x_q \leq p_j \# \#_{q \in A_j} y_q = R_C(j, y).$$

The same ordinal function controls movement inside a stage and across arbitrarily many stages. $\square$

By Lemma 16.1, $h(Y_C) \leq L(C)$. The proof keeps the shared coordinates; it never replaces the strata by totally ordered blocks.

Lemma 17.6 (Block realization; 12 Lemma 3.5, 11 Lemma 4.1). *For every nonempty chain C of T , Y_C contains a chain of order type $L(C)$.*

Proof. If $w_i > 1$, choose $q_i \in E_i$ with $\lambda_{q_i} = w_i$; at stage i , vary its value through all $\xi < w_i$ and keep every other coordinate 0. This is a chain of type w_i . If $w_i = 1$, take the all-zero configuration at that stage alone (at the last stage, a whole largest-length block, or one tuple if A_m is empty). For $i < j$, a coordinate varied at stage i is absent at every later state of C : if it reappeared, (27) with t_{i+1} as the middle state would fail. So all coordinates shared by stages i and j were 0 at stage i , and every chosen configuration at stage i is below every chosen configuration at stage j , strictly because the states differ. The ordered concatenation has type $w_0 + \dots + w_m$. $\square$

A long variation of a coordinate can be placed wholly before the next stage only if that coordinate disappears there; persistent coordinates cannot supply the same long block repeatedly. It remains to pass from one chain of states to the whole system. Let $\text{Dec}(P)$ be the tree of finite strictly decreasing sequences in P , including the empty sequence; its rank at a nonempty sequence is the rank of the last element, and at the root it is $h(P)$.

Lemma 17.7 (Finite unions of prefix-closed trees; 07 Lemma 5.1, 12 Lemma 3.6). *Let $T_1, \dots, T_k$ be well-founded prefix-closed trees on a common alphabet, each containing the empty sequence. Their union T is well-founded, and at every node s of T*

$$r_T(s) = \max_{i: s \in T_i} r_{T_i}(s);$$

in particular the root rank of T is the largest root rank of the T_i .

Proof. An infinite branch of T would have arbitrarily long prefixes in one of the finitely many T_i , and prefix closure would put the whole branch into that tree. For the rank identity induct on s : the rank at s is the supremum, over children u , of $r_T(u) + 1$; substituting the induction hypothesis and exchanging the supremum with a *finite* maximum (successor commutes with finite maxima) gives the right-hand side. Prefix closure ensures that $u \in T_i$ implies $s \in T_i$; a tree in which s has no child contributes 0. $\square$

Proposition 17.8 (Finite monotone-index reduction; 07 Prop. 5.2). *Let $\sigma : X \rightarrow F$ be a monotone map from a well-founded poset to a finite nonempty poset, and for a maximal chain C of F let $X_C = \sigma^{-1}(C)$ with the induced order. Then*

$$h(X) = \max_{C \text{ maximal chain in } F} h(X_C).$$

Proof. The labels of a finite strictly descending sequence of X form, after removing repetitions, a chain of F , which lies in a maximal chain C . Hence $\text{Dec}(X) = \bigcup_C \text{Dec}(X_C)$, a finite union of prefix-closed trees. Apply Lemma 17.7 at the root. $\square$

Completion of the first proof of Theorem 17.4. Apply Proposition 17.8 to the state map $Y \rightarrow T$. Lemmas 17.5 and 17.6 give $h(Y_C) = L(C)$ for every maximal chain; an optimal C exists since there are finitely many, and Lemma 17.6 gives the attaining chain. A shorter chain of states extends to a maximal one, and its configurations form an induced suborder of the latter's; so maximizing over all nonempty chains gives the same value $H(T)$. $\square$

Remark 17.9 (A distinction that prevents a false proof; 12, 07). An arbitrary finite cover of a poset by induced subposets does not make its height the maximum of theirs: two singletons cover a two-point chain, and a union can create descending sequences crossing between pieces. Here the whole descending-sequence tree is covered by prefix-closed subtrees, so every entire descending sequence lies in one piece. Finiteness is essential for the maximum.

17.4 Second proof (11): leading-exponent peeling

This route supplies a recursive upper-bound certificate and the recurrence (37). By convention 3 we may delete coordinates of length 1 and assume every $\rho_q > 0$. For a state t let $\text{depth}(t) = \max\{k : t_0 < \dots < t_k = t\}$; thus $s < t$ implies $\text{depth}(s) < \text{depth}(t)$.

Assume some active set is nonempty and let

$$\Lambda = \max\{\lambda_q : q \in A_t \text{ for some } t\} = \omega^\rho, \quad U^\top = \{q : \lambda_q = \Lambda\}.$$

For $s < t$ let $\delta(s, t) = 1$ if $(A_s \setminus A_t) \cap U^\top \neq \emptyset$ and $\delta(s, t) = 0$ otherwise, and define, along a linear extension of T ,

$$d_\Lambda(t) = \max(\{0\} \cup \{d_\Lambda(s) + \delta(s, t) : s < t\}), \quad (33)$$

$$c_\Lambda = \max_{t \in T} (d_\Lambda(t) + \mathbf{1}_{A_t \cap U^\top \neq \emptyset}), \quad (34)$$

$$Z = \{t \in T : d_\Lambda(t) = c_\Lambda\}. \quad (35)$$

So $d_\Lambda(t)$ is the largest number of Λ -departure events on a path ending at t , and c_Λ also allows a terminal Λ -block. We call Z the terminal region.

Lemma 17.10 (11). $c_\Lambda \geq 1$ and $d_\Lambda \leq c_\Lambda$. The set Z is upward closed in T and no state of Z has an active label of length Λ . If $s < t$ and $d_\Lambda(s) = d_\Lambda(t)$, then $A_s \cap U^\top \subseteq A_t \cap U^\top$.

Proof. A state with an active Λ -label gives $c_\Lambda \geq 1$; the other bound follows from (34). By (33), d_Λ is nondecreasing along the order, so $s \in Z$, $s < t$ give $c_\Lambda = d_\Lambda(s) \leq d_\Lambda(t) \leq c_\Lambda$. An active Λ -label at $t \in Z$ would force $c_\Lambda \geq c_\Lambda + 1$. Finally, deleting a Λ -label along $s < t$ forces $d_\Lambda(t) \geq d_\Lambda(s) + 1$. $\square$

The restriction of the system to Z is again a persistence system, with fewer distinct lengths; write Y_Z for its order. This is the induction parameter. The exponents themselves need not be finite or countable.

Lemma 17.11 (Upper-bound step; 11). If a strictly increasing map $R_Z : Y_Z \rightarrow H(Z)$ is available (the empty map when $Z = \emptyset$), then there is a strictly increasing map $R : Y \rightarrow \Lambda \cdot c_\Lambda + H(Z)$.

Proof. Let κ be the largest length below Λ used anywhere, and $\kappa = 1$ if there is none. For $(t, x) \in Y$ put

$$u(t, x) = \kappa \cdot \text{depth}(t) + \#_{\substack{q \in A_t \\ \lambda_q < \Lambda}} x_q, \quad v(t, x) = \#_{\substack{q \in A_t \\ \lambda_q = \Lambda}} x_q, \quad \vartheta(t, x) = v(t, x) \# u(t, x).$$

The lower-coordinate natural sum is below κ (or absent when $\kappa = 1$), so $u(t, x) < \kappa \cdot (\text{depth}(t) + 1) < \Lambda$, using finiteness of the depth; also $v(t, x) < \Lambda$ and $\vartheta(t, x) < \Lambda$. Define

$$R(t, x) = \begin{cases} \Lambda \cdot d_\Lambda(t) + \vartheta(t, x), & t \notin Z, \\ \Lambda \cdot c_\Lambda + R_Z(t, x), & t \in Z. \end{cases} \quad (36)$$

The first line is below $\Lambda \cdot c_\Lambda$ because $d_\Lambda < c_\Lambda$ off Z ; the second is below $\Lambda \cdot c_\Lambda + H(Z)$.

Strictness. Within one state off Z , a strict coordinate increase strictly raises u or v and lowers neither, so ϑ and R rise. Let $s < t$ both lie off Z . If $d_\Lambda(s) < d_\Lambda(t)$, then $R(s, x) < \Lambda(d_\Lambda(s) + 1) \leq \Lambda d_\Lambda(t) \leq R(t, y)$. If $d_\Lambda(s) = d_\Lambda(t)$, no Λ -coordinate is lost (Lemma 17.10); shared coordinates do not decrease and new ones are nonnegative, so $v(s, x) \leq v(t, y)$, while $u(s, x) < \kappa(\text{depth}(s) + 1) \leq \kappa \text{depth}(t) \leq u(t, y)$. Thus ϑ rises strictly even if lower-length coordinates disappear. Inside Z use the strictness of R_Z and strict monotonicity of ordinary addition in its right argument. A comparison from outside Z into Z crosses the bound $\Lambda \cdot c_\Lambda$, and the reverse direction is impossible because Z is upward closed. $\square$

The finite depth term pays for state changes that erase smaller coordinates while leaving every largest coordinate alive. It is harmless at scale Λ , yet without it the natural sum of surviving coordinates alone need not be monotone.

Lemma 17.12 (Attaining the step; 11). *If $H(Z)$ is attained by a state path whenever $Z \neq \emptyset$, then some state path of T has cost $\Lambda \cdot c_\Lambda + H(Z)$.*

Proof. Let $Z \neq \emptyset$ and choose an attaining path in Z with first state t . As $d_\Lambda(t) = c_\Lambda$, some path ending at t realizes c_Λ departure events of Λ -labels. Keep the predecessor state of each such event, discard the other states except t , and concatenate with the path in Z . Each newly consecutive pair in the kept prefix has exit weight Λ : a Λ -label left right after the earlier state, and (27) prevents its presence at any later kept state. The cost is $\Lambda \cdot c_\Lambda + H(Z)$. If $Z = \emptyset$, a state attaining (34) has an active Λ -label and $d_\Lambda = c_\Lambda - 1$; keep the predecessor states of $c_\Lambda - 1$ departures on a path to it, and let that state contribute the last Λ -block. The cost is $\Lambda \cdot c_\Lambda$. $\square$

Second proof of Theorem 17.4. Induct on the number of distinct lengths active somewhere. If T is empty everything holds with value 0. If no label is active, $Y \cong T$, every weight is 1, $H(T)$ is the largest chain cardinality, the depth map is a strict map into it, and a longest chain attains it. Otherwise form Λ, c_Λ, Z ; no Λ -label is active on Z , so the induction hypothesis applies to Y_Z . Lemma 17.11 and Lemma 16.1 give $h(Y) \leq \Lambda \cdot c_\Lambda + H(Z)$; Lemma 17.12 supplies a path of that cost, which Lemma 17.6 realizes by a chain; and every path cost is at most $h(Y)$ by the same lemma. Hence

$$H(T) = h(Y) = \Lambda \cdot c_\Lambda + H(Z), \quad (37)$$

and the induction supplies the strict map and the attaining chain. $\square$

The recurrence (37) and the path formula come out together; neither is justified by extrapolation from finite approximations.

18 Algorithms and certificates for persistence systems

18.1 An ordinal longest-path computation (11)

For the weights (29) define, along a linear extension,

$$p(t) = \max(\{0\} \cup \{p(s) + e(s, t) : s < t\}), \quad H(T) = \max_{t \in T} (p(t) + t(t)). \quad (38)$$

Here $p(t)$ is the transition part of a path ending at t , before its final state is charged; it starts at 0 so that a path may begin at t .

Proposition 18.1 (11). *The recurrence (38) computes $H(T)$, and storing a maximizing predecessor yields an attaining state path.*

Proof. A path ending at t is t alone or has a penultimate state $s < t$; its transition sum is that of the path to s followed by $e(s, t)$. Ordinary addition is nondecreasing in its left argument, so the largest prefix value at s can be kept without losing a better continuation, and finite maxima are attained. Append the terminal weight and maximize over endpoints. $\square$

All strict comparable pairs are used; no restriction to Hasse edges is claimed, and the code does not assume that an optimal path is saturated. With $m = |T|$ and $\ell = |U|$, all edge weights cost $O(m^2\ell)$ membership and ordinal-comparison operations and the program $O(m^2)$ ordinal additions and comparisons. These are operation counts, not bit-complexity claims for arbitrary notations; building T from a skeleton may itself be exponential, and checking (27) on all triples costs $O(m^3\ell)$ (in the applications it is proved structurally).

18.2 Two complementary certificates (11)

A maximizing path is a *lower* certificate: one checks its state comparisons and the finite list of weights, and Lemma 17.6 turns it into an ordinal-indexed chain. The recursive records $(\Lambda, d_\Lambda, c_\Lambda, Z)$ are an *upper* certificate: one checks (33), (34) and (35) at every stage, and the recipe (36) interprets them as a strict rank map. It explains why every point, not only the points of a maximizing path, receives an ordinal below the proposed height. 11's reference implementation computes both answers, and its peeling routine never calls the weighted-path routine; this separates implementation failure modes but does not replace the proof.

Corollary 18.2 (Finite CNF support; 11 Cor. 5.2). *The positive exponents in $H(T)$ are among the ρ_q of used labels; its finite term, if any, is a natural number at most $|T|$; and the sum of the coefficients of its positive-exponent terms is at most the number of used labels.*

Proof. Each weight of an attaining path is 1 or a pure power ω^{ρ_q} , and ordinary addition of such a list can delete terms or merge equal exponents but creates no new exponent. For every infinite nonterminal weight choose a witnessing label as in Lemma 17.6; it disappears before the next state and never returns, so it witnesses no later weight, the terminal one included. A path has at most $|T|$ weights, so at most that many finite ones survive. $\square$

Corollary 18.3 (Order-pattern invariance with certificates; 11 Cor. 5.3). *Keep the finite system fixed and replace its positive exponents by other positive ordinals with the same equalities and strict order relations. The Cantor-normal-form coefficient pattern of $H(T)$ is unchanged, only the exponents being relabelled, and the same peeling certificate remains valid.*

Proof. At each stage the set of labels of largest remaining length is unchanged, hence so are δ , d_Λ , c_Λ and Z . Induct on the finitely many exponent levels using (37); the final coordinate-free height is unchanged. $\square$

No regularity, cofinality or countability assumption on an exponent is hidden in the theorem, and a symbolic implementation may compute with a finite ordered alphabet of exponent levels and substitute actual ordinals at the end. The lexicographic-sum forms of this invariance are Corollary 24.2 and, for Part III, Theorem 34.1.

19 Arbitrary capacities and exact ranks in persistent systems

19.1 Phase refinement

Keep (27) but let every length γ_q be an arbitrary *nonzero* ordinal. Split its Cantor normal form into a finite ordinal sum of pure blocks $\gamma_q = \sum_{b < B_q} \omega^{\tau_{q,b}}$, equal exponents repeated according to their coefficients, so that each value of the coordinate has a unique block index b and a unique offset below $\omega^{\tau_{q,b}}$. Refine a state t into the states (t, σ) with $\sigma \in \prod_{q \in A_t} B_q$, ordered by

$$(s, \sigma) \leq (t, \varsigma) \iff s \leq t \text{ and } \sigma_q \leq \varsigma_q \text{ for every } q \in A_s \cap A_t. \quad (39)$$

The active labels of (t, σ) are the pairs (q, σ_q) with $q \in A_t$ and $\tau_{q, \sigma_q} > 0$, of length $\omega^{\tau_{q, \sigma_q}}$; a singleton phase carries no coordinate.

Theorem 19.1 (Phase refinement; 11 Thm 8.1, 12 Cor. 6.4). *The refined system is a finite persistence system with pure lengths satisfying (27), and its persistence order is isomorphic to the original one. Hence the height of every finite persistence system with nonzero ordinal lengths, and a chain attaining it, are computed by refinement followed by Theorem 17.4.*

Proof. Relation (39) is a finite partial order by the transitivity argument of Lemma 17.2, now with block indices. If a refined label (q, b) is active at both ends of a chain of three refined states, the original label is active at the middle state by (27), and the block index there is squeezed between two equal indices; so refined lifetimes are convex. Decompose each coordinate value into block and offset. This is bijective because the block intervals partition γ_q . Values in different blocks compare by block index, regardless of offsets; within an infinite block they compare by offset; a singleton block offers no choice. So the original comparison is exactly the refined-state comparison together with inequalities on the common refined active labels. $\square$

Remark 19.2 (Why impure lengths cannot simply be substituted; 12, 11). Two comparable states carrying the same coordinate of length $\omega + 1$ give the order $2 \times (\omega + 1)$ of height $\omega + 2$ (its greatest tuple has natural-sum rank $1 \# \omega = \omega + 1$), while the unrefined cost rule gives $1 + (\omega + 1) = \omega + 1$ (12). One state with two coordinates of length $\omega + 1$ gives $(\omega + 1) \times (\omega + 1)$, of height $\omega \cdot 2 + 1$ (Section 23), while the unrefined terminal maximum gives $\omega + 1$ (11). Phase refinement is a substantive correction, not bookkeeping.

19.2 Exact point ranks in any finite persistent system (11)

Let $z = (t, x)$ be a point of a persistence order with nonzero ordinal lengths γ_q . Let $J = \{s : s \leq t\}$, keep the active sets on J , and cap

$$\gamma'_q = \begin{cases} x_q + 1, & q \in A_t, \\ \gamma_q, & q \notin A_t. \end{cases} \quad (40)$$

Every γ'_q is nonzero.

Lemma 19.3 (Principal-ideal representation; 11). *The inclusive principal ideal $Y_{\leq z}$ is isomorphic to $Y(J, A, \gamma')$.*

Proof. A point (s, y) is below z exactly when $s \leq t$ and $y_q \leq x_q$, that is $y_q < x_q + 1$, for every $q \in A_s \cap A_t$; all other coordinates keep their ranges, and the order on the restricted tuples is unchanged. $\square$

Theorem 19.4 (Exact point ranks in persistent systems; 11 Thm 8.4). *Form the capped system of Lemma 19.3, refine it by Theorem 19.1 and compute its height H_z . Then H_z is a successor and*

$$\boxed{r_Y(z) = \text{Pred}(H_z)}, \quad (41)$$

where $\text{Pred}(\eta + 1) = \eta$.

Proof. The principal ideal has greatest element z , and ranks of its points agree with their ranks in Y because all their predecessors lie in the ideal. So $h(Y_{\leq z}) = r_Y(z) + 1$, and the two preceding results compute this height. $\square$

In executable Cantor normal form the predecessor decrements the positive finite last coefficient (deleting the final term when it reaches zero); a returned limit would signal an implementation error. The algorithm does not claim that the rank maps of Theorem 17.4 are exact: it solves a different finite instance for each point, whose state space may be larger.

20 Support antichains and an exact finite dynamic program

20.1 The support model, including the empty downset

For a finite poset Q let $\text{Ant}(Q)$ be the set of all antichains, including $\emptyset$, under the Hoare order $A \leq_H B$ iff every $a \in A$ lies below some $b \in B$. The map $A \mapsto \downarrow A$ is an isomorphism onto the lattice $\mathcal{J}(Q)$ of downsets of Q , with inverse $I \mapsto \max I$. For an antichain A and $x \in \prod_{q \in A} \alpha_q$ write $\langle A; x \rangle = \downarrow \{(q, x_q) : q \in A\}$, with $\langle \emptyset; \emptyset \rangle = \emptyset$.

Lemma 20.1 (Support coordinates; 12 Lemma 4.1, 07 Lemma 2.2). *Let Q be finite and every $\alpha_q > 0$. The map $(A, x) \mapsto \langle A; x \rangle$ is an order isomorphism from*

$$\left\{ (A, x) : A \in \text{Ant}(Q), x \in \prod_{q \in A} \alpha_q \right\}, \quad (A, x) \leq (B, y) \iff A \leq_H B \text{ and } x_q \leq y_q \ (q \in A \cap B), \quad (42)$$

onto $\mathcal{K}(S_a(Q))$. Moreover, for antichains of any finite poset,

$$A \leq_H B \leq_H C \implies A \cap C \subseteq B. \quad (43)$$

Proof. A finite generating set can be reduced to its maximal elements. They form an antichain of points, no two in one chain fibre, and they are exactly the maximal points of the generated downset; this gives existence and uniqueness, the empty set included. A generator (a, x_a) lies below a generator (b, y_b) exactly when $a < b$, or $a = b$ and $x_a \leq y_b$. If $a \in A \cap B$, no other member of the antichain B lies above a , so the comparison at a must be $x_a \leq y_a$; this proves both directions of the order criterion. For (43) take $a \in A \cap C$ and $b \in B, c \in C$ with $a \leq b \leq c$; since $a, c \in C$, antichainness forces $c = a$ and then $b = a$. $\square$

Inequality (43) is the no-return lemma of 07 (Lemma 7.1) and 11 (Lemma 6.2); maximality of the antichains is not used. When all α_q are pure, the support model is exactly the persistence system of Definition 17.1 with state poset $\text{Ant}(Q)$ and active sets A . No assumption excluding finite cofinal subsets is needed, and singleton fibres ($\alpha_q = 1 = \omega^0$) are legitimate (12). Since there are finitely many supports and each carries a finite product of ordinal chains, $\mathcal{K}(S_a(Q))$ is a WPO; well-foundedness alone also follows because a descending sequence changes support only finitely often (07).

20.2 The ideal-state recurrence (12)

Let Q be a finite poset with pure nonzero weights λ_v (weight 1 allowed). For $I \in \mathcal{J}(Q)$ and v minimal in $Q \setminus I$, $I \cup \{v\}$ is a downset and the active set changes from $\max I$ to $\max(I \cup \{v\})$; the retiring set is

$$E(I, v) = \{a \in \max I : a < v\}. \quad (44)$$

Define

$$c(I, v) = \max(\{1\} \cup \{\lambda_a : a \in E(I, v)\}), \quad (45)$$

$$\tau_Q = \max(\{1\} \cup \{\lambda_a : a \in \max Q\}), \quad (46)$$

put $F(\emptyset) = 0$ and, for every nonempty ideal J ,

$$F(J) = \max_{v \in \max J} (F(J \setminus \{v\}) + c(J \setminus \{v\}, v)). \quad (47)$$

Theorem 20.2 (Exact finite algorithm; 12 Thm 4.2). *For a finite poset Q and pure nonzero ordinal weights,*

$$h(\mathcal{K}(S_\lambda(Q))) = F(Q) + \tau_Q. \quad (48)$$

This includes $Q = \emptyset$, where the answer is 1. An optimizing path in the downset lattice supplies an explicit attaining chain of finitely generated downsets.

Proof. A maximal chain of the finite lattice $\mathcal{J}(Q)$ starts at $\emptyset$, ends at Q and adds one vertex at each step (otherwise an intermediate ideal could be inserted); such chains are the linear extensions of Q . Under the support model the exit weight of the state $\max I$ before adjoining v is (45), and the terminal weight is (46). Theorem 17.4 reduces the height to the greatest path cost plus the terminal cost, and (47) computes that cost: the last vertex added is some $v \in \max J$, ordinary addition is monotone in its left argument, and finite maxima are attained, so keeping only the maximum at each state is valid. The chain is transported through Lemma 20.1. $\square$

The addition in (47) must not be replaced by natural addition: the latter proves rank bounds, it does not concatenate stages ($\omega + \omega^2 = \omega^2$, whereas $\omega \# \omega^2 = \omega^2 + \omega$). The recurrence has the shape of Part I's block recurrence (22) but computes a different invariant with different weights.

20.3 Certificates and complexity (12)

A certificate consists of the expanded finite poset, its pure weights, a value $F(I)$ for every ideal and one optimizing path. A verifier checks that all ideals are present, checks every equality (47), and independently sums the path costs and the terminal cost. The table is the upper certificate; the path is the lower certificate, whose chain is described in Lemma 17.6. With $N = |Q|$ and $J = |\mathcal{J}(Q)|$ there are at most NJ edges and at most NJ ordinal additions and comparisons; finite set operations add their usual costs, and the stored state space is $O(J)$ ordinal entries apart from the certificate. Since J can equal 2^N , this is an exact finite algorithm, not a polynomial-time claim. Enumerating all linear extensions is only an independent small-case test.

Corollary 20.3 (A supplied chain partition; 12 Cor. 4.3). *If the expanded skeleton Q is partitioned into w chains of lengths $n_1, \dots, n_w$, then $|\mathcal{J}(Q)| \leq \prod_{i=1}^w (n_i + 1) \leq (N + 1)^w$, and the ideal-state method uses $O(N(N + 1)^w)$ ordinal operations. Cantor-normal-form refinement (Section 22) does not increase the minimum number of chains in a partition of the original skeleton.*

Proof. An ideal meets each chain in a prefix, so the tuple of prefix lengths determines it; cross-chain order constraints only reduce the count. Replacing each vertex of a chain of the original partition by its consecutive pure blocks gives a chain partition of the refinement with no more chains. $\square$

A fixed supplied chain-partition size is therefore already a tractable case in the expanded input. This is not a bit-complexity result for succinctly encoded large Cantor coefficients.

Corollary 20.4 (Threshold certificates; 12 Cor. 4.4). *For a pure-block input, encode the finitely many exponent scales by their ordered integer labels. Given a threshold γ in Cantor normal form over an explicitly ordered finite list of scales, the problem $h(\mathcal{K}(S_\lambda(Q))) \geq \gamma$ belongs to NP with respect to this finite encoding.*

Proof. If the inequality holds, a winning linear extension of Q is a witness of length N : check that it is a linear extension, compute its retiring sets and stage costs, and compare their ordinary sum with γ . A path has $N + 1$ pure costs, so its normal form has coefficient sum at most $N + 1$ and exponent labels from the input together with 0; all this is polynomial in the finite encoding. Conversely an accepted path gives a chain of its computed type by Theorem 17.4. $\square$

This is an upper bound, not an NP-completeness claim, and it refers to the pure-block encoding: unary expansion of a succinct coefficient is not silently counted as polynomial preprocessing. [Added 29 September 2026, batch 42: the problem is NP-complete; hardness holds already for height-two skeletons with every fibre ω (Theorem 46.2 and Remark 46.5).]

21 The compressed formula for positive pure fibres

Throughout this section Q is finite and nonempty and $\alpha_q = \omega^{\rho_q}$ with $\rho_q > 0$. Such a fibre is a nonzero limit with no greatest element and no finite cofinal subset, so every $K \in \mathcal{K}(S_\alpha(Q))$ has the canonical frontier of Lemma 21.2(ii) below (12's reproof of Part I's Lemma 6.1). The support model of Section 20 is exact and handles finite endpoints; here a smaller state system, the frontier poset $\mathcal{M}(Q)$ of Part I, gives (25).

21.1 Weighted frontiers

Lemma 21.2(i) below (12 Lemma 5.1, reproving Part I's Lemma 5.2) gives $A = \min(Q \setminus D(A))$ and $A \leq B \iff D(A) \subseteq D(B)$. Two more facts are needed (07, 11, 12): if $A < B$ in $\mathcal{M}(Q)$ then $A \setminus B \neq \emptyset$, because one inclusion-maximal antichain cannot properly contain another; and the active-set assignment $A \mapsto A$ on $\mathcal{M}(Q)$ satisfies (27), by (43). For $A < B$ put

$$w(A, B) = \max_{q \in A \setminus B} \alpha_q, \quad t(A) = \max_{q \in A} \alpha_q, \quad (49)$$

and for a nonempty chain $C = (A_0 < \dots < A_m)$ let $W(C) = w(A_0, A_1) + \dots + w(A_{m-1}, A_m) + t(A_m)$ as in (24); for $m = 0$ this is $t(A_0)$.

Theorem 21.1 (Mixed pure frontier formula; 07 Thm 3.1, 11 Thm 1.1, 12 Thm 5.4). *For finite nonempty Q and $\alpha_q = \omega^{\rho_q}$, $\rho_q > 0$,*

$$h(\mathcal{K}(S_\alpha(Q))) = \max_{A_0 < \dots < A_m} \left(\sum_{i < m} \max_{q \in A_i \setminus A_{i+1}} \alpha_q + \max_{q \in A_m} \alpha_q \right). \quad (50)$$

The maximum is over all nonempty strict chains of $\mathcal{M}(Q)$, one-element chains included; it may be restricted to maximal chains, or to saturated chains from $\min Q$ to $\max Q$ (Proposition 21.13). It is attained by a chain in $\mathcal{K}(S_\alpha(Q))$ of exactly that order type.

Part III proves the same formula for well-partial-order fibres whose finitary powersets have pure height (09 Thm 7.2 with Cor. 3.2; Theorem 31.2). Two proofs follow; a third is the special case of Part III, whose upper bound goes through Wolk's theorem.

21.2 First proof (12, 11): two embeddings

Lemma 21.2 (Canonical frontier and comparison across strata; 12 Lemmas 5.1–5.2, 11 Lemma 6.1). Write P_q for the chain fibre α_q over q .

- (i) (12 Lemma 5.1.) For $A, B \in \mathcal{M}(Q)$, $A = \min(Q \setminus D(A))$ and $A \leq B \iff D(A) \subseteq D(B)$.
- (ii) (12 Lemma 5.2.) For $K \in \mathcal{K}(S_\alpha(Q))$ let $D_K = \{q \in Q : P_q \subseteq K\}$ be its set of completed fibres. There is a unique $A \in \mathcal{M}(Q)$ with $D_K = D(A)$, the canonical frontier of K . The downset K contains all fibres in $D(A)$, contains a possibly empty principal downset in every fibre of A , and is empty in all remaining fibres.
- (iii) (11 Lemma 6.1.) Let $K, L \in \mathcal{K}(S_\alpha(Q))$ have canonical frontiers A, B and active sections $K_a = K \cap P_a$, $L_b = L \cap P_b$. Then

$$K \subseteq L \iff A \leq B \text{ and } K_a \subseteq L_a \text{ for every } a \in A \cap B. \quad (51)$$

- (iv) Every tuple of nonempty principal sections at A occurs in the stratum of A .

Proof. (i) (12.) Every predecessor of an element of A is in $D(A)$, so A consists of minimal elements outside $D(A)$. Conversely, a minimal element outside $D(A)$ must be comparable with some element of the maximal antichain A ; it can lie neither below nor strictly above it, so it equals it. Domination implies inclusion of strict-predecessor sets. Conversely, suppose $D(A) \subseteq D(B)$. An $a \in A$ either belongs to $D(B)$, in which case it lies below a member of B , or is minimal outside $D(B)$ (its strict predecessors lie in $D(A) \subseteq D(B)$) and therefore belongs to B . This proves domination.

(ii) (12.) Let A_0 be the support of the maximal generators of K (Lemma 20.1). A fibre is entirely present exactly when some generator has strictly greater index: internal finite generators cannot be cofinal in a limit chain. Thus $D_K = D(A_0)$, where the same strict-predecessor notation is used for a possibly nonmaximal antichain, including $A_0 = \emptyset$. Set $D = D_K$ and $A = \min(Q \setminus D)$. Every member of A_0 is in A , while all predecessors of members of A belong to D . Hence $D(A) = D(A_0) = D$. Any element outside D lies above some member of A ; any element of D lies below a member of $A_0 \subseteq A$. Consequently A is inclusion-maximal. A nonempty nonfull section must lie at the index of a maximal generator, so its index is in A . Uniqueness follows from (i).

(iii) (11.) If $K \subseteq L$, completed fibres stay completed, so $D(A) \subseteq D(B)$ and $A \leq B$ by (i), and shared sections are included. Conversely, if $A \leq B$, every $a \in A \setminus B$ lies below some $b \in B$, strictly since $a \notin B$, so $a \in D(B) = D_L$ and the fibre of a is completed in L ; completed fibres of K remain completed in L because $D(A) \subseteq D(B)$; and the shared active sections are included by hypothesis.

(iv) (12.) The downset generated by one point (a, x_a) for each $a \in A$ has support A , so by the proof of (ii) its completed-fibre set is $D(A)$, and its section at a is $\downarrow x_a$. $\square$

Parts (i) and (ii) are Part I's Lemmas 5.2 and 6.1 for chain fibres, and (iii) and (iv) are Part I's comparison rule and the sufficiency half of (13). The proofs printed here are 12's and 11's reproofs, so the new height theorem does not rest on Part I's arguments. Not every tuple of empty or nonempty sections belongs to the indicated stratum, but every tuple with all active sections nonempty does, and this suffices for matching bounds (12).

Let Y_M be the persistence order with states $\mathcal{M}(Q)$, active sets A and lengths α_q .

Lemma 21.3 (Two embeddings; 12 Lemma 5.3, 11 §6.2). *There are order embeddings $Y_M \hookrightarrow \mathcal{K}(S_\alpha(Q)) \hookrightarrow Y_M$.*

Proof. First map: send (A, x) to the downset generated by the points (a, x_a) , $a \in A$. It contains every fibre over $D(A)$ and the nonempty principal section $\downarrow x_a$ at each $a \in A$; by Lemma 21.2 comparison of two images is frontier domination plus comparison on shared coordinates, which is the order of Y_M . Second map: use the order isomorphisms $\theta_q : \mathcal{K}(\alpha_q) \rightarrow \alpha_q$, $\theta_q(\emptyset) = 0$, $\theta_q(\downarrow \xi) = 1 + \xi$, which exist because $\mathcal{K}(\alpha_q) \cong 1 + \alpha_q = \alpha_q$ for infinite α_q . Send K with canonical frontier A to $(A, (\theta_a(K_a))_{a \in A})$. By (51) this preserves and reflects order. It need not be surjective: empty coordinates are subject to the support condition of Part I's (13), which the proof does not use. $\square$

First proof of Theorem 21.1. By Lemma 16.1(ii) the two embeddings give $h(Y_M) = h(\mathcal{K}(S_\alpha(Q)))$, and Theorem 17.4 computes the first height. Every exit along a strict chain of $\mathcal{M}(Q)$ is nonempty and every state is nonempty, with pure weights greater than 1, so the fallback 1 disappears and the weights are exactly (49). The first embedding transports the attaining chain. $\square$

Remark 21.4 (Height equivalence is not a rank identification; 11). The two embeddings prove equality of heights, not preservation of pointwise rank: the zero tuple of Y_M need not correspond to a rank-zero downset. For exact ranks use the support model (Theorems 19.4 and 22.4), never this compression.

21.3 Second proof (07): normalization and retirement boxes

This route avoids abstract state systems. It normalizes onto maximal supports, covers descending sequences by frontier chains, and computes the height over one chain through a standalone box theorem.

Definition 21.5 (Minimal completion; 07). For an antichain A of Q let $\text{Inc}(A)$ be the set of vertices incomparable with every member of A , and $\text{cl}(A) = A \cup \min \text{Inc}(A)$; thus $\text{cl}(\emptyset) = \min Q$.

Lemma 21.6 (Least maximal completion; 07 Lemma 4.2). *$\text{cl}(A)$ is a maximal antichain, and it is the least member of $\mathcal{M}(Q)$ lying Hoare-above A . Consequently $A \leq_H B$ implies $\text{cl}(A) \leq \text{cl}(B)$, and*

$$A \leq_H B \implies A \setminus B \subseteq \text{cl}(A) \setminus \text{cl}(B). \quad (52)$$

Proof. The members of $\min \text{Inc}(A)$ are pairwise incomparable and incomparable with A ; every vertex of $\text{Inc}(A)$ lies above one of them, and every other vertex is comparable with a member of A ; so $\text{cl}(A)$ is maximal. Let $B \in \mathcal{M}(Q)$ with $A \leq_H B$ and $u \in \min \text{Inc}(A)$, and suppose no member of B lies above u . Maximality of B gives $b \in B$ with $b < u$. Then $b \in \text{Inc}(A)$: $a \leq b$ would give $a < u$, and $b \leq a$ with $a \leq b' \in B$ would force $b = a = b'$, both contrary to $u \in \text{Inc}(A)$. This contradicts the minimality of u . So $\text{cl}(A) \leq_H B$; apply this to $\text{cl}(B)$ for monotonicity. Finally, $a \in A \setminus B$ lies strictly below some $b \in B \subseteq \text{cl}(B)$, so $a \notin \text{cl}(B)$, while $a \in \text{cl}(A)$. $\square$

Remark 21.7 (Editorial: three names for one antichain). For every antichain A , $\text{cl}(A) = \min(Q \setminus D(A))$. Both sets contain A . A member $u \notin A$ of $\min(Q \setminus D(A))$ is not below a member of A (it would lie in $D(A)$) and not above one, say $a < u$ (then $a \notin D(A)$ would contradict minimality); so $u \in \text{Inc}(A)$, and it is minimal there because $\text{Inc}(A)$ is disjoint from $D(A)$. Conversely let $u \in \min \text{Inc}(A)$; then $u \notin D(A)$, and a strict predecessor $v \notin D(A)$ of u would lie in $\text{Inc}(A)$, against minimality, or satisfy $a \leq v < u$ for some $a \in A$, against $u \in \text{Inc}(A)$. Hence 07's completion, the canonical frontier of Lemma 6.1 (for A the support of K) and 09's canonical antichain $A(I) = \min(Q \setminus C(I))$ of Lemma 31.1 (for $A = \max I$) coincide. This identification is an editorial observation of the merge.

Let $\mathcal{K}^{\max}(S)$ be the induced subposet of $\mathcal{K}(S_\alpha(Q))$ on the downsets $\langle A; x \rangle$ with $A \in \mathcal{M}(Q)$ (07's Y); its support map to $\mathcal{M}(Q)$ is monotone. It omits, for example, the empty downset.

Proposition 21.8 (Height-preserving normalization; 07 Prop. 4.3). *There is a strictly order-preserving map $\nu : \mathcal{K}(S_\alpha(Q)) \rightarrow \mathcal{K}^{\max}(S)$, and hence $h(\mathcal{K}(S_\alpha(Q))) = h(\mathcal{K}^{\max}(S))$.*

Proof. Put $\nu(\langle A; x \rangle) = \langle \text{cl}(A); z \rangle$ with $z_q = 1 + x_q$ for $q \in A$ and $z_q = 0$ for $q \in \text{cl}(A) \setminus A$. Left addition $1 + x_q$ is strictly increasing in x_q and stays below α_q because $1 + \alpha_q = \alpha_q$, so ν is defined even on the empty profile. Let $\langle A; x \rangle \subseteq \langle B; y \rangle$. The completions are Hoare ordered (Lemma 21.6). If $q \in A \cap \text{cl}(B)$, choose $b \in B$ with $q \leq b$; both lie in the antichain $\text{cl}(B)$, so $q = b \in B$. Hence a coordinate common to the two completions that is originally active at the source is originally active at the target, and its shifted value does not decrease by Lemma 20.1; a filler coordinate has value 0. So ν is monotone. If the inclusion is strict and the completions differ, the images differ; if they agree, (52) gives $A \subseteq B$, and either $A \neq B$, when some filler 0 becomes a positive $1 + y_q$, or $A = B$ and some original coordinate increases. So ν is strict, and Lemma 16.1 and inclusion give equal heights. $\square$

Remark 21.9 (07). The normalization is a height comparison, not an order isomorphism, and not an argument that cofinal subposets preserve rank (they need not). Its treatment of the empty profile is essential: for a finite singleton fibre, $\mathcal{K}(P)$ has height 2 while its nonempty maximal-support part has height 1. The hypotheses of this section avoid that defect.

Definition 21.10 (Retirement box; 07). Let $G_0, \dots, G_m$ be pairwise disjoint nonempty finite groups of coordinates, coordinate q having capacity $\alpha_q = \omega^{\rho_q}$, $\rho_q > 0$, and let $\lambda_i = \max_{q \in G_i} \alpha_q$. The retirement box $B = B(G_0, \dots, G_m)$ is the union of the stages

$$B_i = \{z : z_q = \alpha_q \text{ for } q \in G_j, j < i; \ z_q < \alpha_q \text{ for } q \in G_j, j \geq i\},$$

ordered coordinatewise. The value α_q is a sentinel for a retired coordinate. The last group never retires: there is no stage $m + 1$.

Theorem 21.11 (Retirement-box height; 07 Thm 6.2). $h(B(G_0, \dots, G_m)) = \lambda_0 + \lambda_1 + \dots + \lambda_m$, and the box contains a chain of this order type.

Proof. Lower bound. In stage i let a coordinate $q_i \in G_i$ with $\alpha_{q_i} = \lambda_i$ run through λ_i , keep every other nonretired coordinate 0 and retired coordinates at their sentinels. Every point of this chain is below every point chosen in a later stage, since the varied coordinate has reached its sentinel and every coordinate still active at both stages was 0. Concatenate.

Upper bound. Induct on the number of groups. Let $\Lambda = \max_i \lambda_i$, $j = \max\{i : \lambda_i = \Lambda\}$ and $k = |\{i : \lambda_i = \Lambda\}|$; call a group large when $\lambda_i = \Lambda$ and small otherwise. For $z \in B_i$ with $i \leq j$ let $n_\Lambda(i) = |\{\ell < i : \lambda_\ell = \Lambda\}|$ and

$$\text{res}_i(z) = \sum_{\ell \geq i} \sum_{q \in G_\ell} z_q \sum_{\substack{\ell < i \\ \lambda_\ell < \Lambda}} \sum_{q \in G_\ell} \alpha_q. \quad (53)$$

The second sum records the full capacities of already retired *small* groups; retired large groups are omitted and counted by $n_\Lambda(i)$. Every summand is below Λ and there are finitely many, so $\text{res}_i(z) < \Lambda$. Put

$$f(z) = \Lambda \cdot n_\Lambda(i) + \text{res}_i(z) \quad (z \in B_i, i \leq j). \quad (54)$$

Since the last large group has not retired in this prefix, $n_\Lambda(i) \leq k - 1$ and $f(z) < \Lambda \cdot k$. If $z < z'$ lie in one stage, an active coordinate rises and none falls, so f rises. If they lie in stages $i < i'$ and a large group retires in between, $n_\Lambda(i') > n_\Lambda(i)$ and $f(z) < \Lambda(n_\Lambda(i) + 1) \leq \Lambda n_\Lambda(i') \leq f(z')$. Otherwise the count is unchanged, the residual replaces at least one active coordinate of a

retiring small group by its full capacity, and no residual term decreases; the replacement is strict. So the prefix has height at most $\Lambda \cdot k$. The stages after j form a retirement box on $G_{j+1}, \dots, G_m$, earlier coordinates being fixed sentinels; by induction its height is at most $\lambda_{j+1} + \dots + \lambda_m$. Refining the order by putting every prefix point below every suffix point gives an ordinal sum, an upper bound by Lemma 16.1. Thus $h(B) \leq \Lambda \cdot k + \lambda_{j+1} + \dots + \lambda_m = \lambda_0 + \dots + \lambda_m$: the terms below Λ before its last occurrence are absorbed and the k occurrences of Λ remain. When $j = m$ the suffix is empty. $\square$

Why the residual uses natural addition (07). Path answers use ordinary addition because coordinate groups retire in a prescribed order; the within-stage bound uses natural addition because several active coordinates can increase independently. Confusing the roles produces exactly the wrong persistent-coordinate calculation. For $(\lambda_0, \lambda_1, \lambda_2) = (\omega, \omega^2, \omega)$ the first two stages have height at most ω^2 , not $\omega^2 + \omega$, because the last large group has not retired and its active values absorb the finite natural-sum contributions of small groups; the final stage contributes a new ω , giving $\omega^2 + \omega$. The prefix map (54) makes this absorption rigorous for finitely many coordinates of any capacities, uncountable ones included.

For a frontier chain $C = (A_0 < \dots < A_m)$ let $\mathcal{K}_C^{\max}(S)$ be the induced subposet of $\mathcal{K}^{\max}(S)$ on those supports. By (43) a vertex in A_i and A_k ($i < k$) lies in every A_j between, so

$$G_i = A_i \setminus A_{i+1} \ (i < m), \quad G_m = A_m \quad (55)$$

are pairwise disjoint nonempty sets partitioning $\bigcup_i A_i$; each vertex lies in the group of its last occurrence (07 Lemma 7.1).

Proposition 21.12 (Exact height over one frontier chain; 07 Prop. 7.2). *For every nonempty frontier chain C , $h(\mathcal{K}_C^{\max}(S)) = W(C)$, and a chain of that type is explicitly constructible in $\mathcal{K}_C^{\max}(S)$.*

Proof. Send $\langle A_i; x \rangle$ to the vector ι on $\bigcup_j A_j$ with $\iota_q = \alpha_q$ if q retired before stage i , $\iota_q = 1 + x_q$ if $q \in A_i$, and $\iota_q = 0$ if q has not yet appeared. By the no-return property these are all cases and the vector lies in stage B_i of the box of the groups (55); shifted active values stay below their capacities. If $\langle A_i; x \rangle \subseteq \langle A_j; y \rangle$ then $i \leq j$; coordinates active at both stages do not decrease (Lemma 20.1), coordinates not yet introduced are 0, retiring coordinates rise to their sentinels and retired ones stay fixed. A proper inclusion in one support raises an active coordinate, and a change of support retires a group and raises a coordinate to its sentinel. So ι is strictly increasing and Theorem 21.11 gives $h(\mathcal{K}_C^{\max}(S)) \leq \max_{G_0} \alpha + \dots + \max_{G_m} \alpha = W(C)$. Conversely, in support A_i let a coordinate of largest capacity in G_i run through its whole fibre with every other active coordinate 0. The varied coordinate is absent from every later support, so all coordinates shared with a later support are 0 in the earlier profile, and each chosen profile at stage i lies strictly below each chosen profile at a later stage. The concatenation has type $W(C)$. $\square$

Second proof of Theorem 21.1, except saturation. By Proposition 21.8, $h(\mathcal{K}(S_\alpha(Q))) = h(\mathcal{K}^{\max}(S))$. Apply Proposition 17.8 to the support map $\mathcal{K}^{\max}(S) \rightarrow \mathcal{M}(Q)$ and Proposition 21.12 to its chain pieces; one of the finitely many chains realizes the maximum, and its constructed chain realizes the height. $\square$

Proposition 21.13 (Saturation does not reduce cost; 07 Prop. 7.3, 09 Lemma 6.1 and end of proof of Thm 7.2). *Every nonempty frontier chain extends to a saturated chain from $\min Q$ to $\max Q$ without decreasing its cost W .*

Proof. If $A < B < C$ then $A \setminus C \subseteq (A \setminus B) \cup (B \setminus C)$, since a vertex of $A \setminus C$ is absent from B or lies in $B \setminus C$. Hence $w(A, C) \leq \max\{w(A, B), w(B, C)\} \leq w(A, B) + w(B, C)$, and replacing an

edge by the two refined edges cannot reduce the cost, ordinary addition being nondecreasing in each argument. Prepending a smaller frontier prepends a nonnegative ordinal. To append B above a terminal A , note $t(A) \leq \max\{w(A, B), t(B)\} \leq w(A, B) + t(B)$, since vertices of A either retire or lie in B . Add the bottom and top frontiers and fill every gap; finiteness makes this terminate. $\square$

This completes the second proof. No step sums the heights of whole adjacent strata; such a sum is used only in the auxiliary upper bound after the last largest retirement group, where it is justified (07).

Corollary 21.14 (Uniform recovery; all four sources). *If $\alpha_q = \alpha = \omega^\rho$ for every q , with $\rho > 0$, then $h(\mathcal{K}(S_\alpha(Q))) = \alpha \cdot h(\mathcal{M}(Q))$.*

Proof. Every transition and terminal weight is α , so a chain with ℓ frontiers costs $\alpha \cdot \ell$; maximize over the finite poset $\mathcal{M}(Q)$. $\square$

This is Theorem 9.1 of Part I, recovered as a special case and not claimed as new (07 Cor. 8.1, 09 Cor. 7.3, 11 Cor. 6.3, 12 Cor. 5.5).

22 Arbitrary ordinal fibres and exact point ranks

22.1 Cantor-normal-form refinement

Write each nonzero fibre length in Cantor normal form,

$$\alpha_q = \omega^{\rho_{q,1}} n_{q,1} + \cdots + \omega^{\rho_{q,k_q}} n_{q,k_q}, \quad \rho_{q,1} > \cdots > \rho_{q,k_q} \geq 0, \quad 0 < n_{q,j} < \omega. \quad (56)$$

Replace q by a chain of $\sum_j n_{q,j}$ vertices carrying, in order, $n_{q,1}$ blocks of weight $\omega^{\rho_{q,1}}$, then $n_{q,2}$ of the next weight, and so on; exponent 0 gives a singleton block. If $q < r$ put every q -block below every r -block; blocks of incomparable vertices stay incomparable; delete zero fibres. Call the result $Q^\#$ with pure weights λ , and let $U^\#$ be its blocks of positive exponent.

Lemma 22.1 (Exact refinement; 12 Lemma 6.1; also 07, 09, 11). $S_\alpha(Q) \cong S_\lambda(Q^\#)$.

Proof. Inside a fibre the ordinal concatenation of its blocks is its Cantor normal form; comparisons between distinct original fibres are unchanged by putting every block below every block when the original indices were comparable. $\square$

Theorem 22.2 (All finite ordinal lexicographic sums; 12 Thm 6.2, 11 Cor. 7.2). *For every finite poset Q and ordinals $\alpha_q \geq 0$,*

$$h(\mathcal{K}(S_\alpha(Q))) = F(Q^\#) + \tau_{Q^\#},$$

with F and τ from (47) and (46), and the answer is attained by a chain. No countability, cofinality, purity or successor restriction is imposed on the fibres.

Proof. An order isomorphism induces an isomorphism of finitely generated downsets. Apply Lemma 22.1 and Theorem 20.2. $\square$

Second route (11 Thm 7.1). Deleting the coordinates of length 1 from the support model of $S_\lambda(Q^\#)$ gives an exact representation

$$\mathcal{K}(S_\alpha(Q)) \cong Y(\text{Ant}(Q^\#), (B \cap U^\#)_{B \in \text{Ant}(Q^\#)}, \lambda|_{U^\#}) : \quad (57)$$

singleton blocks may occur in a state but carry no coordinate. 11 proves it directly: a finitely generated downset has a unique finite antichain of maximal actual points, at most one per fibre; over a singleton block the point is forced, over $u \in B \cap U^\#$ it is a unique $\xi_u < \lambda_u$; one such downset lies in another exactly when every maximal generator of the first lies below one of the second, which for different fibres is the skeleton order and in a shared infinite fibre is $\xi_u \leq \eta_u$ (a shared fibre cannot be matched to a strictly higher generator of an antichain already containing its index); and (43) gives (27). The persistence theorem then gives the height and an attaining chain.

Cantor normal forms are finite for every ordinal, and no recursive notation for the exponents is needed for the statement. An implementation needs a representation and comparison procedure for them, and 11 does not claim a uniform machine representation of all set ordinals. Expanding a coefficient n into n blocks is a unary expansion: an algorithm polynomial in the number of blocks is not polynomial in the bit length of a large coefficient, and the ideal state space may already be exponential; no stronger complexity claim is made (12).

Corollary 22.3 (Nonzero limit fibres; 07 Thm 9.1, 09 Thm 9.2). *If every α_q is a nonzero limit ordinal, the last exponent of each Cantor normal form is positive, every block of $Q^\#$ is infinite, and Theorem 21.1 applied to $Q^\#$ computes the height and constructs an attaining chain. The reduction is not limited to countable ordinals or to ordinals below ε_0 .*

22.2 The empty, finite and successor cases

If all fibres are zero, $Q^\# = \emptyset$, $\text{Ant}(Q^\#) = \{\emptyset\}$ and the answer is 1, as required for $\mathcal{K}(\emptyset) = \{\emptyset\}$. If the whole sum is finite with N points, $Q^\#$ has N singleton blocks, the state order is the finite downset lattice, and

$$h(\mathcal{K}(S)) = N + 1 \quad (58)$$

(each of the N edges of a maximal ideal chain and the terminal stage cost 1). The fallback weight 1 recovers exactly this case and must not be dropped merely because the main application has infinite fibres (11). For one chain, $\mathcal{K}(\alpha) \cong 1 + \alpha$, which is α for infinite α and $n + 1$ for finite n . Adjoining a greatest point adds one greatest element: $h(\mathcal{K}(P + 1)) = h(\mathcal{K}(P)) + 1$; for the chain $\omega + 1$ the expanded skeleton is an ω block below a singleton, the stage costs are 1, ω , 1, and the answer is $\omega + 1$ (12). Successor endpoints must be kept explicitly; the maximal-frontier compression, which assumes no internally completed fibre, is not the right formula for such inputs.

07 left finite fibres and successor tails outside its limit-fibre reduction and observed that exponent 0 cannot simply be admitted there: for two incomparable singleton fibres $\mathcal{K}(P)$ is the four-element Boolean lattice of height 3, whereas the maximum-fibre value would be 1. The support model handles this case: (58) gives $2 + 1 = 3$.

22.3 Exact point ranks of finitely generated downsets (12)

Let $K = \downarrow F \in \mathcal{K}(S_\alpha(Q))$, normalized to its maximal generators with support A and positions $x_a < \alpha_a$, and set

$$\beta_q = \begin{cases} \alpha_q, & q < a \text{ for some } a \in A, \\ x_q + 1, & q \in A, \\ 0, & \text{otherwise} \end{cases} \quad (59)$$

(disjoint cases, A being an antichain).

Theorem 22.4 (Exact point-rank algorithm; 12 Thm 6.3). *Let $H_K = h(\mathcal{K}(S_\beta(Q)))$, computed by Theorem 22.2. Then H_K is a successor and $r_{\mathcal{K}(S_\beta(Q))}(K) + 1 = H_K$. More precisely, in the expansion $Q_K^\#$ of $S_\beta(Q)$ every maximal block is a singleton, so*

$$r_{\mathcal{K}(S_\beta(Q))}(K) = F(Q_K^\#).$$

This includes the empty downset, of rank 0.

Proof. As an induced poset, K is $S_\beta(Q)$. A finitely generated downset of S inside K is generated by points of itself, hence of K , and a finite subset of K generates the same downset in K and in S because K is downward closed; so $\{L \in \mathcal{K}(S) : L \subseteq K\} \cong \mathcal{K}(K)$. Ranks in a principal lower interval agree with ambient ranks, and this interval has greatest element K , so its height is $r(K) + 1$. Every maximal nonzero fibre in (59) is a successor $x_a + 1$ whose last Cantor term has exponent 0, so its last block is a singleton; every completed nonmaximal fibre lies below one of them. Hence all maximal blocks of $Q_K^\#$ have weight 1, or $Q_K^\#$ is empty, and $\tau = 1$ in both cases. $\square$

This computes the rank of each specified compact downset exactly, not merely an upper ranking function. In executable Cantor normal form the predecessor of H_K is obtained by subtracting one from its nonzero finite final coefficient, never by commutative ordinal subtraction (12). 11 reaches the same ranks through the exact model (57) and Theorem 19.4; only exact constructions, never the compressed model, are used for point ranks.

23 Worked examples and diagnostic counterexamples

23.1 The persistent-coordinate obstruction, now resolved (07, 11, 12)

Let Q consist of an isolated x and a chain $y < z$, with $(\alpha_x, \alpha_y, \alpha_z) = (\omega^2, \omega, \omega)$, the example of (23). Its frontiers are $\{x, y\} < \{x, z\}$; both states have largest length ω^2 , but only y retires at the transition, so $w = \omega$ and $t(\{x, y\}) = t(\{x, z\}) = \omega^2$. The two-state path costs

$$\underbrace{\omega}_{y \text{ retires}} + \underbrace{\omega^2}_{\text{terminal frontier}} = \omega^2,$$

one-state paths cost ω^2 as well, and

$$h(\mathcal{K}(\omega^2 \sqcup (\omega + \omega))) = \omega^2, \quad (60)$$

in agreement with (23). The obstruction of Part I is a direct instance of the new rule, not an exception to it; replacing the retirement cost by the whole first-stratum height would give the wrong value $\omega^2 + \omega^2$. An attaining chain first varies the y -coordinate through ω at $x = 0$, then the x -coordinate through ω^2 in the next frontier at $z = 0$; its type is $\omega + \omega^2 = \omega^2$, and the long x -coordinate is never reset. In the peeling calculation the top label x never departs, so $d_\Lambda = 0$ on both states, $c_\Lambda = 1$, $Z = \emptyset$, and the rank bound has the form $\theta < \omega^2$ despite the state change and the two lower fibres. The same holds with ω^2 replaced by ω^η and the lower fibres by ω^σ , for any $0 < \sigma < \eta$ (11).

At ideal-state resolution (12), the six ideals and optimal prefix values are

I	$\emptyset$	$\{x\}$	$\{y\}$	$\{x, y\}$	$\{y, z\}$	$\{x, y, z\}$
$\max I$	$\emptyset$	$\{x\}$	$\{y\}$	$\{x, y\}$	$\{z\}$	$\{x, z\}$
$F(I)$	0	1	1	2	ω	$\omega + 1$

At the last state the two final additions give the prefix costs $2 + \omega = \omega$ and $\omega + 1$, and the latter wins; with terminal cost ω^2 , $F(Q) + \tau_Q = (\omega + 1) + \omega^2 = \omega^2$. Maximizing prefixes is safe even when different prefixes are later absorbed to the same value: the recurrence records the actual greatest prefix rather than relying on future absorption. Part III's weight-0 table for the same example is Remark 32.2.

23.2 The order of transitions matters (07)

Example 23.1 (07 Ex. 3.3). Let Q be the two disjoint chains $a < c$ and $b < d$ with $(\alpha_a, \alpha_b, \alpha_c, \alpha_d) = (\omega^2, \omega, \omega, \omega)$. The frontier poset is a diamond, and its two saturated paths cost

$$\begin{aligned} \{a, b\} \rightarrow \{b, c\} \rightarrow \{c, d\} : \quad & \omega^2 + \omega + \omega = \omega^2 + \omega \cdot 2, \\ \{a, b\} \rightarrow \{a, d\} \rightarrow \{c, d\} : \quad & \omega + \omega^2 + \omega = \omega^2 + \omega. \end{aligned}$$

The first is larger. A rule that forgets the order of transitions loses information even when every edge weight is known.

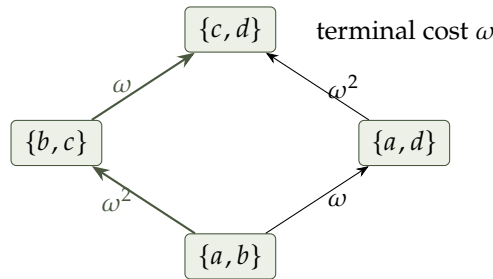


Figure 2: The frontier diamond of Example 23.1; edges carry the weight $w(A, B)$ of the retiring coordinate. The thick path is optimal.

23.3 The weighted N -poset (07, 11, 12)

For $a < c, b < c, b < d$ the frontiers form the chain $\{a, b\} < \{a, d\} < \{c, d\}$ of (11). The transition weights are α_b and α_a and the terminal weight is $\max(\alpha_c, \alpha_d)$, so for all positive pure fibres

$$\boxed{h(\mathcal{K}(S_\alpha(N))) = \alpha_b + \alpha_a + \max(\alpha_c, \alpha_d),} \quad (61)$$

an ordered sum: b retires first, then a . Paths that skip a frontier and one-state paths are also in the maximum, but never exceed this one; both transition weights are forced, and a persistent terminal coordinate can absorb the first (07).

For the weighted example of Section 7, realized by the fibres $(\alpha_a, \alpha_b, \alpha_c, \alpha_d) = (\omega^3, \omega, \omega, \omega^2)$, the height is $\omega + \omega^3 + \omega^2 = \omega^3 + \omega^2$: this supplies the height missing from Part I for that example (12). For $(\omega^2, \omega^3, \omega, \omega^2)$ it is $\omega^3 + \omega^2 + \omega^2 = \omega^3 + \omega^2 \cdot 2$, whereas skipping the middle frontier would give only $\omega^3 + \omega^2$: intermediate retirement opportunities matter (12). Further values on this one skeleton (11, 07): In the third row the ω^2 coordinate retires first, leaving two independent opportunities for later ω blocks; in the second, the early ω block is absorbed by the later ω^2 . The difference is the noncommutativity of ordinary addition, not a numerical approximation (11).

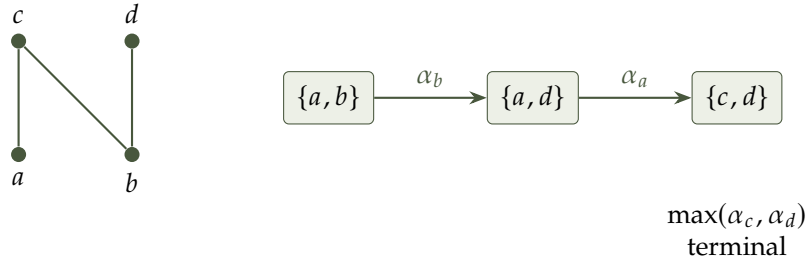


Figure 3: The weighted N -poset and its retirement costs $(12, 11)$. An edge is weighted by a disappearing coordinate, not by the height of its source stratum.

α_a	α_b	α_c	α_d	$h(\mathcal{K}(S))$
ω	ω	ω	ω	$\omega \cdot 3$
ω^2	ω	ω	ω	$\omega^2 + \omega$
ω	ω^2	ω	ω	$\omega^2 + \omega \cdot 2$
ω^3	ω	ω	ω^2	$\omega^3 + \omega^2$
ω^2	ω	ω	ω^2	$\omega^2 \cdot 2$
ω^2	ω^3	ω	ω	$\omega^3 + \omega^2 + \omega$

Table 5: Nonuniform heights on the N -skeleton. The first four rows are 11's table, the fifth is 07's, and the last is 09's example (Section 35) in the labelling of this report.

23.4 Stratum heights do not determine the height

Corollary 23.2 (Stratum types do not determine the answer; 07 Cor. 8.4). *The frontier poset together with the isomorphism type of each maximal-support stratum does not determine $h(\mathcal{K}(S_\alpha(Q)))$, even for pure fibres on a three-vertex skeleton.*

Proof. On the skeleton $\{x, y < z\}$ compare $P_X = \omega^2 \sqcup (\omega + \omega)$ and $P_Y = \omega \sqcup (\omega^2 + \omega^2)$. Both have the two-element frontier chain, and every maximal-support stratum of either is isomorphic to $\omega^2 \times \omega$ (interchanging coordinates if needed). Yet $h(\mathcal{K}(P_X)) = \omega + \omega^2 = \omega^2$ and $h(\mathcal{K}(P_Y)) = \omega^2 + \omega^2 = \omega^2 \cdot 2$. The difference is which coordinate is shared and which retires. $\square$

The first member of the pair is Part I's counterexample. 12 gives a second pair, comparing stratum heights only: the complete two-level poset $a, b < c, d$ with weights $(\omega^2, \omega, \omega^2, \omega)$ also has two comparable frontiers, each stratum of height ω^2 , but both lower coordinates retire and the height is $\omega^2 + \omega^2 = \omega^2 \cdot 2$. Each claimed stratum height follows without the main formula: the stratum is bounded by a product whose largest pure coordinate is ω^2 (natural-sum rank below ω^2) and contains a copy of that coordinate with the other active coordinates fixed nonempty; this does not assert that the strata are isomorphic. The input "frontier poset plus stratum heights" is thus identical while the outputs differ; the missing data are which coordinates persist. Part III shows that the resulting ambiguity is unbounded (Theorem 34.3). An exact finite invariant must retain not merely sizes or heights of pieces but the capacities of coordinates lost along each transition (07).

23.5 Successor endpoints carry real information (11)

For two disjoint chains of type $\omega + 1$, $\mathcal{K}$ commutes with the finite disjoint union, so $\mathcal{K}((\omega + 1) \sqcup (\omega + 1)) \cong (\omega + 1) \times (\omega + 1)$, of height

$$\omega \cdot 2 + 1. \quad (62)$$

Directly: $(\xi, \eta) \mapsto \xi \# \eta$ is strictly increasing with largest value $\omega \cdot 2$ at (ω, ω) , bounding the height by $\omega \cdot 2 + 1$, and the chain $(0, 0) < (1, 0) < \dots < (\omega, 0) < (\omega, 1) < \dots < (\omega, \omega)$ has exactly that type. In the expanded skeleton each fibre is an ω block below a singleton block, and the last state consists of singleton blocks only and contributes the final 1. Using only the maximal antichains of the expanded skeleton would discard essential earlier states; all antichains retain them. Similarly $h((\omega^2 + 1) \times (\omega + 1)) = \omega^2 + \omega + 1$: the natural-sum bound has maximum $\omega^2 + \omega$, and the chain through $(\xi, 0)$ for $\xi < \omega^2$, then (ω^2, η) for $\eta < \omega$, then the greatest point attains its successor.

These are heights of products $\mathcal{K}(\alpha_1 \sqcup \alpha_2) \cong (1 + \alpha_1) \times (1 + \alpha_2)$, not of finitary powersets of products: the neighbouring report on intrinsic ordinal approximations computes $h(\mathcal{K}((\omega + 1) \times (\omega + 1))) = \omega^2 + 1$ for the latter.

23.6 A genuinely mixed Cantor-normal-form example (12)

On the N -skeleton put $\alpha_a = \omega^2 + \omega \cdot 2 + 1$, $\alpha_b = \omega + 2$, $\alpha_c = \omega^2 + 1$, $\alpha_d = \omega \cdot 2$. Expansion gives eleven pure blocks and a downset lattice with thirty-six states, and

$$h(\mathcal{K}(S_a(N))) = \omega^2 \cdot 2 + \omega \cdot 2. \quad (63)$$

Adding all blocks of a , then of b , then of c , then of d is an optimizing path; its stage costs, the last terminal, are $1, \omega^2, \omega, \omega, 1, \omega, 1, 1, \omega^2, 1, \omega, \omega$, whose ordinary sum is (63). The recurrence table in `data/05-ranks-mixed-CNF-certificate.json` evaluates every state and certifies that no path is larger. This is an exact symbolic consequence of Theorem 22.2, not an extrapolation from finite approximations to ω .

23.7 Exact ranks inside infinite orders

In the persistent example let K be generated by the point at position $\omega + 2$ of the x -fibre and the point at position 3 of the z -fibre. The whole y -fibre is present and $K \cong (\omega + 3) \sqcup (\omega + 4)$ as an induced order; its finitary powerset has height $\omega \cdot 2 + 6$, so $r_{\mathcal{K}(S)}(K) = \omega \cdot 2 + 5$. This agrees with the classical product point rank $(\omega + 2) \# (\omega + 3) = \omega \cdot 2 + 5$; the local-rank algorithm does not need to recognize the example as a product (12).

In $X = \omega^2 \times \omega^2$ take $z = (\omega + 1, \omega)$. Its capped principal ideal is $(\omega + 2) \times (\omega + 1)$, of height $\omega \cdot 2 + 2$, so $r_X(z) = \omega \cdot 2 + 1$. Directly, the natural-sum bound at z is $(\omega + 1) \# \omega = \omega \cdot 2 + 1$, and the chain through $(n, 0)$, then (ω, n) , then (ω, ω) and $(\omega + 1, \omega)$ has type $\omega \cdot 2 + 2$ (11).

23.8 Transfinite exponents are not a new combinatorial case (12)

On the N -skeleton take $(\alpha_a, \alpha_b, \alpha_c, \alpha_d) = (\omega^{\omega+1}, \omega^\omega, \omega^\omega, \omega^{\omega+1})$. Formula (61) gives $\omega^\omega + \omega^{\omega+1} + \omega^{\omega+1} = \omega^{\omega+1} \cdot 2$. The proof uses only the order of the exponents, not a finite-exponent approximation; Corollary 24.2 makes this precise.

23.9 Products of chains: an independent check

The following classical product-height calculation is recorded as a separate verification route for antichain skeletons; it is not a new-priority claim (07, 12). The finite Milner–Rado expression is given, e.g., in [5, Theorem 2.20].

Lemma 23.3 (Ranks in finite products of chains; 12 App. A, 07 Lemma 9.2). *For nonzero ordinal chains $\theta_1, \dots, \theta_k$ the rank of a tuple in the componentwise product is $r(x_1, \dots, x_k) = x_1 \# \dots \# x_k$. Let ρ_i be the smallest exponent in the Cantor normal form of θ_i and $\rho = \max_i \rho_i$; let B_i be θ_i with one copy of its final ω^ρ removed when $\rho_i = \rho$, and θ_i truncated to exponents at least ρ when $\rho_i < \rho$. Then*

$$h\left(\prod_i \theta_i\right) = \left(\# B_i\right) + \omega^\rho. \quad (64)$$

The empty product has height 1; a product with a zero factor has height 0.

Proof (12). Write $z = \#_i x_i$. Strict monotonicity gives $\#_i y_i + 1 \leq z$ when one coordinate is decreased and the others fixed. Conversely fix $\gamma < z$, align the exponents of z and γ , and let δ be the greatest exponent at which their coefficients differ; that of z is larger. Choose x_j contributing to the coefficient at δ , keep its terms above δ , lower that coefficient by one and replace its tail below δ by the tail of γ below δ . The resulting $y_j < x_j$, and the natural sum with this one coordinate changed agrees with z above δ , has coefficient at δ at least that of γ and, when these are equal, lower tail at least that of γ ; so it is at least γ . This proves $z = \sup\{\#_i y_i + 1 : \text{one } y_i < x_i, \text{ the others } y_i = x_i\}$, and well-founded induction gives the rank formula. Hence $h(\prod_i \theta_i) = \sup_{x_i < \theta_i} (\#_i x_i + 1)$. For $\rho = 0$ every factor is a successor with greatest element B_i , and the formula is the rank of the greatest tuple plus one. For $\rho > 0$ every $x_i < \theta_i$ is bounded by some $B_i + t_i$ with $t_i < \omega^\rho$, and natural-sum closure gives the upper bound. For the lower bound, set the coordinates with $\rho_i < \rho$ equal to B_i (allowed, since a positive lower tail remains), set all but one of the others to B_i and vary the last as $B_i + t$, $t < \omega^\rho$; these natural sums are cofinal in the right side. $\square$

Second route for limit factors (07). If every $\theta_i = \beta_i$ is a nonzero limit, write $\beta_i = b_i + \omega^{\tau_i}$ by removing one copy of the final monomial and put $\Lambda = \max_i \omega^{\tau_i}$; then (64) reads $h(\prod_i \beta_i) = (b_1 \# \dots \# b_r) + \Lambda$ (terms of $\# b_i$ below the exponent of Λ are absorbed either way). The product of the final intervals $[b_i, b_i + \omega^{\tau_i})$ is cofinal, and since natural addition is monotone the supremum of values plus one may be restricted to it. There $x_i = b_i + u_i = b_i \# u_i$ with $u_i < \omega^{\tau_i}$, because the exponents of b_i are at least τ_i , so the rank is $B \# \#_i u_i$ with $B = \# b_i$. All variable summands are below Λ , one of them varies cofinally in Λ , and aligning Cantor normal forms gives $\sup((B \# \#_i u_i) + 1) = B + \Lambda$. The rank recursion used for $\#_i x_i$ is the one proved above: at a successor reduce a coordinate with a nonzero finite term by one, and at a limit reduce inside the last monomial of a coordinate of least trailing exponent. $\square$

Since $\mathcal{K}(\sqcup_i \alpha_i) \cong \prod_i \mathcal{K}(\alpha_i) \cong \prod_i (1 + \alpha_i)$, the benchmark uses $\theta_i = 1 + \alpha_i$, not α_i ; the distinction matters for finite and zero fibres. For infinite fibres $1 + \beta = \beta$, and the lemma checks Corollary 22.3 independently on antichain skeletons. For example (07),

$$\beta_1 = \omega^2 + \omega, \quad \beta_2 = \omega^3 + \omega^2 \implies h(\mathcal{K}(\beta_1 \sqcup \beta_2)) = \omega^3 + \omega^2 \cdot 2.$$

23.10 Several tempting generalizations are false (11)

The persistent example disproves adding full state heights. Example 17.3 shows that disappearance and reappearance can destroy transitivity. The successor product shows that a maximum

of impure capacities is not the right product height. Replacing ordinary by natural sums also fails: it would turn $\omega + \omega^2$ into $\omega^2 + \omega$ in (60). Finally, replacing every infinite fibre by a long finite chain makes the whole order finite, and then (58) forces height “number of points plus one”. That is not an approximation scheme for the transfinite formula: finite fibres have endpoints and pure infinite fibres do not. The finite tests of Section 25 check representations and algorithms under their actual hypotheses, not a limit of finite height counts.

23.11 Summary of 07’s worked certificates

Skeleton and fibres	Maximizing weight word	Exact height
Isolated ω^2 beside $\omega + \omega$	ω, ω^2	ω^2
Isolated ω beside $\omega^2 + \omega^2$	ω^2, ω^2	$\omega^2 \cdot 2$
N skeleton, all fibres ω	ω, ω, ω	$\omega \cdot 3$
N skeleton, fibres $(\omega^2, \omega, \omega, \omega^2)$	$\omega, \omega^2, \omega^2$	$\omega^2 \cdot 2$
Two disjoint two-chains, Example 23.1	ω^2, ω, ω	$\omega^2 + \omega \cdot 2$
Two incomparable limit fibres (above)	computed after block expansion	$\omega^3 + \omega^2 \cdot 2$

Table 6: The examples whose inputs and certificates 07 ships, as `data/02-retire-*.json`.

24 Structural consequences

Corollary 24.1 (Small output and universal bounds; 12 Cor. 8.1, 07 Cor. 8.2). *Let $Q^\#$ have N pure blocks. Every exponent of $F(Q^\#) + \tau_{Q^\#}$ is 0 or the exponent of an input block, and the sum of its Cantor-normal-form coefficients is at most $N + 1$. If Q is nonempty and all fibres are positive pure, $\alpha_q = \omega^{p_q}$, then the coefficient sum is at most $|Q|$, no finite term occurs, the leading exponent is $\max_q p_q$, and*

$$\max_{q \in Q} \alpha_q \leq h(\mathcal{K}(S_\alpha(Q))) \leq \sum_{q \in Q} \alpha_q. \quad (65)$$

Both bounds are sharp.

Proof. A winning ideal path has $N + 1$ pure stage costs; ordinary addition can merge equal exponents and erase smaller tails, but cannot introduce an exponent or increase the number of pure summands. For the compressed formula choose one coordinate attaining each retirement cost and one attaining the terminal cost; a chosen retiring coordinate never appears later, by convexity, so these are at most $|Q|$ distinct coordinates, all of positive exponent. The cost is an ordinary sum of their monomials, hence at most their natural sum and at most the right side of (65). The principal downsets of one fibre form a chain of its full type, giving the left inequality, and the leading exponent follows from the two bounds. An antichain skeleton attains the left bound; a chain skeleton with fibre exponents in nonincreasing order attains the right one, where natural and ordinary sums agree (07). $\square$

The optimization may have many states while the answer has a short normal form; the difficulty is choosing the order of useful retirements, not generating new exponent expressions (12).

Corollary 24.2 (Exponent-scale invariance; 12 Cor. 8.2, 07 Cor. 8.3). *Fix a finite pure-block skeleton and its pattern of equal and unequal exponents. Replace the distinct exponents by any strictly increasing list of ordinals, keeping 0 at 0 when it occurs and positive exponents positive. The*

output keeps its Cantor-normal-form coefficient pattern, with the exponents relabelled in the same way, and an optimizing path remains optimizing. In particular, if all $\rho_q > 0$ (07's standing hypothesis) and $\phi : \{\rho_q\} \rightarrow \text{Ord} \setminus \{0\}$ is strictly increasing and $h(\mathcal{K}(\sum_q^{\text{lex}} \omega^{\rho_q})) = \omega^{\sigma_1} c_1 + \dots + \omega^{\sigma_s} c_s$ with $\sigma_1 > \dots > \sigma_s$, then $h(\mathcal{K}(\sum_q^{\text{lex}} \omega^{\phi(\rho_q)})) = \omega^{\phi(\sigma_1)} c_1 + \dots + \omega^{\phi(\sigma_s)} c_s$.

Proof. Include exponent 0 among the scale labels to account for default costs. On finite ordinal polynomials in these labels a strictly increasing relabelling preserves comparison, maximum and ordinary addition: the algorithms use only equality and relative order of exponents and finite coefficient arithmetic (the addition rule deletes smaller trailing exponents and merges equal boundary exponents). So it commutes with every step of (47), including tie comparisons and terminal addition (12), and equally with every path evaluation of (24) and their finite maximum (07). $\square$

The hypothesis $\rho_q > 0$ of the last sentence cannot be dropped, since ϕ may move the exponent 0 that the first sentence keeps fixed. For a single vertex with $\rho = 0$ the height is $h(\mathcal{K}(1)) = 2 = \omega^0 \cdot 2$, while $0 \mapsto 1$ would predict $\omega \cdot 2$ for $h(\mathcal{K}(\omega)) = \omega$. (This example is an editorial addition of the merge.)

The formula is thus a finite combinatorial calculation on the *weak order pattern* of the exponents; their magnitudes, cardinalities and cofinalities are not extra inputs, though the actual cardinalities and cofinalities of the output of course change after relabelling (07). This is why integer exponent ranks suffice in 07's implementation. For arbitrary fibres, apply the statement to the expanded blocks; the block skeleton is unchanged as long as the ordering and equalities of exponents and all finite coefficients are kept. This gives a parameter-uniform family of identities, not separate computations for finitely many integer exponents (12). Corollary 18.3 is the version for abstract systems, with its certificate.

Proposition 24.3 (A sufficient extension to non-chain coordinates; 12 Prop. 8.3). *In Definition 17.1 replace each pure coordinate λ_q by a well-founded poset P_q with a least element, height $\lambda_q = \omega^{\rho_q}$, and an explicitly given chain of type λ_q , and use the order of P_q on shared coordinates. Then the formula of Theorem 17.4 remains valid.*

Proof. For the upper bound replace x_q in (32) by its rank $r_{P_q}(x_q) < \lambda_q$; ranks are nondecreasing on weak and strictly increasing on strict comparisons, so every step of Lemma 17.5 holds. For the lower bound vary a retiring coordinate along its given chain, the others at their least elements. The finite-tree argument is unchanged. $\square$

This does not assert that the finitely generated downsets of arbitrary non-chain fibres have the same exact support representation; 12 names that representation question as an obstacle to a broad WPO extension. Part III supplies such a representation for WPO fibres whose finitary powersets have pure height.

24.1 The frontier-graph algorithm (07)

For $A \in \mathcal{M}(Q)$ let $F^M(A)$ be the largest cost of a nonempty frontier chain starting at A . By Theorem 21.1,

$$F^M(A) = \max\left(t(A), \max_{A < B} (w(A, B) + F^M(B))\right), \quad h(\mathcal{K}(S_\alpha(Q))) = F^M(\min Q), \quad (66)$$

the inner maximum omitted at $\max Q$. Reverse topological evaluation is well-defined on the finite poset, left addition of the fixed ordinal $w(A, B)$ preserves finite maxima, and Proposition 21.13 allows cover edges only. A certificate is an attaining saturated path, its

retirement groups, one largest-capacity vertex per group and the ordinary sum of those capacities. Checking it confirms the lower-bound chain recipe; it does not show that no better path exists unless the finite program is also checked.

Listing 2: Height computation after the frontiers are enumerated (07).

```

for A in reverse_topological_order(frontiers):
    F[A] = largest_fibre(A)
    path[A] = [A]
    for B in successors(A):
        candidate = largest_fibre(A minus B) + F[B] # ordinal +
        if candidate > F[A]:
            F[A] = candidate
            path[A] = [A] + path[B]
return F[minimal_vertices(Q)], path[minimal_vertices(Q)]

```

With d distinct fibre exponents a cost is a decreasing list of exponent–coefficient pairs; to add a right operand with leading exponent τ , erase the left operand’s terms below τ , merge coefficients at τ and append the rest; no floating point is used. With $M = |\mathcal{M}(Q)|$, all frontier comparisons cost $O(M^2|Q|)$ with plain set representations, the program $O(M^2d)$ exponent comparisons and coefficient operations on the full graph, naive cover extraction $O(M^3)$ (not automatically preferable), and enumerating antichains by subsets $O(2^{|Q|}|Q|^2)$. The method is polynomial in an explicit frontier graph, *not* asserted to be polynomial in the size of the skeleton. For arbitrary ordinal exponents these are operation counts relative to an oracle deciding their comparisons; 07’s code uses positive integers, interpreted by Corollary 24.2 as order labels for any finite set of positive exponents. Limit-fibre expansion has $\sum_{q,j} n_{q,j}$ vertices, so large coefficients make naive expansion expensive; a compressed implementation is a separate problem.

25 Implementation, verification and proof audit of Part II

Each of 07, 11 and 12 ships an independent standard-library Python package; 09’s is described in Section 36. The files are shipped under the prefixes of Table 3 in `code/` and `data/`, and the delivered text of their scripts, Makefiles and audits still uses the delivered names (`code/verify.py`, `examples/...`, `data/...`). The commands below are quoted in that delivered layout; the report’s `README.md` explains how to recreate it on a scratch copy. Every suite rewrites its own evidence files and must not be run inside the repository. The recorded runs used Python 3.13.5. Before placement every suite was rerun on a copy; fresh outputs equalled the recorded ones apart from line endings, interpreter versions and timings, and a 152-instance comparison fed through all four programs found no disagreement (placement commit 1a1396d4d; that comparison script is not shipped).

25.1 12: ideal-state heights and point ranks

The ordinal class stores hereditary Cantor normal forms below ε_0 ; a finite ordinal is a nonnegative JSON integer and a nonfinite one a list of `[exponent, coefficient]` pairs in strictly decreasing exponent order, so $\omega^2 + \omega \cdot 3 + 2$ is `[[2, 1], [1, 3], [0, 2]]`. Arithmetic is exact; cyclic inputs, invalid normal forms and out-of-range generator positions are rejected, and a default cap of twenty expanded vertices is a safety limit for the exponential state space, not a mathematical restriction. The principal commands of the delivered layout are `python code/verify.py` and `python code/heights.py examples/{weighted_N,mixed_CNF,point_rank}.json`; an input with a `generators` field requests a point rank. The recorded run passed 63,270 assertions; its ledger is shipped as `data/05-ranks-verification.json`. The census is 1, 1, 2, 7, 40, 357

naturally labelled posets on zero through five vertices (408), covering every isomorphism type at these sizes but not isomorphism-reduced.

Validation family (12)	Assertions or cases
Dynamic program versus all linear extensions	2,196
Full recurrence-and-path certificate checks	2,196
All-support versus maximal-frontier formula	860
Recovery of the uniform frontier theorem	814
Disjoint-chain cases versus the classical product-height formula	820
Finite support-order pairs versus generated downset inclusion	12,585
Exact local ranks versus directly computed finite ranks	1,603
Finite support injectivity / direct heights / size-plus-one checks	291 each
Ordinary / natural addition associativity on a fixed ordinal grid	19,683 each
Strict absorption inequalities	1,809
Exponent-scale relabelling checks	100

Table 7: Principal families in 12’s deterministic run; the ledger also records normal-form, example and invalid-input tests.

The path oracle enumerates linear extensions and updates active sets itself, without the production recurrence or cost helper; the finite-rank oracle builds actual finite downsets and ranks them by strict inclusion; the Cartesian oracle uses a separate Milner–Rado calculation. They share the separately tested ordinal representation, so they are “not three independent proof assistants” (12). No test replaces an infinite ordinal by a large integer and reads off a trend; infinite examples are evaluated symbolically after the theorems reduce them to finite calculations. The ledger certifies the recorded execution of the finite and symbolic suite, not novelty, absence of proof errors, or Lean kernel checking.

25.2 07: retirement paths

The evaluator code/02-retire-retirement.py accepts cover edges or any acyclic relation with the intended closure and rejects cycles, missing fibre values, nonpositive exponents and finite tails in limit-fibre mode. Its outputs give a height, a saturated maximizing path, groups and pivot vertices, and for limit fibres the original owner of each expanded vertex. The recorded run of its verify.py passed 3,210,888 assertions (data/02-retire-verification_results.json) on the 1, 2, 7, 40, 357, 4824 naturally labelled posets on one through six vertices (5,231): every exponent assignment in $\{1, 2, 3\}^n$ through five vertices (90,201 weighted instances) and eight deterministic assignments per six-vertex order, 128,793 weighted instances in all. The full-comparison program agrees with a cover-edge program on all of them, and on the 90,201 smaller ones both agree with explicit enumeration of all nonempty frontier chains evaluated by a separate monomial-word routine. Minimal completion, Hoare monotonicity, strict retirement, disjoint retirement groups, uniform recovery, exponent relabelling, coefficient budgets and the universal bounds are checked throughout; sampled profile maps are checked for strict monotonicity through four vertices; limit-fibre expansion is compared with ordinary sums on chain skeletons and with Lemma 23.3 on antichain skeletons (258 instances each); and sampled prefix maps (54) are checked for range and strictness. These tests do not compute the rank of an infinite descending-sequence tree, prove anything for all ordinals, establish consistency, or certify novelty; they look for finite combinatorial errors, wrong ordinal arithmetic, wrong boundary conventions and disagreements between implementations.

25.3 11: persistent systems and local ranks

11 ships an ordinal engine (code/04-peel-ordinals.py, below ε_0), the solver code/04-peel-persistent_height.py (compact pure-frontier construction, all-ordinal expanded construction, weighted-path and peeling algorithms, phase refinement, exact point ranks), and two test programs. The solver rejects cyclic skeletons, invalid normal forms and violations of (27); general expansion has a default guard of 18 blocks (configurable) and phase refinement one of 4,096 refined states, and the algorithm stays exponential in the worst case. The main verifier passed 129,593 assertions (data/04-peel-verification.json) on the same labelled census 1, 2, 7, 40, 357, 4824.

Test family (11)	Cases or comparisons
Pure weighted skeletons through four vertices, exponents in $\{1, 2, 3\}$	3,450
Uniform skeletons with five or six vertices	5,181
Accepted abstract two-label persistent systems through four states	9,714
Rejected abstract assignments violating (27)	1,011
Finite all-antichain representation bijections	147
Finite representation order-pair comparisons	47,221
Finite ordinal-fibre height cases, including zero fibres	3,451
Worked ordinal examples	10

Table 8: Selected families of 11's run; rows do not sum to the assertion total.

Weighted-path answers were compared with independent peeling answers and with exhaustive path enumeration on the smaller cases, and pure maximal-frontier answers with exact all-antichain answers; arithmetic used an independent coefficient-array model below ω^4 . The local-rank program checked 940 finite coordinate systems and 6,132 individual finite point ranks against direct enumeration, plus 37 symbolic regressions (data/04-peel-local_rank_verification.json). These are not a Lean formalization, not an evaluation of infinite bad-sequence trees and not a replacement for the rank-map proof; agreement of two algorithms alone would not prove that either computes the height.

25.4 Proof dependencies, delicate points and formalization routes

25.4.1 Dependency map

The principal chain of 12 is: Cantor normal form $\Rightarrow$ strict absorption $\Rightarrow$ fixed-chain upper bound; convex lifetimes $\Rightarrow$ attaining chain; finite tree union plus these bounds $\Rightarrow$ finite persistence theorem; support antichains $\Rightarrow$ ideal-state recurrence; finite CNF refinement $\Rightarrow$ arbitrary ordinal fibres; principal lower intervals $\Rightarrow$ exact point ranks. The compressed frontier theorem is a separate embedding-sandwich corollary. Neither the product-height benchmark nor Part I's survivor formula is a premise. 07's route and 11's peeling route are summarized below.

Where a shortcut would change the result (11). First, (27) is used both for transitivity and when concatenating coordinate blocks. Second, state changes that preserve every highest coordinate still need a strictly increasing lower-scale term, which is why $\kappa \cdot \text{depth}(t)$ occurs in (36). Third, the terminal region is $Z = \{d_\Lambda = c_\Lambda\}$, not the set of states whose active maximum happens to be small. Fourth, the fallback weight is one, which is essential for endpoints and local ranks. In the compact frontier proof the two embeddings give equality of heights, not an isomorphism; only the exact mixed-block and phase-refinement isomorphisms are used for point ranks.

Result (07)	Depends on	Sensitive point
Profile criterion	finite generators	equality constraints only on shared vertices
Maximal-support normalization	minimal completion; left shift	empty profile and strictness, not cofinality
Finite-chain cover	prefix-closed descending trees	finitely many chain pieces
Retirement-box theorem	natural-sum residual; finite induction	retired large groups counted once, not per coordinate
Fixed-path theorem	no return; box theorem	active shared coordinates never reset
Global height formula	normalization; chain cover; path theorem	height, not maximal order type
Limit-fibre extension	finite Cantor normal forms	no exponent-zero blocks

Table 9: 07's proof-dependency audit.

07's checklist for an independent reader.

- (1) In a profile comparison, a vertex active in both supports cannot be dominated by a different target vertex; hence exactly the shared-coordinate inequalities of Lemma 20.1.
- (2) In the completion proof, a hypothetical $b < u$ must be shown incomparable with *every* old support vertex before minimality of u is used.
- (3) In the normalization use $1 + x$, not $x + 1$: left addition is strictly increasing and stays below every infinite pure fibre, also at limit values.
- (4) In the tree cover, cover complete descending sequences, not points.
- (5) In the prefix residual, record retired small coordinates at full capacity and omit retired large groups; count large groups, not their coordinates.
- (6) At a path transition, an old pivot lies below a strictly higher support vertex and never returns; shared old coordinates were kept at 0.
- (7) In the block extension, check that the final Cantor exponent is positive; finite tails are outside that reduction (and inside Theorem 22.2).

None of these checks invokes the status or correctness of any other repository report.

No theorem of Part II imports a large-cardinal assertion or an admitted repository or Lean declaration; the repository is the source of the target and of credited examples, not a logical dependency. The only background facts are elementary ordinal arithmetic, well-founded rank induction and finite-poset arguments. No proof assistant has checked any of it, and no placeholder formalization is supplied that could be mistaken for one (07); no Lean source is delivered as though checked (12).

25.4.2 Formalization routes proposed by the sources

All three sources separate finite combinatorics from ordinal ranks and define height by well-founded rank, never as the output of the proposed algorithm.

- 12: formalize strict absorption and the fixed-chain rank function, then finite tree union and the support-antichain isomorphism (the pure-block theorem), then CNF refinement, recurrence correctness and the principal-lower-interval rank identity. A certificate checker should be proved sound against the recurrence rather than make its JSON output an

axiom; proof-relevant certificates need only the finite poset, weights, all ideal values and a path, with diagrams and formatted ordinals outside the trusted interface. It can be staged without the repository's large-cardinal material or its unrefereed research theorems.

- *07*: finite antichains, Hoare order, completion and no-return first (testable on finite skeletons), then support profiles and the shifted normalization; for the transfinite layer, strict-map monotonicity, the tree union, and the box theorem, which needs only finite products of ordinals, natural sums, sentinels and induction on the group list. The first milestone is the prefix map (54), the one place where a subtle nonuniform upper bound enters; a second is strictness of the normalization on the empty profile; finally, connect an executable frontier-graph evaluator to the mathematical maximum.
- *11*: define the tagged product over a finite state poset and label type, prove the order axioms from (27) and well-foundedness separately; formalize departure indicators, d_Λ , c_Λ and upward closure of Z by natural-number induction; in the ordinal layer, natural-sum monotonicity, closure below a pure power and Lemma 17.11, with the induction on the number of active lengths; then the exact all-antichain and phase-refinement isomorphisms, the compression as two embeddings (never coerced into an isomorphism), the principal-ideal lemma, and a small checked interpreter for CNF expressions. This is a proposed plan, not a completed verification.

Part III. Nonuniform heights: WPO fibres of pure finitary-powerset height

This part is manuscript 09. Its hypothesis is incomparable with those of Part II: the fibres need not be chains, but their finitary powersets must have positive pure height, and finite or successor fibres are excluded. It keeps its own hypotheses and 09's weight-0 convention for empty retirements.

26 Setting, hypotheses and classical inputs

26.1 The hypothesis

Let Q be a finite nonempty poset and let each fibre P_q be a nonempty well-partial-order, and write $S = \sum_{q \in Q}^{\text{lex}} P_q$ as in Problem 1.1. Throughout this part assume

$$h_q := h(\mathcal{K}(P_q)) = \omega^{\rho_q}, \quad \rho_q > 0. \quad (67)$$

09 writes α_q for h_q . It is a height of a finitary powerset, *not* a fibre length and *not* $h(P_q)$: for an ordinal chain fibre α , $h_q = h(\mathcal{K}(\alpha)) = 1 + \alpha = \alpha$ when α is infinite, but in general $h(P_q)$ alone does not suffice. A *positive pure power* means ω^ρ with $\rho > 0$: infinite, a limit, additively indecomposable, and closed under finite natural sums of smaller ordinals. The restriction $\rho > 0$ is deliberate; the finite ordinal 1 is not admitted as a weight in this part. Part I's hypothesis concerns maximal order type, $\text{otp}(\mathcal{K}(P_q)) = \omega^{\beta_q}$; the hypothesis here concerns height, and neither is assumed from the other.

26.2 Classical inputs

A WPO is a partial order with no infinite descending sequence and no infinite antichain. If P is a WPO, so is $\mathcal{K}(P)$ [1]: enumerate each finite set as a word and apply Higman's lemma, as in Section 2. Finite disjoint unions of WPOs are WPOs, and adding cross-fibre comparabilities cannot create a bad sequence, so S is a WPO.

Theorem 26.1 (Classical maximal-chain theorem [6, 9]; 09 Thm 2.1). *A WPO X has a chain of order type $h(X)$; hence $h(X) = \max\{\text{otp}(C) : C \subseteq X \text{ a chain}\}$.*

A compactness proof of the classical input (09). For $\beta < r_X(x)$ there is $y < x$ of rank exactly β : among predecessors of x of rank at least β choose one of least rank; were its rank larger than β , the rank recurrence would give a lower-rank predecessor of rank at least β . So each level $L_\beta = \{x : r_X(x) = \beta\}$, $\beta < h(X)$, is nonempty, and it is finite, since equal-rank elements are incomparable and a WPO has no infinite antichain. Give the levels the discrete topology; their product is compact in ZFC. For $\beta < \gamma < h(X)$ impose the closed condition $x_\beta < x_\gamma$. Finitely many conditions involve finitely many levels; choose an element at the greatest of them and use the predecessor observation repeatedly to obtain a chain through the others. So the conditions have the finite intersection property, and compactness gives a family satisfying all of them, a chain of type $h(X)$. Conversely rank is strictly increasing on chains, so no chain is longer. $\square$

This is the form of Wolk's theorem (Theorem 9 of [9]; stated as Theorem 1.2 of [6]) used here. Maximal order type *among chains* is not maximal order type *among linear extensions*; only the former is used. An arbitrary well-founded poset need not admit this chain interpretation of rank, so the WPO assumption must not be dropped.

26.3 Support ideals

Write $\mathcal{J}(Q)$ for the downsets (“ideals”, without directedness) of Q . For $I \in \mathcal{J}(Q)$ call $\max I$ its *active set* (09’s “active frontier” $F(I)$, which need not be a frontier) and $C(I) = I \setminus \max I$ its *completed part*; $C(I) = D(\max I)$ in Part I’s notation, and $C(I)$ concerns vertices of Q , not a chain. Both are empty for $I = \emptyset$.

27 The exact formula

For $I \subseteq J$ in $\mathcal{J}(Q)$ define, with the *empty maximum equal to 0*,

$$W^0(I, J) = \max\{h_q : q \in \max I \setminus \max J\}, \quad T^0(I) = \max\{h_q : q \in \max I\}, \quad T^0(\emptyset) = 0. \quad (68)$$

A linear extension $\pi = (q_1, \dots, q_n) \in \text{LE}(Q)$ gives the saturated ideal chain $I_j = \{q_1, \dots, q_j\}$, $I_0 = \emptyset$, and the value

$$V^0(\pi) = W^0(I_0, I_1) + W^0(I_1, I_2) + \dots + W^0(I_{n-1}, I_n) + T^0(Q), \quad (69)$$

every plus sign ordinary ordinal addition.

Theorem 27.1 (Pure-height substitution; 09 Thm 3.1). *Under (67),*

$$\boxed{h\left(\mathcal{K}\left(\sum_{q \in Q}^{\text{lex}} P_q\right)\right) = \Phi(Q; h) := \max_{\pi \in \text{LE}(Q)} V^0(\pi).} \quad (70)$$

For each π there is an explicitly described chain in the powerset of order type $V^0(\pi)$, once height-attaining chains in the component powersets have been chosen; in particular the maximum is attained.

Corollary 27.2 (Ordinal fibres; 09 Cor. 3.2). *For arbitrary positive ordinals ρ_q , $h(\mathcal{K}(\sum_q^{\text{lex}} \omega^{\rho_q})) = \Phi(Q; (\omega^{\rho_q})_q)$, with no restriction to countable exponents or to a notation system.*

Proof. The nonempty finitely generated downsets of an ordinal chain α are the initial intervals $[0, \xi]$, so $\mathcal{K}(\alpha)$ has type $1 + \alpha = \alpha$ for infinite α . Apply Theorem 27.1. $\square$

This recovers the chain-fibre formula of Part II by an independent route, and Section 31 shows that Φ equals the frontier formula (50) with α_q replaced by h_q .

28 A normal form for finitely generated downsets

Lemma 28.1 (No internally completed fibre; 09 Lemma 4.1). *A fibre satisfying (67) has no finite cofinal subset; its underlying set is not an element of $\mathcal{K}(P_q)$.*

Proof. A finite cofinal subset would generate P_q , giving $\mathcal{K}(P_q)$ a greatest element and a successor height, contrary to the limit ω^{ρ_q} . $\square$

This is the height analogue of Part I’s remark that purity of $\text{o}(\mathcal{K}(P_q))$ excludes finite cofinal subsets.

Proposition 28.2 (Support-profile normal form; 09 Prop. 4.2). *Every nonempty $D \in \mathcal{K}(S)$ is uniquely represented by a nonempty ideal $I \in \mathcal{J}(Q)$ and nonempty $B_q \in \mathcal{K}(P_q) \setminus \{\emptyset\}$ for $q \in \max I$, with*

$$D \cap P_q = \begin{cases} P_q, & q \in C(I), \\ B_q, & q \in \max I, \\ \emptyset, & q \notin I, \end{cases} \quad (71)$$

and every such profile represents an element of $\mathcal{K}(S)$. For profiles $(I, (B_q))$ and $(J, (E_q))$,

$$D \subseteq E \iff I \subseteq J \text{ and } B_q \subseteq E_q \ (q \in \max I \cap \max J). \quad (72)$$

Proof. Take a finite generating set of D . Its support $I = \{q : D \cap P_q \neq \emptyset\}$ is a downset, since a point of a strictly higher fibre lies above every point of a lower one. For $q \in \max I$ no generator in a strictly higher fibre contributes to $D \cap P_q$, which is therefore a nonempty finitely generated downset of P_q , proper by Lemma 28.1. For $q \in C(I)$ some $r \in \max I$ satisfies $q < r$, and one point of $D \cap P_r$ forces the whole q -fibre into D . This proves (71) and uniqueness. Conversely the union of finite generating sets of the B_q generates exactly (71): every vertex of $C(I)$ lies below a vertex of $\max I$, and every B_q is nonempty. For inclusion, support inclusion and inclusion on common active fibres are necessary; they suffice because a formerly active fibre that is no longer active is now completed and completed fibres stay completed. $\square$

A fixed-support stratum is thus a finite product of nonempty component powersets. Different strata are *not* arranged as an ordinary lexicographic sum: common active coordinates keep the comparison conditions of (72), and ignoring them is exactly the incorrect resetting operation that the transition formula avoids.

Lemma 28.3 (Nonempty height-attaining component chains; 09 Lemma 4.3). *For each q there is a chain $\mathbf{C}_q \subseteq \mathcal{K}(P_q) \setminus \{\emptyset\}$ of order type h_q ; it has a least element b_q .*

Proof. $\mathcal{K}(P_q)$ is a WPO of height h_q ; apply Theorem 26.1. If the chain contains $\emptyset$, delete that first element; removing one first element from an infinite additively indecomposable type leaves the type unchanged. $\square$

29 The ordinal mechanism: bounded progress is absorbed later

Lemma 29.1 (A finite product bound; 09 Lemma 5.1). *Let a chain vary in finitely many coordinates whose rank values lie below positive pure powers h_q , partitioned into unrestricted coordinates R and bounded coordinates B , every $p \in B$ being bounded throughout the chain by an element of rank $\beta_p < h_p$. Put $\lambda = \max_{q \in R} h_q$ and $\gamma = \#_{p \in B} (\beta_p + 1)$ (empty maximum and sum 0). Then the chain has order type at most $\lambda \# \gamma$.*

Proof. Map each tuple to the natural sum of its coordinate ranks; along a strict product comparison one coordinate rank rises and none falls, so the map is strictly increasing. The unrestricted part is below λ (or 0) and the bounded part below γ (or 0), so the chain embeds in $\lambda \# \gamma$ whenever there is a coordinate; the coordinate-free case is not needed for nonempty strata. $\square$

Lemma 29.2 (Deferred absorption; 09 Lemma 5.2). *Let each of $\lambda_0, \dots, \lambda_k$ be 0 or a positive pure power, with $\lambda_k > 0$; for $i < k$ put $L_i = \max_{j>i} \lambda_j$, and suppose $\gamma_i < L_i$ for $i < k$ and $\gamma_k = 0$. Then*

$$(\lambda_0 \# \gamma_0) + \dots + (\lambda_k \# \gamma_k) = \lambda_0 + \dots + \lambda_k. \quad (73)$$

Proof. Work from the right. If the suffix after position i is already $S_i = \lambda_{i+1} + \cdots + \lambda_k$, its leading monomial is L_i (its coefficient may exceed one). Every exponent of γ_i is below that of L_i , and ordinary addition of S_i erases all terms of smaller exponent from the preceding summand; if λ_i has smaller exponent it is erased too, and otherwise inserting γ_i does not affect its surviving terms. So $(\lambda_i \# \gamma_i) + S_i = \lambda_i + S_i$; the last step uses $\gamma_k = 0$. $\square$

The lemma lets a persistent large coordinate contribute a bounded amount before it is finally used at full scale; it does not let that coordinate's full pure power be added at every earlier stage. The strict inequality $\gamma_i < L_i$ is the substantive point. It plays the role of Part II's strict-margin Lemma 16.2, applied to a chain after the fact rather than inside a rank map.

30 Proof of the pure-height substitution theorem

Lemma 30.1 (Refinement inequalities; 09 Lemma 6.1). *For ideals $I \subseteq J \subseteq K$, $W^0(I, K) \leq W^0(I, J) + W^0(J, K)$, and for a nonempty ideal I , $T^0(I) \leq W^0(I, Q) + T^0(Q)$. Consequently the value of any finite strictly increasing path of nonempty support ideals, with terminal weight at its last ideal, is at most the value of some saturated ideal path from $\emptyset$ to Q .*

Proof. If $q \in \max I \setminus \max K$, then either $q \notin \max J$, contributing to $W^0(I, J)$, or $q \in \max J \setminus \max K$, contributing to $W^0(J, K)$; an ordinary sum is at least its larger summand. A vertex witnessing $T^0(I)$ either stays active at Q or lies in $\max I \setminus \max Q$. Prepending $\emptyset$ adds a zero weight, the last ideal extends to Q by the second inequality, and intermediate ideals are inserted one vertex at a time by the first, using monotonicity of ordinary addition inside a finite sum. A saturated path is a linear extension. $\square$

30.1 Upper bound: an arbitrary chain

Let $\mathbf{C} \subseteq \mathcal{K}(S)$ be a chain. If it has no nonempty member, its type is at most 1, below the infinite bound. Otherwise its nonempty supports form a finite chain $I_0 \subsetneq \cdots \subsetneq I_k$; let $\mathbf{C}_i$ be the nonempty convex segment with support I_i , and for $i < k$ put

$$R_i = \max I_i \setminus \max I_{i+1}, \quad B_i = \max I_i \cap \max I_{i+1}, \quad \lambda_i = \max_{q \in R_i} h_q,$$

with $\lambda_k = T^0(I_k)$; R_i and B_i partition $\max I_i$. Choose a member $E^{(i+1)}$ of $\mathbf{C}_{i+1}$. By (72), for $p \in B_i$ its coordinate $E_p^{(i+1)}$ bounds every p -coordinate in $\mathbf{C}_i$. Put $\beta_{i,p} = r_{\mathcal{K}(p)}(E_p^{(i+1)})$ and $\gamma_i = \#_{p \in B_i} (\beta_{i,p} + 1)$, $\gamma_k = 0$; as h_p is a limit, $\beta_{i,p} + 1 < h_p$. Lemma 29.1 gives $\text{otp}(\mathbf{C}_i) \leq \lambda_i \# \gamma_i$, and for $i = k$ all coordinates are unrestricted.

Each persistent coordinate is charged in full later. A vertex $p \in B_i$ stays active until it first stops being maximal in a later support and never becomes active again; if it retires at a transition $j > i$ then $h_p \leq \lambda_j$, and if it never retires then $p \in \max I_k$ and $h_p \leq \lambda_k$. So $h_p \leq L_i = \max_{j>i} \lambda_j$ for $p \in B_i$, every summand of γ_i is below the pure power L_i , and $\gamma_i < L_i$. By Lemma 29.2,

$$\text{otp}(\mathbf{C} \setminus \{\emptyset\}) = \sum_{i \leq k} \text{otp}(\mathbf{C}_i) \leq \sum_{i \leq k} (\lambda_i \# \gamma_i) = \lambda_0 + \cdots + \lambda_k. \quad (74)$$

The right side is a nonzero finite sum of positive infinite monomials (possibly with zero terms), so adding one on the left leaves it unchanged, and including $\emptyset$ does not enlarge the bound. By Lemma 30.1 it is at most $\Phi(Q; \mathbf{h})$. Finally $\mathcal{K}(S)$ is a WPO and Theorem 26.1 converts the bound on chains into $h(\mathcal{K}(S)) \leq \Phi(Q; \mathbf{h})$.

30.2 Lower bound: spend a coordinate just before it retires

Fix π with support ideals I_j and choose $\mathbf{C}_q, b_q$ as in Lemma 28.3. At a step $I_{j-1} \subsetneq I_j$ of nonzero weight choose $r_j \in \max I_{j-1} \setminus \max I_j$ with $h_{r_j} = W^0(I_{j-1}, I_j)$; at support I_{j-1} vary the r_j -coordinate through $\mathbf{C}_{r_j}$ and hold every other active coordinate at its base b_q . This block has type $W^0(I_{j-1}, I_j)$; skip zero-weight steps. At support Q choose $r_* \in \max Q$ with $h_{r_*} = T^0(Q)$ and vary it through $\mathbf{C}_{r_*}$ with the others at their bases. The blocks concatenate: a coordinate varied in an earlier block is completed at the next step and fully included later, and a coordinate active in an earlier and a later block had not yet been selected in the earlier one, so its earlier value is its base, below its later value in its chain. So (72) holds across every pair of blocks, strictly since the supports differ. The component chains need not be cofinal, since a strictly higher fibre dominates a whole lower fibre. The resulting chain has type (69), and maximizing over the finitely many π proves the theorem. $\square$

Remark 30.2 (Why this is not a reset argument; 09). Only a coordinate about to be fully dominated is varied through its whole component chain; a persistent coordinate stays at its base in the lower bound. In the upper bound it may vary, but only below an actual later coordinate value. These are two sides of one constraint.

31 Compression to inclusion-maximal antichains

Lemma 31.1 (Canonical completion of a support frontier; 09 Lemma 7.1). *For $I \in \mathcal{J}(Q)$ put $A(I) = \min(Q \setminus C(I))$. Then $A(I) \in \mathcal{M}(Q)$, $\max I \subseteq A(I)$, and $C(I) = \{q : (\exists a \in A(I)) q < a\} = D(A(I))$. If $I \subseteq J$ then $A(I) \leq A(J)$, and for a one-vertex enlargement $J = I \cup \{v\}$,*

$$A(I) \setminus A(J) = \max I \setminus \max J. \quad (75)$$

Proof. $C(I)$ is a downset. Every strict predecessor of a vertex of $\max I$ lies in $C(I)$, so $\max I \subseteq A(I)$. Each point of $C(I)$ lies below a member of $\max I$, and each point outside $C(I)$ lies above a minimal point of the complement, so every vertex is comparable with a member of the antichain $A(I)$. Minimality gives one inclusion for $C(I)$, and the fact that completed vertices lie strictly below $\max I$ the other. If $I \subseteq J$ then $C(I) \subseteq C(J)$; a member a of $A(I)$ either lies in $C(J)$, hence below a member of $A(J)$, or has all strict predecessors in $C(I) \subseteq C(J)$ and so lies in $A(J)$. For a one-vertex enlargement the new vertex is maximal in J , the newly completed vertices are exactly $\max I \setminus \max J$, and a member of $A(I)$ leaves $A(J)$ exactly when it enters $C(J)$. $\square$

By Remark 21.7, $A(I)$ is also 07's completion $\text{cl}(\max I)$ and Part I's canonical frontier. With w and t as in (49), but built from the heights h_q :

Theorem 31.2 (Maximal-frontier formula; 09 Thm 7.2).

$$\Phi(Q; \mathbf{h}) = \max_{\substack{A_0 < \dots < A_k \\ A_0 = \min Q, A_k = \max Q}} (w(A_0, A_1) + \dots + w(A_{k-1}, A_k) + t(A_k)). \quad (76)$$

When $\min Q = \max Q$ the length-zero path is allowed. It suffices to maximize over saturated chains of $\mathcal{M}(Q)$, though all comparable pairs are also correct.

Proof. Map a saturated support path to its canonical antichains $A(I_j)$. By Lemma 31.1 they are nondecreasing from $\min Q$ to $\max Q$ and their transition weights equal the support weights; repeated antichains are zero-weight steps and can be deleted. Conversely a strict chain of frontiers gives the ideals $I_i = \downarrow A_i$ with $\max I_i = A_i$; prepending $\emptyset$ costs 0, the last ideal is Q , and Lemma 30.1 refines it to a saturated support path of at least the same value. For $A < B < C$, $w(A, C) \leq w(A, B) + w(B, C)$, so inserting intermediate frontiers cannot decrease a value. $\square$

Corollary 31.3 (Uniform formula, independently recovered; 09 Cor. 7.3). *If every $h_q = \alpha = \omega^\rho$, $\rho > 0$, then $h(\mathcal{K}(\sum_q^{\text{lex}} P_q)) = \alpha \cdot h(\mathcal{M}(Q))$.*

For chain fibres this is Theorem 9.1; for WPO fibres with a common pure powerset height it is new relative to Part I, whose height theorem assumes ordinal fibres. No unrefereed repository theorem is imported.

32 First-successor scheduling and exact finite algorithms

32.1 A second interpretation of the weights

For $\pi \in \text{LE}(Q)$ with positions pos_π , define the retirement time $\tau_\pi(q) = \min\{\text{pos}_\pi(r) : q < r\}$ for $q \notin \max Q$ and $\tau_\pi(q) = n + 1$ for $q \in \max Q$. A vertex first stops being maximal in the current support when its first strict successor is inserted, so

$$V^0(\pi) = \sum_{j=1}^{n+1} \max\{h_q : \tau_\pi(q) = j\}, \quad (77)$$

chronologically, with empty groups contributing 0; the final group is $\max Q$. This needs no frontiers and is the independent linear-extension calculation of 09's verification suite.

32.2 The support-ideal and compressed programs

Let $d^0(\emptyset) = 0$ and, for nonempty ideals J ,

$$d^0(J) = \max_{v \in \max J} [d^0(J \setminus \{v\}) + W^0(J \setminus \{v\}, J)], \quad \Phi(Q; h) = d^0(Q) + T^0(Q). \quad (78)$$

Proposition 32.1 (Correctness of the recurrence; 09 Prop. 8.1). *$d^0(J)$ is the greatest transition sum of a saturated ideal path from $\emptyset$ to J , without terminal term, and recording a maximizing predecessor reconstructs an extension attaining (78).*

Proof. The last vertex of a saturated path to J lies in $\max J$, and deleting it leaves a saturated path to $J \setminus \{v\}$; conversely every such path extends by v . For a fixed edge weight η the map $\xi \mapsto \xi + \eta$ is nondecreasing, so a greatest predecessor value gives a greatest value after the edge, even though distinct values may become equal. Induct on $|J|$. $\square$

There are at most 2^n ideals with at most n incoming edges each, so after finite preprocessing the program uses $O(n2^n)$ ordinal additions and comparisons, and $O(n^2 2^n)$ elementary set operations with naive frontier handling: an exponential algorithm, not a polynomial-time claim. The compressed program sets $g(\min Q) = 0$ and $g(B) = \max_{A < B} (g(A) + w(A, B))$ for $B \neq \min Q$, with $\Phi = g(\max Q) + t(\max Q)$; cover edges suffice, and with $m = |\mathcal{M}(Q)|$ it uses $O(m^2)$ ordinal updates beyond finding the frontiers. 09's implementation deliberately uses all comparable pairs so that its correctness does not depend on a transitive-reduction routine. In pseudocode (09):

```

Input: finite poset Q, positive pure weights h[q]
D[empty] := 0
for each nonempty ideal J, in increasing cardinality:
  D[J] := maximum over v in Max(J) of
    D[J - {v}] + max(h[q] : q in Max(J - {v}) - Max(J))
return D[Q] + max(h[q] : q in Max(Q))
# Empty maxima are 0. Every + above is ordinary ordinal addition.
# Empty Q is a separate case with answer 1.

```

Neither program infers component heights (the weights h_q are inputs justified outside the computation) or decides equality of arbitrary ordinal notations.

Remark 32.2 (The two empty-exit conventions on one example). For the persistent example (x isolated, $y < z$, heights $(\omega^2, \omega, \omega)$; 09's vertices $(x, a, b) = (0, 1, 2)$), 09's support program gives

bitmask	0	1	2	3	6	7
I	$\emptyset$	$\{x\}$	$\{y\}$	$\{x, y\}$	$\{y, z\}$	$\{x, y, z\}$
$d^0(I)$	0	0	0	0	ω	ω

and $\omega + \omega^2 = \omega^2$; its compressed certificate needs the two states $\{x, y\}, \{x, z\}$ and one edge of weight ω . Part II's weight-1 program gives 0, 1, 1, 2, $\omega, \omega + 1$ on the same ideals (Section 23) and the same final value. For positive pure weights the answers always agree, because the terminal weight is infinite and absorbs every earlier finite term; the tables, and therefore the certificates, differ. The weight-0 convention is wrong as soon as a fibre is finite: a singleton fibre has a two-element finitary powerset. 09 excludes such fibres, so this is a scope boundary, not an error.

32.3 Finite certificates (09)

A support certificate lists every ideal, its active set, its value $d^0(I)$ and a selected predecessor for each nonempty ideal. A checker verifies $d^0(\emptyset) = 0$, $d^0(I) + W^0(I, J) \leq d^0(J)$ for every unit edge, and equality on each selected edge; these prove both finite path bounds. It also validates a linear extension whose full value is the claimed answer. The supplied checker does not call the optimizer but shares its exact arithmetic and poset routines, so it is not an independent formal kernel; the first-successor calculation tests the shared choices from another direction. A certificate proves the *finite optimization* relative to Theorem 27.1; it formalizes neither that theorem, nor Wolk's theorem, nor the input height claims.

32.4 A bounded-width consequence (09)

If Q has width at most b , every ideal is generated by an antichain of size at most b , so $|\mathcal{I}(Q)| \leq \sum_{j=0}^b \binom{n}{j}$, and enumerating those antichains and their downclosures gives a finite-state algorithm in time $n^{O(b)}$, transition generation included. This is an XP bound in b , not an FPT bound. The default program enumerates subsets directly; the bound describes a proved algorithmic variant, not a performance claim about that implementation.

33 Substitution invariance and limit-ordinal fibres

Corollary 33.1 (Pure-height congruence; 09 Cor. 9.1). *If (P_q) and (P'_q) are families of WPOs with $h(\mathcal{K}(P_q)) = h(\mathcal{K}(P'_q)) = \omega^{\rho_q}$, $\rho_q > 0$, then $h(\mathcal{K}(\sum_q^{\text{lex}} P_q)) = h(\mathcal{K}(\sum_q^{\text{lex}} P'_q))$.*

Proof. Both sides are the same finite optimization in the common weights. $\square$

This is a genuine loss of dependence on the internal structure of a fibre, but only once its own finitary-powerset height is supplied. Replacing that input by $h(P_q)$, by its maximal linear-extension type, or by its width is not justified; and knowing $h(\mathcal{K}(P_q))$ does not give $h(\mathcal{K}(\mathcal{K}(P_q)))$: unrestricted iteration is a different problem.

Theorem 33.2 (All nonzero limit-ordinal fibres; 09 Thm 9.2). *For nonzero limit ordinals β_q , expanded as in (56) (all exponents positive) into $Q^\#$ with pure weights $\alpha_v^\#$, $h(\mathcal{K}(\sum_q^{\text{lex}} \beta_q)) = \Phi(Q^\#; (\alpha_v^\#)_v)$. Finite*

ordinal normal forms and finite ordinal comparisons determine the answer exactly; the algorithm expands coefficients literally, and its size is measured in expanded blocks.

Proof. Lemma 22.1 and Corollary 27.2. □

No recursive expansion of an exponent's own normal form is needed: a block ω^{ε_0} is one pure block with exponent label ε_0 . A zero fibre is deleted and the induced skeleton used; the all-empty case is separate, $\mathcal{K}(\emptyset)$ being a singleton of height 1. Successor fibres have finite terminal terms and are not covered by this theorem; for chain fibres they are covered by Theorem 22.2 of Part II. This agrees with Corollary 22.3.

34 Exponent patterns, bounds and lost information

Theorem 34.1 (Exponent-order invariance; 09 Thm 10.1). *Fix Q and two families of positive exponents $(\rho_q), (\sigma_q)$ with the same equality and strict-order pattern. Replacing each ρ by the corresponding σ preserves the Cantor-normal-form coefficients of every path value, every program value and the final height, and preserves exactly the set of maximizing linear extensions and the set of maximizing frontier chains.*

Proof. Each transition takes the largest exponent in a finite subset; a path adds the selected monomials in order, which deletes strictly smaller trailing exponents and merges equal ones; comparing two results depends only on ordered exponents and finite coefficients. All of this commutes with a strictly increasing relabelling of the finite exponent set, for paths, maxima and both recurrences. □

The parameter space thus splits into finitely many weak-order patterns (“chambers”), each completely represented by integer priorities. A priority is not automatically an ordinal exponent: 09’s optional symbolic labels may denote arbitrary positive exponents in a declared order, and checking that order is outside the program’s scope. This theorem adds the preservation of maximizer sets to Corollaries 24.2 and 18.3.

Proposition 34.2 (Support and size of the answer; 09 Prop. 10.2). *Every exponent of $\Phi(Q; \mathbf{h})$ occurs among the ρ_q ; its leading exponent is $\max_q \rho_q$; its coefficient sum is at most $n = |Q|$; it has no finite tail; and $\max_q h_q \leq \Phi(Q; \mathbf{h}) \leq \sum_q h_q$.*

Proof. The retirement groups of (77), terminal group included, partition Q . One maximizer per nonempty group gives at most n distinct monomials whose ordinary sum is the path value; normalization neither introduces exponents nor raises the coefficient count, and the value is at most their natural sum. A vertex of globally largest weight lies in one group, whose selected weight is that maximum, and no later summand can erase it; so every schedule has the stated leading exponent and at least the largest input monomial. □

Even for enormous input exponents the answer is thus a finite coefficient vector on the input exponent labels.

34.1 Unbounded ambiguity of local stratum heights

For a frontier A the unrestricted product of active coordinates has height $t(A) = \max_{q \in A} h_q$ (natural-sum rank bound above; vary a largest coordinate through its attaining chain below), and so has the fixed-support stratum at $\downarrow A$. Thus $t(A)$ is exactly the local height label one might keep from a stratification.

Theorem 34.3 (Local labels do not determine the height; 09 Thm 10.3). *For every $m \geq 1$ and pure powers $\beta < \alpha$ with positive exponents there is one finite skeleton Q_m with m weight assignments for which the labelled frontier poset $(\mathcal{M}(Q_m), t)$ is identical while the global heights are the distinct values $\alpha, \alpha \cdot 2, \dots, \alpha \cdot m$.*

Proof. Take an isolated x of weight α beside a chain $a_1 < \dots < a_m$ whose vertices get weight α or β . The frontiers are exactly $A_i = \{x, a_i\}$, an m -element chain, and every $t(A_i) = \alpha$. For $j \in \{1, \dots, m\}$ give exactly $j - 1$ of $a_1, \dots, a_{m-1}$ weight α and the other chain vertices weight β . The compressed formula has successive weights $h_{a_1}, \dots, h_{a_{m-1}}$ and terminal weight α ; each smaller term is absorbed by a later α , and the sum is $\alpha \cdot j$. $\square$

The deficiency of a stratum-only formula is therefore unbounded in the number of stages: even the whole frontier poset with every local height label is insufficient, and the identities of the coordinates leaving between frontiers carry indispensable information. This strengthens Part I's single counterexample and Corollary 23.2.

35 Worked examples and a height-two reduction

35.1 Chains, antichains and the persistent coordinate

If Q is a chain $q_1 < \dots < q_n$, its frontiers are its singletons and $\Phi = h_{q_1} + \dots + h_{q_n}$. If Q is an antichain, $\mathcal{M}(Q) = \{Q\}$ and $\Phi = \max_q h_q$. This does not contradict the natural-sum bound for product ranks: the supremum of natural sums of finitely many coordinates strictly below pure limits is the largest such limit, not their natural sum. For the persistent skeleton $\{x, y < z\}$ the frontiers are $\{x, y\} < \{x, z\}$ and

$$\Phi = h_y + \max(h_x, h_z); \quad (79)$$

$(h_x, h_y, h_z) = (\omega^2, \omega, \omega)$ gives $\omega + \omega^2 = \omega^2$, and changing only h_y to ω^2 gives $\omega^2 \cdot 2$, although both frontier labels remain ω^2 .

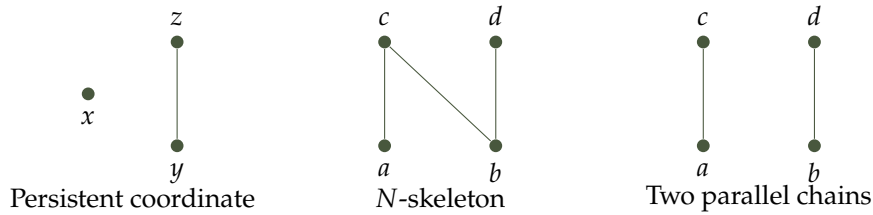


Figure 4: The skeletons of 09's examples, in this report's labelling (09 draws the N -skeleton as $a < c, a < d, b < d$). The persistent coordinate contributes fully only at the terminal stage; the parallel chains permit different retirement orders.

35.2 A genuinely non-series-parallel skeleton

The N -skeleton $a < c, b < c, b < d$ has connected comparability and incomparability graphs, so it is neither a disjoint union nor an ordinal sum and is not generated from singletons by those operations. Its frontiers form the chain (11), and $\Phi = h_b + h_a + \max(h_c, h_d)$. For $(h_a, h_b, h_c, h_d) = (\omega^2, \omega^3, \omega, \omega)$ this is $\omega^3 + \omega^2 + \omega$ (09's weights $(\omega^3, \omega^2, \omega, \omega)$ in its labelling). A path jumping from the first frontier to the last would charge only $\max(h_a, h_b)$ before the terminal term; refining a transition can keep an additional lower-order term, so grouping matters even when the leading exponent is fixed.

35.3 Optimization depends on the exponent pattern

For two disjoint chains $a < c$ and $b < d$, either a or b may retire first. With $t_{cd} = \max(h_c, h_d)$,

$$\Phi = \max(h_a + h_b + t_{cd}, h_b + h_a + t_{cd}) = (h_a \# h_b) + t_{cd}, \quad (80)$$

because putting the larger of two monomials first produces their natural sum. For $(\omega^2, \omega^3, \omega, \omega)$ the result is $\omega^3 + \omega^2 + \omega$; retiring a before b would lose the ω^2 , and swapping the source weights swaps the preferred order. The limit fibres $\beta_a = \omega^2 + \omega$ and $\beta_b = \omega \cdot 2$ on a two-point antichain expand to exactly this pair of two-step chains with block weights (ω^2, ω) and (ω, ω) , so

$$h(\mathcal{K}((\omega^2 + \omega) \sqcup (\omega \cdot 2))) = \omega^2 + \omega \cdot 2; \quad (81)$$

both $\max(\beta_a, \beta_b) = \omega^2 + \omega$ and $\beta_a \# \beta_b = \omega^2 + \omega \cdot 3$ miss it. Purity matters in the simple antichain formula, and block expansion handles this nonpure limit case exactly.

35.4 Height-two skeletons as first-neighbour schedules

(09 writes $L, U, Z, N(\ell)$ and $G_i(\sigma)$ for the sets and groups renamed here, to avoid the label set U and the terminal region Z of Part II and the N -poset.)

Let Q be the disjoint union of Q_{lo} , Q_{up} and Q_{iso} , every strict relation having the form $\ell < u$ with $\ell \in Q_{\text{lo}}$, $u \in Q_{\text{up}}$; every point of Q_{lo} has an upper neighbour, every point of Q_{up} a lower neighbour, and Q_{iso} is the set of isolated vertices. Write $N(\ell) \subseteq Q_{\text{up}}$ for the nonempty upper-neighbour set of ℓ ; all of $Q_{\text{up}} \cup Q_{\text{iso}}$ is maximal.

Theorem 35.1 (Height-two scheduling formula; 09 Thm 11.1). *For a permutation $\sigma = (u_1, \dots, u_s)$ of Q_{up} let $\Gamma_i(\sigma) = \{\ell \in Q_{\text{lo}} : u_i \in N(\ell), u_j \notin N(\ell) \text{ for } j < i\}$. Then*

$$\Phi(Q; h) = \max_{\sigma} \left[\sum_{i=1}^s \max_{\ell \in \Gamma_i(\sigma)} h_{\ell} + \max_{q \in Q_{\text{up}} \cup Q_{\text{iso}}} h_q \right], \quad (82)$$

the case $Q_{\text{up}} = \emptyset$ being the antichain case.

Proof. Restrict a linear extension to Q_{up} . A lower vertex retires at its first upper neighbour in this order, so its retirement group is the corresponding $\Gamma_i(\sigma)$, and inserting lower or isolated vertices retires nothing already active. Every permutation of Q_{up} is realized by inserting all of $Q_{\text{lo}} \cup Q_{\text{iso}}$ first. So (77) becomes (82) with no lost schedules. $\square$

For example take $Q_{\text{lo}} = \{0, 1, 2\}$, $Q_{\text{up}} = \{3, 4, 5\}$, $N(0) = \{3, 4\}$, $N(1) = \{4, 5\}$, $N(2) = \{3, 5\}$, source weights $(\omega^3, \omega^2, \omega)$ and upper weights ω . The order $(3, 5, 4)$ retires $\{0, 2\}$, then $\{1\}$, then nothing new, giving $\omega^3 + \omega^2 + \omega$ with the terminal term, and the shipped certificate `data/03-wpo-height_two_overlap_certificate.json` establishes optimality. This is a concrete finite set-system scheduling problem; its computational complexity is not classified here. [Added 29 September 2026, batch 42: Part IV classifies it. With all weights equal, the largest number of nonempty groups is the maximum uniquely restricted matching number of the incidence graph, and deciding the height threshold is NP-complete already when every fibre is ω (Theorems 43.3, 45.1 and 46.2); for unequal weights (82) is an ordered-matching maximum (Theorem 47.1).]

36 Implementation, verification and foundations of Part III

09's module `code/03-wpo-frontier_heights.py` contains exact finite Cantor-normal-form arithmetic, transitive closure and cycle rejection, both programs, limit-fibre expansion and a

certificate checker, with the standard library only and no floating point; answers are decreasing priority-coefficient lists. It takes pure weights or limit-fibre normal forms; the persistent input is `{"n": 3, "edges": [[1, 2]], "priorities": [2, 1, 1]}`, vertices numbered from 0 and edges any acyclic generating relation. Without labels priorities are read as finite exponents; an optional dictionary such as `{"1": "omega", "2": "omega+1", "3": "epsilon_0"}` labels increasing positive exponents, and the program does not prove that the strings denote ordinals in the declared order; the justification for relabelling is Theorem 34.1. An input-size guard rejects more than 22 vertices, expanded blocks included: a practical guard for an exponential implementation, neither a restriction of the theorem nor a promise that every 22-vertex instance is cheap. In the delivered layout the commands are `python code/verify.py` and `python code/frontier_heights.py` applied to `examples/nonuniform_N.json` (with `-output` and `-check`) and `examples/limit_fibers.json`; the verifier overwrites only its own results and generated examples and certificates.

The recorded run (Python 3.13.5, seed 24092026) enumerated, for each $n \leq 6$, every transitive subrelation of the natural strict order on $\{0, \dots, n-1\}$: 1, 1, 2, 7, 40, 357, 4824 posets for $n = 0, \dots, 6$, 5,232 in all and 5,231 nonempty, covering every isomorphism type but not an unlabelled count. Uniform checks cover the census; weighted checks use all three-priority assignments for $n \leq 4$, all assignments from $\{1, 3\}$ for $n = 5$, and one seeded four-priority assignment for each of the first 256 six-vertex posets; every weighted instance enumerates *all* its linear extensions and values them by the first-successor rule.

Check or measured quantity (09)	Count
Weighted instances	15,131
Independently valued linear extensions	249,057
Uniform finite certificates	5,232
Weighted finite certificates	15,131
Weighted program versus all-extension maximum	15,131
Weighted support/compressed program agreement	15,131
Uniform maximal-antichain height identities	5,231
Sampled profile inclusion comparisons	39,371
Canonical frontier membership checks	407
Compressed retirement identities	555
Deferred-absorption CNF checks	20,000
Priority relabelling / vertex relabelling / tamper rejection	369 each
Total recorded assertions, including further checks	167,920

Table 10: Selected categories of 09’s ledger, shipped as `data/03-wpo-verification.json`. The total also includes witness values, uniform program agreement, coefficient and natural-sum bounds and the limit-fibre example. Counts are not proof strength.

The sampled profiles use a small concrete representation of full and partial fibres and check the inclusion criterion, not the rank of a truncation standing in for an infinite rank. The absorption tests use bounded coefficients; the proof of Lemma 29.2, not their range, establishes the identity. The recorded runtime, about 9.2 seconds, is not a performance guarantee. No finite test computes an uncountable order type from samples. The checks do not establish Wolk’s theorem, the input component heights, the transfinite upper bound or novelty, and “two programs can share an incorrect mathematical premise” (09): the chain construction and the deferred-absorption proof are the justification for the infinite theorem.

36.1 Foundational scope and a formalization route (09)

All inputs of an instance are sets; the fibre family, the set of component heights and the set of linear extensions are finite. The proof works for uncountable fibres and exponents without large-cardinal hypotheses; it is a theorem scheme over set-sized ordinals, not a claim about a set of all ordinals or a proper-class linear extension. For ordinal fibres the lower-bound chains are explicit initial-segment chains. For WPO fibres the finite schedule is computable from the weights, but selecting height-attaining chains in the components uses the classical maximal-chain theorem: a nonuniform input, not an algorithm, so the computability phenomena of [6] are not bypassed.

A formal development can separate the finite optimization from the transfinite mathematics: first the support-profile theorem, then the finite natural-sum bound and deferred absorption, then the main upper proof as a finite induction over the support values of a chain, and the lower proof as an ordinal-indexed concatenation of component chains with checked cross-block inclusions. The classical input should be stated exactly as “a WPO has a chain attaining its rank-defined height”, never as a replacement of rank by a supremum of chain types for arbitrary well-founded orders. A separate finite layer should prove (78), certificate soundness and relabelling invariance; a checker verified there would certify values conditional on the transfinite theorem and the component-height lemmas. No Lean file is included or claimed to have been compiled; a future formalization must report which classical inputs are proved in its library and which, if any, are admitted.

The boundaries are concrete. The skeleton is finite, since the number of support changes and the sums of bounded terms must be finite. The component powerset heights are positive pure powers, which supplies additive indecomposability, limits and the absence of internally completed fibres. Components are WPOs, which supplies closure and chain attainment. The order is the standard Hoare/finitely-generated-downset order; a different powerset ordering need not satisfy the profile rule. The empty skeleton needs its own case (the formula’s empty maximum would be 0, while $\mathcal{K}(\emptyset)$ has height 1), and allowing the finite weight 1 without correction is false, since a singleton has a two-element finitary powerset. 09’s proof audit lists the same boundaries: empty Q is separate; finite or successor components are not admitted to the pure-height theorem; general nonpure WPO components are not covered by the ordinal CNF expansion; infinite skeletons need new limit-stage arguments; the theorem computes height, not width or maximal linear-extension type; it does not justify repeated application of $\mathcal{K}$ without new component-height calculations; and set-sized instances in ZFC are used, with no proper-class rank object.

09’s dependency audit, in brief: the profile normal form needs the finite skeleton and no finite cofinal subset in a fibre (a completed fibre need not be compact internally); the product chain bound needs strictly increasing coordinate ranks, finite natural addition and persistent coordinates bounded by actual later elements; deferred absorption needs finite Cantor normal forms and a strictly smaller bounded term; the main upper bound combines profile comparisons, the finite support partition, the product bound, absorption, refinement and Wolk’s theorem; the lower bound needs nonempty height-attaining component chains, varying a coordinate only just before it disappears; the compressed formula needs the canonical antichains and exact equality of retirement sets at one-vertex steps; the limit-fibre theorem is a finite order isomorphism by literal block expansion, finite and successor terminal blocks excluded; exponent chambers need only the finite exponent preorder; and the certificates check every state and edge, with no transfinite theorem machine-checked.

Closing the addition

37 Research questions of the addition

The four sources propose nine (07), twelve (09), eight (11) and nine (12) further questions. None of them is asserted to be a previously recognized or numbered open problem in the literature, except for the broader published extension programme already cited; 09 adds that its directions are “not all asserted to be previously unstudied” and that each should first be compared with existing results. Where one source’s question is answered by another, it is re-scoped here rather than printed as open.

37.1 Questions answered or partly answered inside the addition

- *Finite tails and successor fibres* (07 Question 1; 09 §14.1, with the milestone “fibres of the form $\omega^p c + k$ ”): **answered** for ordinal-chain fibres by Theorem 22.2 (12) and the exact model (57) (11). The “extra rank data attached to a frontier” that 07 asked for are the all-antichain states with fallback weight 1; the empty-downset and greatest-element corrections that 09 anticipated are exactly the states that maximal frontiers of the expanded skeleton would discard (Section 23).
- *Rank-level refinement and a direct point-rank formula* (07 Question 9, second half; 09 §14.3): **answered** for ordinal-chain fibres by Theorems 22.4 (12) and 19.4 (11), the latter for any finite persistent system. 09’s hope that a point-rank formula could remove the appeal to chain attainment in the ordinal-fibre upper bound is met differently: the proofs of Part II never use chain attainment. For WPO fibres the point-rank question stays open.
- *Non-chain fibres* (07 Question 2; 12 Problem 10.6; 11 Problem 12.1): **partly answered** by Part III, for WPO fibres whose finitary powersets have positive pure height. 12’s Proposition 24.3 gives an abstract non-chain version without a support representation. Still open: nonpure component heights (09 §14.2: does equality of $h(\mathcal{K}(P_q))$ still determine the output when those heights are not pure; if not, find a small counterexample and a replacement invariant describing bounded initial segments; Theorem 33.2 does not answer this, since it expands actual ordinal fibres), and 11’s question whether a finite family of principal-ideal height profiles could replace the ordinal-coordinate phases, with finite products of ordinal chains carrying their local rank functions as a first target. 07 asked whether a pure value of $h(\mathcal{K}(P_q))$ together with “suitable uniform upper-cone rank information” suffices: Part III shows the pure value alone suffices for WPO fibres, at the price of Wolk’s theorem. 12 asked whether pure height, a chosen least element and cofinal-chain data suffice on a nontrivial class; any positive answer must prove the representation conditions separately, since equal scalar heights do not identify which coordinates survive.
- *Complexity* (07 Question 4; 09 §14.4; 12 Problem 10.3): **partly addressed** by Corollary 20.4, NP membership of the threshold problem for the explicit pure-block encoding. A polynomial-time algorithm or an NP-hardness result, even for two positive scales, would sharpen it; the short output of Corollary 24.1 does not make the path problem easy. 07 asked in particular about deciding whether the leading or a specified lower coefficient exceeds a threshold when Q is given only by its order relation and exponent pattern, before enumerating frontiers; 09 asks whether the uniform case (largest number of nonempty retirement groups) and the mixed case (a lexicographically significant coefficient vector) are known ordered-set or set-system optimization problems, and to separate input-

comparison cost from combinatorial cost before any hardness claim. Any hardness claim needs a separate reduction, not supplied here. [Added 29 September 2026, batch 42: the two preceding sentences describe the state before Part IV, which **answers** this item for the hardness alternative and supplies the reduction. The threshold problem is NP-complete for height-two skeletons with every fibre ω , one scale (Theorem 46.2), hence for the pure-block encoding of Corollary 20.4 (Remark 46.5); 09's uniform case is a uniquely restricted matching number, its mixed case an ordered-matching objective, and 07's leading-coefficient threshold is NP-complete at one scale (Section 54).]

- *The report's own next target* (Section 14): **answered**, by Theorems 21.1, 22.2 and 27.1.

37.2 Questions that remain open

Problem 37.1 (Coefficient-compressed algorithms; 12 Problem 10.1, 07 Question 3, 09 §14.6). Compute the height directly on Cantor-normal-form coefficients given in binary, without expanding a coefficient n into n blocks, and determine the complexity in the coefficient bit lengths and the size of the original skeleton.

Block expansion proves computability but can blow up before ideal enumeration even begins. A first target is a fixed skeleton with arbitrary coefficients and a fixed number of exponent scales (12); even an efficient rule for two interacting repeated chains would help, and it must keep lower-order terms that survive only for particular retirement interleavings (09). The issue is succinct finite combinatorics, not ordinal notation; chains and antichains give closed-form test cases (07). A compressed algorithm must account for finite tails rather than discard them (11).

Problem 37.2 (Structural and fixed-parameter algorithms; 12 Problem 10.2, 09 §14.5, 11 Problem 12.5). Beyond Corollary 20.3 and the $n^{O(b)}$ bound of Section 32, find algorithms of the form $f(k)N^{O(1)}$ for a parameter k : the width of Q or of $\mathcal{M}(Q)$, the number of distinct exponent priorities or capacity levels, a decomposition width of the comparability or cover graph (bounded treewidth, series-parallel skeletons), or the number of distinct upper-neighbour sets, avoiding the full antichain or phase expansion.

The relevant parameter may be the number of simultaneously persistent coordinates rather than the number of frontiers; a proof must identify which partial-state information permits safe merging of histories under noncommutative ordinal addition (12). The present solvers are polynomial in an explicit state graph that may be exponentially larger than the input (11). [Added 29 September 2026, batch 42: for height-two skeletons Part IV gives algorithms of the form $2^a N^{O(1)}$ in the number a of distinct upper-neighbour sets, and FPT algorithms in the number of distinct upper neighbourhoods and in the vertex-cover number (Corollaries 50.3, 50.2 and 50.4); the number of distinct exponent priorities admits no FPT or XP algorithm unless $P = NP$ (Corollary 46.3). The problem stays open for the other parameters and for taller skeletons.]

Problem 37.3 (Ordinal width and maximal linear-extension type; 12 Problem 10.4, 07 Question 5, 09 §14.8, 11 Problem 12.2). Compute the ordinal width (and, 09, the maximal linear-extension type) of the same mixed-fibre finitary powersets, beginning (11) with persistent systems with two distinct positive capacity levels and state posets of bounded width. In particular (07), can the finite weighted frontier data (49) that compute the height also compute the ordinal width, or must they be enriched by information about concurrent rather than sequential coordinate changes?

The height proofs concern decreasing sequences; antichain sequences need not have chain-ordered labels, so the finite chain cover is unavailable in that form and a chain certificate no

longer controls an antichain rank tree (07, 11, 12). Vialard's treatment of widths of Cartesian products of ordinals [8] is a starting point (11). Theorem 34.3 already shows that coarse local data lose information; for these invariants the replacement for bounded persistent progress may involve antichain or bad-sequence residuals (09). Part I's own width question (Section 14) is unchanged.

Problem 37.4 (Repeated finitary powersets; 12 Problem 10.5, 07 Question 6, 11 Problem 12.3). Find a finite representation, containing finite ordinal lexicographic sums and closed under $\mathcal{K}$, on which $h(\mathcal{K}^r(S))$ is computable for every finite r , or at least $h(\mathcal{K}(\mathcal{K}(S)))$ on a substantial subclass without reduction to finitely many numerical samples.

The first output generally has product-like pieces rather than chain fibres, so it is not another finite lexicographic sum of chains; this is the exact obstruction to an iteration theorem from Theorem 22.2 (11, 12). The output height is not a sufficient description of the first powerset, and even stratum isomorphism types are insufficient without their attachments (Corollary 23.2); a symbolic presentation of profiles and transitions (07), or of finite antichains of persistent tuples (11, whose state component need not remain finite), is a plausible start. Proposition 24.3 suggests building blocks but not the required support representation.

Problem 37.5 (Infinite skeletons and infinite state orders; 12 Problem 10.7, 07 Question 7, 09 §14.7, 11 Problem 12.7). Extend the theorems to well-founded or well-quasi-ordered infinite skeletons, or to infinite persistent-state systems, under explicit compactness, bounded-frontier or controlled-support-evolution hypotheses.

Finiteness is used for the maximum of finitely many tree ranks, for keeping finitely many smaller monomials below a pure power, for the existence of a largest active capacity, and for the finite depth correction in (36); infinite unions of well-founded trees can acquire infinite branches, finitely bounded progress can accumulate at a limit stage, and a support need not be captured by a finite chain of states. Replacing a maximum by a supremum is not a proof and may not be the right statement (07, 09, 11, 12); increasing ordinal skeletons are a test class (09).

Problem 37.6 (Minimal presentations, chambers and schedules; 12 Problem 10.8, 07 Question 8, 09 §§14.10–14.11, 11 Problems 12.4 and 12.6). Characterize which weighted finite frontier graphs arise from finite skeletons with ordinal fibres (07). Characterize when two finite persistence presentations, or two weighted frontier graphs arising from finite skeletons, give the same height for every order-preserving exponent substitution, and find minimal or canonical presentations; find a stronger equivalence that also preserves every point rank. Classify, for a fixed skeleton, all output coefficient vectors as the exponent preorder varies, and represent them compactly without enumerating all ordered partitions of the labels; determine which skeletons admit one schedule optimal in every chamber (09). Describe equivalence classes of linear extensions with the same retirement groups or value.

Exponent invariance removes the numerical names of scales; a further reduction must preserve lifetime intersections and retirement opportunities, and equality of active maxima or of stratum heights is too weak (11, 12). Edges that never lie on a maximizing path may be redundant, and deciding universal redundancy is a finite symbolic optimization (07). The compact frontier model shows that height-only compression can exist without a pointwise-rank identification, so the two goals should be studied separately (11). Two parallel chains show that a schedule optimal in every chamber need not exist, so the question is which skeletons admit one; an intrinsic characterization could say more than enumerating all weak-order patterns (09). Zero-weight insertions and absorbed lower terms create redundancy that a quotient of schedules could remove, giving smaller certificates (09).

Problem 37.7 (Formal verification and constructive scope; 12 Problem 10.9, 07 Question 9, 09 §14.12, 11 Problem 12.8). Formalize the persistence theorem (or the retirement-box theorem and the normalization, or the compressed-frontier theorem and the support certificate checker) together with a certificate checker, and identify the weakest set-theoretic or type-theoretic setting in which the rank maps, attaining chains and checker can be developed uniformly for effective notations below a fixed bound.

The arguments use finite choices and well-founded ordinal recursion, not a large-cardinal principle, and the ε_0 bound of the code is a notation boundary, not an expected theorem boundary (12). For WPO components one should identify the minimum component-witness interface that turns a finite schedule into a represented infinite chain, keeping classical assumptions and algorithmic content visible (09). No reverse-mathematical equivalence is claimed, and for applications to termination a separate norm and control function would be needed: ordinal height alone does not bound all finite execution lengths (11).

Problem 37.8 (Noetherian spaces and Hoare powerspaces; 09 §14.9). Which parts of the support-profile argument survive for a finite gluing of Noetherian spaces under a lower-Vietoris or Hoare construction?

The combinatorial normal form and the chain-attainment step should be audited separately: a topological rank need not share every chain property of a WPO, so the present proof does not establish a topological generalization. The neighbouring report on Hoare powerspace statutes treats a different, topological question.

Problem 37.9 (Editorial: relation between Part III and Proposition 24.3). Can Theorem 27.1 be derived from Proposition 24.3, applied with coordinates $\mathcal{K}(P_q)$ (least element $\emptyset$), Wolk's theorem and the canonical frontier of Lemma 6.1? The lower bound would need 09's base-point chains, since $\mathcal{K}(P_q) \setminus \{\emptyset\}$ has no least element. This question was raised during the merge and has not been checked; none of the four sources observes it.

38 What the addition does not claim

Every limitation stated by any of the four sources is kept. Parts II and III compute height and individual ranks; they give no formula for ordinal width or for maximal order type or maximal linear-extension type of the powersets, no theorem for iterated or repeated finitary powersets, no theorem for infinite skeletons or infinite state systems, and no large-cardinal consistency or inconsistency conclusion. They do not solve the unrestricted finitary-powerset problem for arbitrary well-partial-orders (07) or settle the published programme (12). Part III covers WPO fibres only under pure positive powerset height; general nonpure WPO components and finite or successor WPO components are not covered, and replacing $h(\mathcal{K}(P_q))$ by $h(P_q)$, a maximal linear-extension type or a width is not justified. Complexity statements count ordinal operations on explicit state graphs; none is a polynomial-time bound in the skeleton size or in a binary encoding of Cantor coefficients, and NP membership is not NP-completeness. [Added 29 September 2026, batch 42: Part IV proves NP-completeness (Theorem 46.2 and Remark 46.5); the rest of the preceding sentence stands.] There is no bound on lengths of controlled executions, and no claim that the first rank maps of Theorem 17.4 are pointwise exact. The executable notations stop below ε_0 (07's code takes positive integer exponents and is "not a general parser for arbitrary ordinal notations"; 09's symbolic labels are unchecked), with guards of 20 (12), 22 (09) and 18 blocks or 4,096 refined states (11).

No elementary supporting lemma is claimed with priority (07), and the product-height arithmetic, Wolk's theorem and the uniform formula are credited to their sources. Each source's

literature search was targeted and uneven, and non-discovery is not evidence of novelty. Earlier preprint versions of Vialard’s lexicographic-product paper have different authorship and carry an author-disagreement notice; they must not be conflated with the published record [4] (07), and no theorem is borrowed from them (11). Each source’s audit is a self-audit, not a referee review (12). The sources state that they made no repository write, and none of them assumes the correctness of the unrefereed theorems of Part I; their repository comparisons are report-relative, and no review of every repository theorem, or build of the repository, is asserted (09, 11, 12). The computations certify finite identities and implementations only; they are not proof-assistant checks, not computations of infinite descending-tree ranks, and not certificates of novelty.

39 Provenance of the addition and choices made in merging

The addition was written from four manuscripts (Table 3), all pinned to e21766d04, batch-36 manuscripts 07, 09, 11 and 12, arrived in e13affd32 and placed in 1a1396d4d. Their contributions: 12 is the base of Part II (persistence theorem, ideal-state program, arbitrary fibres, point ranks, NP membership, chain-partition bound, scale invariance, non-chain coordinates, product benchmark); 11 supplies the peeling proof and upper certificate, the pair-based program, the exact all-antichain isomorphism, phase refinement as a theorem, point ranks in any persistent system, and the successor and local-rank examples; 07 supplies the normalization, the chain-cover proposition, the retirement-box theorem and its second proof of the frontier formula, saturation, the paired stratum-types example, the diamond, universal bounds, the product lemma for limit factors and the frontier-graph program; 09 is Part III.

The merge had to choose in the following places.

1. *Base and parts.* 12 is the base for chain fibres because it is the most general there (all ordinal fibres, point ranks) and uses no chain-attainment theorem. 09’s theorem is incomparable with 12’s and keeps its own part and hypotheses.
2. *Duplicates.* A result proved by several sources is printed once with its provers named. Genuinely different proofs are kept as marked routes: 11’s peeling proof of Theorem 17.4, 07’s normalization-and-box proof of Theorem 21.1, 11’s direct proof of (57), 11’s route to point ranks, and 07’s proof of the product lemma for limit factors. The tree-union lemma, block realization, the no-return inequality, uniform recovery, exponent invariance and the coefficient bounds were merged across sources.
3. *Conventions.* Empty exits weigh 1 in Part II and 0 in Part III, with the difference displayed in Remark 32.2. The maximization domain is stated for all nonempty chains, with the saturated and maximal restrictions proved equivalent. Symbols follow Part I, and every renamed symbol is listed in Table 4; 09’s α_q became h_q to keep heights and fibres apart. 09’s examples were relabelled to this report’s N -poset; the persistent example uses Part I’s $x, y < z$ throughout.
4. *Hidden hypotheses checked when splicing.* 07’s normalization uses $1 + x < \alpha_q$ and so needs infinite pure fibres, which Section 21 assumes; 11’s peeling assumes $\rho_q > 0$, handled by convention 3; 09’s lower bound needs Wolk chains; 12’s non-chain proposition needs least elements.
5. *Editorial additions.* Remark 21.7, Problem 37.9 and the example after Corollary 24.2 showing that its positivity hypothesis is needed are the merge’s own and are marked as such. The placement-time cross-check of the four programs is reported from the placement commit; its script is not shipped.

6. *Bibliography*. Entries were merged into Part I's list. 11's entry for Vialard's lexicographic-product paper, which gave no DOI, is replaced by Part I's published record [4]. The Hilton, Schmidt, Wolk, Marcone–Montalbán–Shore and Vialard width entries are new.
7. *Part I*. Its labelled text is unchanged, and so are its section, theorem and equation numbers with one exception: equations are numbered continuously and Part I's appendices now follow Parts II–III, so the truncation formula of Appendix A, equation (24) in the pinned article, is now (83). Sentences in Part I that said no nonuniform height formula was known or claimed were kept as provenance of the pinned state and given dated pointers to this addition. The delivered data files `data/weighted_N_result.json` and `data/mixed_height_counterexample_result.json` still say "No nonuniform height formula is asserted."; they predate the addition and stay byte-identical.

40 Conclusion of the addition

The missing quantity in a mixed-fibre height calculation is not the height of a stratum but the length of a coordinate that can be exhausted before leaving it (12). Charging a coordinate only when it retires gives an exact formula because two mechanisms match: a natural-sum rank bound that keeps the still-active coordinates, and a concatenated chain that exhausts one disappearing coordinate at each stage; a finite cover of the descending-sequence tree then makes the global maximum exact. Three further routes reach the same formula: a leading-exponent peeling induction (11), a normalization onto maximal supports with a box theorem (07), and, for WPO fibres, deferred absorption of bounded persistent progress together with Wolk's theorem (09). The result is a complete height and point-rank calculus for finitely generated downsets of finite lexicographic sums of ordinal chains, including finite and successor fibres, an exact extension to WPO fibres of pure powerset height, and a precise explanation of Part I's persistent counterexample. Width, iteration, non-pure general fibres and infinite skeletons remain separate research tasks. The manuscripts are unrefereed; independent checking and a fuller priority search remain necessary before the addition is treated as a refereed or historically certified contribution.

Part IV. Uniquely restricted matchings: the height-two problem is NP-complete

This part is batch-42 manuscript 04, added on 29 September 2026. It answers the complexity item of Section 37 and the remark closing Section 35, and partly answers Problem 37.2. Its uniform theorem is proved without any theorem of Parts I–III; its nonuniform transfinite readings rest on Theorem 35.1. Equations of this part are numbered within sections, so that no equation number of Parts I–III or of Part I's appendices moves.

41 Provenance, conventions and the answer

41.1 The question

Theorem 35.1 (Part III) expresses the height for a height-two skeleton as an optimization over first upper neighbours, and the text after it said that the computational complexity of this finite set-system scheduling problem was “not classified here”. The complexity item of Section 37 asked whether the uniform and nonuniform problems correspond to established ordered-set or set-system optimization problems, and asked for a polynomial-time algorithm or an NP-hardness result, even for two positive scales. Problem 37.2 asks for fixed-parameter algorithms in structural parameters, including the number of distinct upper-neighbour sets.

The answer of this part (manuscript 04) is that the uniform optimization is exactly a classical matching invariant. Thus the hoped-for unrestricted polynomial algorithm would imply $P = NP$. The obstruction does not require complicated ordinal notation, many scales, large coefficients, or a tall skeleton: it already occurs with a 0–1 incidence matrix and fibres all equal to ω . At the same time, the weighted problem has additional structure not captured by one maximum-cardinality matching. Its lower ordinal terms can distinguish maximum matchings that are indistinguishable at the leading scale.

Main results (04). For a finite nonempty skeleton Q of height at most two, with bipartite incidence graph G_Q , and a uniform fibre $\alpha = \omega^\rho$, $\rho > 0$,

$$h(\mathcal{K}(S_\alpha(Q))) = \alpha \cdot (v_{\text{ur}}(G_Q) + 1) \quad (\text{Theorem 45.1}),$$

where v_{ur} is the maximum uniquely restricted matching number. The proof is a direct ranking and an attaining chain. With the published NP-completeness of bipartite uniquely restricted matching [12], deciding $h(\mathcal{K}(S_\omega(Q))) \geq \omega \cdot (k + 1)$ is NP-complete (Theorem 46.2). For nonuniform pure priorities the finite objective of Theorem 35.1 is an ordered-matching maximum (Theorem 47.1) and obeys an exact priority-layer recurrence (Theorem 48.1).

41.2 Contribution and dependency ledger

The logical boundaries of this part are as important as the formulas.

Result	What is proved or imported
Uniform height formula	A direct ranking and an explicit chain are proved here; no appeal to the report's general height theorems.
Matching characterization	The classical alternating-cycle characterization is reproved. Its application to retirement schedules is made explicit.
NP-completeness	The reduction to ordinal height is proved here. Bipartite uniquely restricted matching NP-completeness is the published input of Golumbic–Hirst–Lewenstein.
Weighted ordered matchings and priority layers	Finite combinatorial theorems proved here, with no transfinite assumption.
Nonuniform height interpretation	Uses Theorem 35.1, restated precisely in Remark 42.2.
Algorithms and certificates	Exact finite correctness proofs plus executed Python tests; no proof-assistant certification.

These distinctions prevent two misleading conclusions. First, identifying a repository question with a known NP-complete problem is not a new proof of the historical NP-completeness theorem itself. Second, tests of finite schedules do not verify transfinite ranks. The uniform rank proof below supplies its own mathematical justification; the more general weighted application has the separate inherited dependency just stated.

41.3 Literature and novelty positioning

Golumbic, Hirst and Lewenstein introduced and studied uniquely restricted matchings, including their bipartite NP-completeness and their relation to matrix-rank bounds [12]. Legal set-covering sequences and the Grundy covering number are also established notions; Brešar, Henning and Rall explain their connection with dominating sequences [10]. The unweighted set-system parameter below is therefore not proposed as a new invariant.

The contribution of 04 is a proof-level connection to this report's ordinal height problem, a direct uniform transfinite proof, the ordered rather than unordered weighted formulation, and the residual priority calculus. These are continuations of the inspected report. A targeted literature search does not establish that every formulation is unprecedented. The claim is a resolution of the identified repository question with fully stated dependencies, “not a claim of a previously unrecognized historical breakthrough” (04), and “not an unspecified famous open problem in matching theory” (04's proof audit).

41.4 Source, pin and placement

No.	Manuscript (date)	Archive	Prefix
04	<i>Ordinal Heights and Uniquely Restricted Matchings: NP-completeness, priority layers, and exact certificates</i> (29 Sep. 2026)	ProveIt_Ordinal_Heights_Matchings	06-ur-matchings-

Table 11: The manuscript of Part IV. The number is its batch-42 manuscript number; the prefix continues this report's local file sequence after the 02--05- of the batch-36 addition. The manuscript is an AI-assisted research manuscript prepared for Vladimir Reshetnikov.

Part IV was written from one manuscript (Table 11). It is pinned to ProveIt commit 9754e83603e223811b7515eb892f22c847144593 (abbreviated 9754e8360) and names the Git blob c48fb7fea57f8efd4c705d858c33e58bdbd9a867 of the article.tex it read; that blob is the report as it stood when this part was added (Parts I–III), and the directory then differed

from the pinned one only by the 27 files staged at placement. 04 read this directory's `README.md` and `article.tex`, in particular Theorem 35.1 (its “Theorem 35.1”), the complexity discussion of Section 37 (its “Section 37”) and Problem 37.2; its references to those numbers are replaced here by cross-references. The archive arrived in commit 8315d24e3 and was placed in commit 3609d0473, which staged its code, data, proof audit and source record under the prefix `06-ur-matchings-`. Its manuscript, delivery `README`, PDF and bibliography files (`references.tex`, `references.bib`) are not shipped; they survive in the arrival commit, and the three literature entries of its bibliography were merged into this report's list. 04 notes that the pinned snapshot also contained newly arrived report archives that it did not unpack or compare, and that absence of a keyword (“uniquely restricted”) from an indexed repository search is not proof that no related material exists elsewhere. 04 states that it changed nothing in the repository.

41.5 Notation

Symbols follow Part I and Theorem 35.1. Every symbol renamed from 04 is listed in Table 12, together with the meaning it must not be confused with elsewhere in this report. All ordinal sums and products without a special symbol are ordinary ordinal operations; $\#$ is the natural (Hessenberg) sum, which 04 prints as a circled plus, and $\#$ its finite iterate.

41.6 A placement-time consistency check

Before this part was written, the uniform objective of Theorem 35.1 (the largest number of nonempty first-neighbour groups over all orders of the upper vertices) was compared by brute force with the maximum uniquely restricted matching number, computed directly from the definition (a matching whose saturated vertices induce a subgraph with exactly one perfect matching). All 4,056 bipartite incidence matrices with one to three lower rows and one to four upper vertices, every lower row nonempty, were enumerated: the two values agreed in every case, with 0 mismatches. This is Theorem 43.3 on those instances, computed independently of 04's programs. The check was run during placement; its script is not shipped.

42 Ordinals, fibres and first-neighbour schedules

42.1 Height and finite generation

For a well-founded partially ordered set P put

$$\text{rk}_P(x) = \sup\{\text{rk}_P(y) + 1 : y < x\}, \quad \text{h}(P) = \sup_{x \in P}(\text{rk}_P(x) + 1),$$

as in Section 2. In particular, the empty order has height 0 and a one-element order has height 1. $\mathcal{K}(P)$ is the set of finitely generated downsets of P , including $\emptyset$, ordered by inclusion; thus $\mathcal{K}(\emptyset)$ is a singleton. The following elementary facts are used repeatedly.

Lemma 42.1 (04). *Let $\alpha = \omega^\rho$ with $\rho > 0$.*

- (i) *A natural sum of finitely many ordinals less than α is less than α and is strictly increasing in each coordinate.*
- (ii) *Ordinary addition is weakly increasing in each argument.*
- (iii) *A finite sum of monomials with exponents strictly below ρ is less than ω^ρ and is absorbed when ω^ρ is added on its right.*
- (iv) *Every final segment of α starting below α has order type α .*

Part IV	Meaning; <i>not to be confused with</i>	04
$Q_{\text{lo}}, Q_{\text{up}}, Q_{\text{iso}}$	nonmaximal minimal, nonminimal maximal and isolated vertices of a height-two skeleton, as in Theorem 35.1; <i>not</i> Part II's label set U or terminal region Z	L, U, Z
$\mathcal{N}(\ell), \mathcal{N}(X)$	upper neighbours of a lower vertex; lower neighbours of a set $X \subseteq Q_{\text{up}}$; <i>not</i> the N -poset, and <i>not</i> the input length N	$N(\ell), N(S)$
X, X'	sets of upper vertices; <i>not</i> the lexicographic sums $S_\alpha(Q)$ (Part I) and S (Part III)	S, S'
$S_\alpha(Q)$	the uniform lexicographic sum $\sum_{q \in Q}^{\text{lex}} \alpha$ of Part I	P
$\nu_{\text{ur}}(G)$	maximum uniquely restricted matching number; <i>not</i> 07's normalization map ν of Section 21	$\nu_{\text{ur}}(G)$
ρ_τ	terminal exponent, $\omega^{\rho_\tau} = T^0(Q)$; <i>not</i> Part II's terminal cost τ_Q (an ordinal), $\tau(q)$ of Appendix A, or the retirement time $\tau_\pi(q)$ of Part III	τ
$\Psi(G, \rho; \rho_\tau)$	finite weighted objective; equals $\Phi(Q; \mathbf{h})$ when $G = G_Q$, $h_q = \omega^{\rho_q}$ (Remark 42.2)	$\Psi(G, \rho; \tau)$
$\text{ncl}(X)$	neighbourhood closure of $X \subseteq Q_{\text{up}}$; <i>not</i> $\text{cl}(A)$, 07's minimal completion of an antichain	$\text{cl}(S)$
$\text{prog}(X)$	finite progress function; <i>not</i> d_Λ (Part II) or $\mathbf{d}^0$ (Part III)	$d(S)$
$\Theta(D), \theta_q(D)$	rank map and cuts of a downset D ; D is a downset of the lexicographic sum, <i>not</i> Part I's predecessor set $D(A)$	$R(D), \theta_q(D)$
$Q_{\text{hi}}, G_{\text{hi}}, k_{\text{hi}}, \delta_{\text{hi}}$	highest-priority rows, their subgraph, its ν_{ur} , and the upper endpoint sets of its maximum uniquely restricted matchings; <i>not</i> $H(T)$ of Part II	$H, G_H, K_H, \mathcal{S}_H$
$Q_{\text{lo}}^{(X)}, G^{(X)}$	residual rows and residual graph	L_S, G_S
$\mathbf{f}(X), \mathbf{c}(C)$	subset and covered-row dynamic programs; <i>not</i> F (Part II)	$F(S), D(C)$
$\mathbf{v}(X), \mathbf{p}(X)$	certificate table and selected last column; <i>not</i> $\mathbf{p}(t)$ (Part II)	$V(S), p(S)$
M	a matching; <i>not</i> the frontier poset $\mathcal{M}(Q)$	M
$\mathcal{K}, \mathbf{h}, \mathcal{M}(Q)$	finitary powerset, height, frontier poset, as in Part I	$K, \mathbf{h}, \mathcal{M}(Q)$

Table 12: Symbols of Part IV. Every symbol of 04 not listed keeps its name; in particular $\Gamma_i(\sigma)$ has the same meaning as in Theorem 35.1, and N is an input length, as in Problem 37.2.

Proof. Cantor normal form proves the first and third assertions: a finite collection of exponents below ρ has a largest member below ρ . The second follows from the recursive definition of ordinal addition. For the fourth, if $\alpha = \beta + \gamma$ with $\beta < \alpha$, then $\gamma \leq \alpha$; if also $\gamma < \alpha$, additive indecomposability would imply $\beta + \gamma < \alpha$. Hence $\gamma = \alpha$. $\square$

42.2 Height-at-most-two skeletons

Let Q be a finite nonempty poset with no chain of three distinct elements. There is a canonical decomposition

$$Q = Q_{\text{lo}} \sqcup Q_{\text{up}} \sqcup Q_{\text{iso}},$$

where Q_{lo} consists of nonmaximal minimal elements, Q_{up} of nonminimal maximal elements, and Q_{iso} of isolated elements. Every strict relation has the form $\ell < u$ with $\ell \in Q_{\text{lo}}$ and $u \in Q_{\text{up}}$, and both endpoints of every edge have a neighbour. Set

$$G_Q = (Q_{\text{lo}}, Q_{\text{up}}, E), \quad \ell u \in E \iff \ell < u, \quad N(\ell) = \{u \in Q_{\text{up}} : \ell u \in E\}.$$

N also denotes the usual neighbourhood of a set on the other side: for $X \subseteq Q_{\text{up}}$, $N(X) = \bigcup_{u \in X} N(u) \subseteq Q_{\text{lo}}$. For a general finite bipartite graph G the two sides are still called lower rows and upper columns, and the same letters are used.

For ordinals $(\alpha_q)_{q \in Q}$ the lexicographic sum $\sum_{q \in Q}^{\text{lex}} \alpha_q$ has elements (q, β) with $\beta < \alpha_q$. Elements in one fibre are ordered by their ordinal coordinates; elements in different fibres are ordered exactly when their indices are ordered in Q . No relation is added between incomparable fibres.

42.3 The finite weighted objective

Give each lower vertex a positive ordinal exponent ρ_ℓ and give a positive terminal exponent $\rho_\top$. For a permutation $\sigma = (u_1, \dots, u_s)$ of Q_{up} define the retirement groups

$$\Gamma_i(\sigma) = \{\ell \in Q_{\text{lo}} : u_i \in N(\ell), N(\ell) \cap \{u_1, \dots, u_{i-1}\} = \emptyset\}, \quad (42.1)$$

exactly as in Theorem 35.1. The groups are disjoint and their union is Q_{lo} . An index is *productive* when its group is nonempty. Set

$$b_i(\sigma) = \max_{\ell \in \Gamma_i(\sigma)} \omega^{\rho_\ell}, \quad \Psi(G, \rho; \rho_\top) = \max_{\sigma} (b_1(\sigma) + \dots + b_s(\sigma) + \omega^{\rho_\top}), \quad (42.2)$$

with an empty maximum equal to 0 (the convention of Part III). The empty permutation has value $\omega^{\rho_\top}$. Columns with no neighbours are allowed for this finite objective; they simply contribute 0.

The terminal term is a permanent boundary condition, not an extra lower vertex. When passing to a residual graph later, it must be retained unchanged, including when the residual graph has no vertices.

Remark 42.2 (Theorem 35.1 in this notation). For a nonempty height-at-most-two skeleton Q let $\alpha_q = \omega^{\rho_q}$ with $\rho_q > 0$ for every $q \in Q$, and put $\rho_\top = \max\{\rho_q : q \in Q_{\text{up}} \cup Q_{\text{iso}}\}$. Since $h(\mathcal{K}(\omega^\rho)) = \omega^\rho$ for $\rho > 0$, these fibres satisfy the hypotheses of Part III with $h_q = \alpha_q$, and Theorem 35.1 states

$$h\left(\mathcal{K}\left(\sum_{q \in Q}^{\text{lex}} \alpha_q\right)\right) = \Phi(Q; h) = \Psi(G_Q, (\rho_\ell)_{\ell \in Q_{\text{lo}}}; \rho_\top). \quad (42.3)$$

This is the precise inherited result that 04 uses for nonuniform transfinite heights. It is an unformalized and unrefereed theorem of this report. The finite results about Ψ in this part are independent of it. The uniform instance of (42.3), and hence the ordinal-height NP-completeness result, is proved directly in Section 45.

Remark 42.3 (Encoding versus mathematics). The mathematical results allow arbitrary set-sized positive ordinal exponents. The executable input uses positive integers in binary. More generally a finite list of exponent labels with a supplied correct total order suffices for the finite optimization. No algorithm deciding the order of arbitrary ordinal descriptions is assumed. The empty skeleton is excluded from (42.3); its powerset height is 1, not an empty maximum.

43 Triangular matchings and productive schedules

Definition 43.1. A matching M in a finite graph is *uniquely restricted* if the subgraph induced by its saturated vertices has M as its unique perfect matching. Write $v_{\text{ur}}(G)$ for the largest cardinality of such a matching, the empty matching allowed. In a bipartite graph, an ordered matching $((\ell_1, u_1), \dots, (\ell_k, u_k))$ is *admissible* or *triangular* if

$$\ell_i u_j \notin E \quad \text{whenever } j < i. \quad (43.1)$$

An ordinary maximum matching need not be uniquely restricted. The distinction is precisely the possibility of switching along an alternating cycle.

Lemma 43.2 (Classical characterization, with proof; 04). *For a bipartite matching M the following are equivalent:*

- (i) M is uniquely restricted;
- (ii) G contains no M -alternating cycle;
- (iii) M admits a triangular ordering.

Proof. An alternating cycle permits exchanging its matched and unmatched edges, producing another perfect matching on the same saturated vertices. Conversely, the symmetric difference of two different perfect matchings on those vertices is a disjoint union of alternating cycles. This proves (i) $\Leftrightarrow$ (ii).

Make a directed graph whose vertices are the matched edges $e_i = (\ell_i, u_i)$, putting $e_i \rightarrow e_j$ for $i \neq j$ when $\ell_i u_j \in E$. A directed cycle gives an alternating cycle by inserting the matching edges between its cross-edges. Every alternating cycle gives a directed cycle in this way, after choosing its orientation. Thus (ii) is equivalent to acyclicity of this directed graph. A topological ordering has no cross-edge from a later row to an earlier column, which is exactly (43.1). Conversely such an ordering forbids a directed cycle. $\square$

The characterization is part of the classical theory [12]; the proof is included to fix the orientation used in all subsequent constructions.

Theorem 43.3 (Productive schedules are uniquely restricted matchings; 04). *The maximum number of productive indices in a permutation of Q_{up} equals $v_{\text{ur}}(G)$.*

Proof. Given a schedule, keep its productive columns in their original order. From each corresponding group choose one lower vertex. Distinct groups give distinct lower vertices, and freshness says that a chosen row is adjacent to none of the earlier chosen columns. The resulting matching is triangular. Thus every schedule has at most $v_{\text{ur}}(G)$ productive indices.

Conversely, order a uniquely restricted matching triangularly, schedule its upper endpoints first in that order, and then schedule the remaining upper vertices arbitrarily. Each matched row is fresh when its own upper endpoint is scheduled. Hence the schedule has at least $|M|$ productive indices. Maximize over M and use the first inequality. $\square$

Lemma 43.4 (Coverage by a maximum matching's upper endpoints; 04). *Suppose every lower vertex has a neighbour. If M is a maximum uniquely restricted matching and X is its set of upper endpoints, then $N(X) = Q_{\text{lo}}$.*

Proof. If $\ell \notin \mathcal{N}(X)$, choose $u \in \mathcal{N}(\ell)$. Necessarily $u \notin X$, and ℓ is not a matched row. Append (ℓ, u) to a triangular ordering of M . The new row is adjacent to no previous matched column, so this is a larger triangular matching, a contradiction. $\square$

43.1 The established set-system interpretation

Consider the family of subsets $(\mathcal{N}(u))_{u \in Q_{\text{up}}}$ of Q_{lo} . A legal covering sequence chooses sets, each of which introduces a new element, until their union is Q_{lo} . Its maximum possible length is the Grundy covering number in the terminology of [10]. Removing nonproductive steps from a full permutation leaves such a sequence; a legal sequence can be extended until it covers all of Q_{lo} . Thus Theorem 43.3 identifies this covering number with $\nu_{\text{ur}}(G)$. This answers the request of the complexity item of Section 37 (09's "largest number of nonempty retirement groups") for a connection with an established finite optimization problem, rather than introducing a new name for an old unweighted parameter.

44 Closure systems and maximal frontiers

The matching identification also explains the finite frontier height in Part I's uniform formula (Theorem 9.1). The correspondence is explicit. For $X \subseteq Q_{\text{up}}$ put

$$\text{ncl}(X) = \{u \in Q_{\text{up}} : \mathcal{N}(u) \subseteq \mathcal{N}(X)\}. \quad (44.1)$$

This is a closure operator: it is extensive and monotone, and $\mathcal{N}(\text{ncl}(X)) = \mathcal{N}(X)$ proves idempotence. In this section $G = G_Q$ has no isolated upper vertices, so $\text{ncl}(\emptyset) = \emptyset$.

Theorem 44.1 (04). *The closed subsets of Q_{up} , ordered by inclusion, are isomorphic to the inclusion-maximal antichains of Q , ordered by Hoare domination. The isomorphism is*

$$X \mapsto A_X = (Q_{\text{lo}} \setminus \mathcal{N}(X)) \cup X \cup Q_{\text{iso}}. \quad (44.2)$$

Consequently, for the frontier poset $\mathcal{M}(Q)$,

$$h(\mathcal{M}(Q)) = \nu_{\text{ur}}(G_Q) + 1. \quad (44.3)$$

Proof. The set in (44.2) is an antichain. An omitted lower vertex is comparable with a member of X . An omitted upper vertex can be added precisely when all its neighbours are in $\mathcal{N}(X)$, that is, when it belongs to $\text{ncl}(X)$. Thus A_X is maximal exactly when X is closed. Conversely, a maximal antichain with upper part X must contain every isolated vertex and every lower vertex outside $\mathcal{N}(X)$, and no other lower vertex; the same observation forces closure.

Recall that $A \leq B$ in the Hoare order means that every element of A is below or equal to some element of B . If $A_X \leq A_{X'}$, every maximal upper vertex of X must itself belong to X' , so $X \subseteq X'$. If $X \subseteq X'$, a lower vertex in A_X either remains in $A_{X'}$ or acquires a neighbour in X' ; all upper and isolated elements of A_X remain. This proves the order isomorphism.

A productive sequence of length k gives $k + 1$ strictly increasing closed sets by taking the closures of its successive prefixes, including the empty prefix. Freshness makes the neighbourhood unions strictly increase. For the converse, extend a longest chain of closed sets to include bottom and top, and write it as $X_0 \subsetneq \cdots \subsetneq X_k$. For each $i \geq 1$ choose $u_i \in X_i \setminus X_{i-1}$. Since X_{i-1} is closed, u_i has a neighbour outside $\mathcal{N}(X_{i-1})$, hence outside the union of the neighbourhoods of the previously chosen columns. The sequence is productive. Theorem 43.3 now gives (44.3); for a finite poset the rank height is the largest number of elements in a chain. $\square$

The result is compatible with Part I's uniform identity $h(\mathcal{K}(S_\alpha(Q))) = \alpha \cdot h(\mathcal{M}(Q))$ (Theorem 9.1). For the N -poset, a path on four vertices, $v_{ur} = 2$ and (44.3) recovers the three-element frontier chain (11). The next section does not use Part I's identity: it supplies a separate proof from the actual downsets.

Remark 44.2 (A second route through Part I; merge). Theorem 9.1 and (44.3) together give the formula of Theorem 45.1 at once. 04 deliberately does not take this route, so that its uniform theorem, and with it the NP-completeness of Theorem 46.2, depends on no unrefereed theorem of this report. The two routes are independent proofs of the same statement; the observation that they combine is the merge's own.

45 A direct proof of the uniform height

Theorem 45.1 (Uniform height, independently proved; 04). *Let Q be finite, nonempty and of height at most two. For every $\alpha = \omega^\rho$, $\rho > 0$,*

$$\boxed{h(\mathcal{K}(S_\alpha(Q))) = \alpha \cdot (v_{ur}(G_Q) + 1)}. \quad (45.1)$$

The multiplication on the right is ordinary multiplication by a positive finite ordinal on the right.

45.1 Profiles of finitely generated downsets

For $D \in \mathcal{K}(S_\alpha(Q))$ let

$$X(D) = \{u \in Q_{up} : D \text{ meets the } u\text{-fibre}\}.$$

Every lower fibre in $\mathcal{N}(X(D))$ is wholly included in D . Every other lower fibre, every isolated fibre and every upper fibre is either empty or a principal initial segment, because a finite collection of generators inside an ordinal chain has a largest member. No upper or isolated fibre can be wholly included: α is a limit ordinal and has no finite cofinal subset. A lower fibre not below $X(D)$ cannot be wholly included either.

The active coordinates are therefore

$$A(X) = (Q_{lo} \setminus \mathcal{N}(X)) \cup X \cup Q_{iso}.$$

For $q \in A(X(D))$ let $\theta_q(D) < \alpha$ be the order type of the initial segment of its fibre present in D . It is 0 or a successor; for $q \in X(D)$ it is positive. This gives a complete profile of D . Notice that $A(X)$ need not be a maximal antichain when X is not closed. The rank argument must use all actual supports, not merely the maximal frontiers.

45.2 A finite progress function

For $X \subseteq Q_{up}$ let $\text{prog}(X)$ be the greatest number of productive steps in an ordering of X alone, with freshness still measured against all lower rows. Then $\text{prog}(\emptyset) = 0$ and

$$\text{prog}(X) \leq v_{ur}(G_Q), \quad (45.2)$$

$$X \subseteq X' \implies \text{prog}(X) \leq \text{prog}(X'), \quad (45.3)$$

$$X \subseteq X', \mathcal{N}(X') \neq \mathcal{N}(X) \implies \text{prog}(X') \geq \text{prog}(X) + 1. \quad (45.4)$$

The first statement follows by extending an ordering to all of Q_{up} and applying Theorem 43.3. For the other two, take an optimal ordering of X and append the columns in $X' \setminus X$. If a new lower neighbour is acquired, at least one appended step must be productive.

45.3 The ordinal ranking

Define

$$\Theta(D) = \alpha \cdot \text{prog}(X(D)) + \sum_{q \in A(X(D))} \theta_q(D). \quad (45.5)$$

The natural-sum part is less than α by Lemma 42.1. Thus $\Theta(D) < \alpha \cdot (v_{\text{ur}}(G_Q) + 1)$.

We claim that $D \subsetneq D'$ implies $\Theta(D) < \Theta(D')$. Write $X = X(D)$ and $X' = X(D')$. Necessarily $X \subseteq X'$. If $\mathcal{N}(X') \setminus \mathcal{N}(X)$ is nonempty, (45.4) gives

$$\Theta(D) < \alpha \cdot (\text{prog}(X) + 1) \leq \alpha \cdot \text{prog}(X') \leq \Theta(D').$$

Otherwise no old active coordinate disappears. The old cuts weakly increase, and the new upper coordinates, if any, are positive. If $\text{prog}(X') > \text{prog}(X)$ the same ordinal block comparison works. If $\text{prog}(X') = \text{prog}(X)$, strict inclusion of the profiles forces either a strict increase of an existing cut or a newly positive upper cut. Strict monotonicity of the finite natural sum proves $\Theta(D) < \Theta(D')$.

A strictly increasing map into the ordinals proves well-foundedness. Moreover, transfinite induction in the definition of rank gives $\text{rk}_{\mathcal{K}(S_\alpha(Q))}(D) \leq \Theta(D)$: every predecessor has rank plus one at most $\Theta(D)$. Taking suprema proves

$$h(\mathcal{K}(S_\alpha(Q))) \leq \alpha \cdot (v_{\text{ur}}(G_Q) + 1). \quad (45.6)$$

This proof uses neither a maximal-chain theorem for well-partial-orders nor the persistence theorems of Parts II–III.

45.4 An attaining chain

Let $k = v_{\text{ur}}(G_Q)$ and choose a productive sequence $(u_1, \dots, u_k)$ of maximum length, with fresh witnesses $\ell_1, \dots, \ell_k$. It covers all lower vertices: otherwise it could be extended by a column meeting an uncovered row. For $1 \leq i \leq k$ and $\beta < \alpha$ let

$$D_{i,\beta} = \downarrow\{(u_j, 0) : j < i\} \cup \downarrow\{(\ell_i, \beta)\}.$$

For fixed i these downsets form a strict chain of order type α . Freshness ensures that the ℓ_i -fibre is not already full. Every member of block i is strictly below every member of block $i + 1$: the newly inserted upper generator $(u_i, 0)$ completes the previous witness's whole fibre and is itself absent from every member of block i .

After the last block, start with the downset generated by all $(u_j, 0)$. Choose any maximal vertex $q \in Q_{\text{up}} \cup Q_{\text{iso}}$ and increase the cut in its fibre through principal initial segments above its current cut. This is another strict chain of type α , by Lemma 42.1(iv), and all its members can be chosen strictly above all preceding blocks. If $k = 0$, the skeleton is an antichain and this final block alone supplies the construction. The union of these $k + 1$ blocks is a chain of type $\alpha \cdot (k + 1)$.

For completeness, if $(x_\xi)_{\xi < \delta}$ is a strictly increasing chain in a well-founded order, induction gives $\text{rk}(x_\xi) \geq \xi$, so its height is at least δ . The constructed chain and (45.6) prove Theorem 45.1.

Remark 45.2 (The terminal block and the missing empty case; 04). The final $+1$ inside the finite coefficient records a genuinely unbounded maximal fibre, not the empty downset as a separate successor. This is why a nonempty antichain has height α , while a single relation has height $\alpha \cdot 2$. For an empty skeleton the height is instead 1.

46 NP-completeness with one exponent scale

Definition 46.1. The decision problem UNIFORM-HEIGHT-TWO takes a finite nonempty height-at-most-two poset Q and a nonnegative integer k , and asks whether

$$h(\mathcal{K}(S_\omega(Q))) \geq \omega \cdot (k + 1).$$

The order relation is given explicitly, or equivalently by its bipartite incidence graph and its isolated vertices. The integer k may be binary encoded.

Theorem 46.2 (04). *UNIFORM-HEIGHT-TWO is NP-complete. The same holds with any fixed positive pure ordinal ω^p replacing ω , when the threshold is specified by its finite coefficient.*

Proof. By Theorem 45.1 the answer is yes exactly when $v_{\text{ur}}(G_Q) \geq k$. A certificate is a list of k disjoint incidence edges with a triangular order. Distinctness, incidence and all forbidden earlier cross-edges can be checked in polynomial time. A yes-instance has $k \leq \min(|Q_{\text{lo}}|, |Q_{\text{up}}|)$, so the certificate is polynomial in the input length.

For hardness, use the established NP-completeness of the bipartite uniquely restricted matching threshold problem [12]. Given its instance (G, k) , form the poset with one element per vertex and the relation $\ell < u$ exactly on its edges. Isolated graph vertices are isolated elements of the poset and do not affect v_{ur} . Add one isolated element if necessary to guarantee a nonempty poset; adding such an element does not affect the matching number. Assign the single fixed fibre ω to every element and ask the displayed height question with coefficient $k + 1$. This is a polynomial transformation, and Theorem 45.1 proves equivalence of the answers. The same reduction works for a fixed ω^p . $\square$

This is an explicit reduction from an established NP-complete problem, not a claim to reproduce the original reduction establishing the matching problem's hardness. It suffices to close the complexity question of Section 37 and is independent of Theorem 35.1.

Corollary 46.3 (04). *The number of distinct exponent priorities alone cannot yield an FPT or even an XP algorithm for the unrestricted height-two problem unless $P = NP$. In particular the leading-coefficient threshold problem is NP-complete at one scale.*

Proof. Both an FPT bound $f(d)N^c$ and an XP bound $N^{f(d)}$ become polynomial at the fixed parameter value $d = 1$, contrary to Theorem 46.2 unless $P = NP$. $\square$

Proposition 46.4 (04). *With positive integer exponents in binary and a threshold in finite Cantor normal form, deciding whether Ψ is at least that threshold is NP-complete. Under Theorem 35.1 this is also the corresponding nonuniform pure-fibre ordinal-height threshold problem.*

Proof. A permutation of the upper vertices is a polynomial-size certificate. Compute its groups, their maximum exponents, and the ordinary sum with the terminal monomial. A sum of at most $|Q_{\text{lo}}| + 1$ nonzero monomials has at most that many surviving units in its coefficient sum. Exponents are copied from the input, never numerically expanded. Standard comparisons and additions of binary integers therefore give polynomial-time verification, including comparison with the supplied finite threshold. Hardness is the one-scale restriction in Theorem 46.2. $\square$

Remark 46.5 (The threshold problem of Part II; merge). UNIFORM-HEIGHT-TWO is a special case of the threshold problem of Corollary 20.4: every fibre ω is a pure block, there is one exponent scale, and $\omega \cdot (k + 1)$ is a threshold in Cantor normal form. Hence that problem, which Corollary 20.4 (12) places in NP for the explicit pure-block encoding, is NP-complete already for height-two skeletons and one scale. The hardness half rests on Theorem 46.2 and [12]; the

membership half for general skeletons rests on Corollary 20.4 and so on Theorem 17.4. This combination is the merge's observation; 04 states NP-completeness for height-two skeletons only.

47 An exact ordered-matching formula for all priorities

The unweighted identification is only the first layer. To preserve lower-order terms one must retain an admissible *order* of the matching edges.

Theorem 47.1 (Weighted ordered-matching variational principle; 04). *For every finite instance of (42.2),*

$$\Psi(G, \rho; \rho_\tau) = \max_{\substack{M \text{ uniquely restricted} \\ ((\ell_1, u_1), \dots, (\ell_k, u_k)) \text{ admissible order of } M}} (\omega^{\rho_{\ell_1}} + \dots + \omega^{\rho_{\ell_k}} + \omega^{\rho_\tau}). \quad (47.1)$$

The empty matching contributes ω^{ρ_τ} .

Proof. Given a full schedule, discard its nonproductive columns and choose, in every productive group, a lower row of maximum exponent. The chosen edges form a triangular matching by freshness. Their ordered monomials are exactly the nonzero summands of the schedule, so the schedule value occurs on the right side of (47.1).

Conversely, start with an admissibly ordered matching. Schedule its columns first in that order. Each corresponding matched row is still fresh, so the maximum exponent of that retirement group is at least the matched row's exponent. Weak monotonicity of ordinary addition in each argument shows that the value of this prefix is at least the sum of the matched-row monomials. Append all remaining columns. Adding their monomials before the terminal term cannot decrease the result: $\eta + \xi \geq \eta$ and hence $(\eta + \xi) + \omega^{\rho_\tau} \geq \eta + \omega^{\rho_\tau}$. The full schedule is therefore at least the candidate matching value. Taking maxima in both directions proves equality. $\square$

Corollary 47.2 (Coefficient budget and priority invariance; 04). *Every exponent in Ψ is one of the lower exponents or the terminal exponent; its total Cantor coefficient sum is at most $|Q_{\text{lo}}| + 1$. Strictly increasing relabellings of the finite ordered set of exponent values preserve the coefficient vector and the set of maximizing schedules and ordered matchings.*

Proof. Every candidate has at most $|Q_{\text{lo}}| + 1$ monomials. Ordinary addition deletes smaller trailing exponents and merges equal terminal exponents; it introduces neither new exponents nor additional coefficient units. The deletion, merging and lexicographic comparison rules depend only on equality and order among the exponents, so commute with a strictly increasing relabelling. $\square$

The invariance also follows from the report's exponent-order invariance (Theorem 34.1; compare Corollary 24.2 and Proposition 34.2); here it is immediate from the matching formula. A short output does not imply a short search, as Theorem 46.2 shows.

47.1 Why natural-sum edge weights are wrong

Example 47.3 (Forced absorption; 04). Let $Q_{\text{lo}} = \{a, b\}$, $Q_{\text{up}} = \{c, d\}$ and

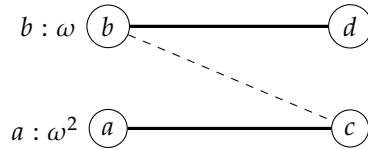
$$N(a) = \{c\}, \quad N(b) = \{c, d\}, \quad (\rho_a, \rho_b) = (2, 1), \quad \rho_\tau = 1.$$

The two-edge matching $\{(a, c), (b, d)\}$ is uniquely restricted. Its only admissible order puts (b, d) before (a, c) , since scheduling c first retires b . Its value is

$$\omega + \omega^2 + \omega = \omega^2 + \omega.$$

Both full schedules attain that value. In contrast, replacing the ordered sum by the natural sum of matched weights and then adding the terminal term gives $\omega^2 + \omega \cdot 2$, a false answer.

The skeleton is this report's N -poset $a < c, b < c, b < d$, with the same labels. The Part III formula for it, $\Phi = h_b + h_a + \max(h_c, h_d)$ (Section 35), gives the same value $\omega + \omega^2 + \omega = \omega^2 + \omega$.



The cross-edge forces d before c .
The earlier ω is absorbed by ω^2 .

Figure 5: Example 47.3 (04). The thick edges are a uniquely restricted matching. The dashed edge is not in the matching but imposes the order that determines the ordinal sum.

The example is deliberately small: a weighted cardinality objective or a maximum natural-sum matching loses precisely the precedence information on which lower ordinal terms depend. The directed graph in Lemma 43.2 is therefore not just a recognition device; it records an essential part of the weighted problem.

48 Priority layers and residual optimization

48.1 The recurrence

The leading term is simpler than the whole ordinal. It depends only on the highest-priority rows. The lower terms record what can be saved after choosing the high-priority endpoints.

Suppose $Q_{lo} \neq \emptyset$, let $p = \max_{\ell \in Q_{lo}} \rho_\ell$, and put

$$Q_{hi} = \{\ell \in Q_{lo} : \rho_\ell = p\}, \quad G_{hi} = G[Q_{hi}, Q_{up}], \quad k_{hi} = \nu_{ur}(G_{hi}).$$

Let $\mathcal{S}_{hi}$ be the family of upper endpoint sets of maximum uniquely restricted matchings in G_{hi} . Thus every $X \in \mathcal{S}_{hi}$ has cardinality k_{hi} . For each such X define

$$Q_{lo}^{(X)} = \{\ell \in Q_{lo} \setminus Q_{hi} : \mathcal{N}(\ell) \cap X = \emptyset\}, \quad G^{(X)} = G[Q_{lo}^{(X)}, Q_{up} \setminus X]. \quad (48.1)$$

Every row in $G^{(X)}$ still has a nonempty neighbourhood. The terminal exponent remains ρ_τ ; it is not recomputed from the residual columns.

Theorem 48.1 (Exact priority-layer calculus; 04). *The following rules determine Ψ recursively.*

(i) If $Q_{lo} = \emptyset$ or $p < \rho_\tau$, then $\Psi = \omega^{\rho_\tau}$.

(ii) If $p = \rho_\tau$, then

$$\Psi = \omega^{\rho_\tau} \cdot (k_{hi} + 1). \quad (48.2)$$

(iii) If $p > \rho_\tau$, then

$$\Psi(G, \rho; \rho_\tau) = \omega^p \cdot k_{\text{hi}} + \max_{X \in \mathcal{S}_{\text{hi}}} \Psi(G^{(X)}, \rho|_{Q_{\text{lo}}^{(X)}}; \rho_\tau). \quad (48.3)$$

Each residual value on the right is strictly below ω^p .

Proof. If there are no lower rows the definition gives the terminal term. If all row exponents are below ρ_τ , every finite sum of their monomials is absorbed by that term, proving (i).

For an arbitrary full schedule, the steps at which at least one row of Q_{hi} retires are exactly the productive steps of the same upper order applied to G_{hi} . Their maximum number is k_{hi} by Theorem 43.3. If $p = \rho_\tau$, all smaller intervening terms are absorbed by subsequent ω^{ρ_τ} terms, including the final one. The value is therefore ω^{ρ_τ} times one plus the number of high productive steps, proving (ii).

Now suppose $p > \rho_\tau$. Every schedule has at least one high productive step. If it has k such steps, its Cantor normal form is

$$\omega^p \cdot k + \eta, \quad \eta < \omega^p;$$

all lower terms before the last high step are absorbed. Schedules with k_{hi} high steps exist, so an optimal schedule must have exactly k_{hi} . Choose a maximum-priority witness at each productive step of such a schedule, as in Theorem 47.1. The high witnesses alone form a maximum uniquely restricted matching in G_{hi} , with some upper endpoint set $X \in \mathcal{S}_{\text{hi}}$.

Consider the witness edges strictly after the last high step. Their rows have lower priority and are adjacent to no earlier chosen high column; thus they lie in $Q_{\text{lo}}^{(X)}$. Their columns lie outside X . In their inherited order they form a triangular matching in $G^{(X)}$. Applying Theorem 47.1 to the residual graph bounds their ordered sum plus the unchanged terminal term by $\Psi(G^{(X)}, \rho|_{Q_{\text{lo}}^{(X)}}; \rho_\tau)$. Earlier lower terms were absorbed. This proves the upper bound in (48.3). This argument deliberately uses the matching witnesses: simply moving all low columns to a later position could introduce unwanted retirements and is not a valid substitute.

For the reverse bound, fix $X \in \mathcal{S}_{\text{hi}}$ and triangularly order a maximum high matching with upper endpoints X . Schedule those columns first. Every one retires a fresh high row and contributes ω^p . By Lemma 43.4 applied to G_{hi} , these columns cover all high rows. They also retire exactly the low rows adjacent to X . The remaining live rows are precisely $Q_{\text{lo}}^{(X)}$, so following with an optimal schedule on $Q_{\text{up}} \setminus X$ gives the value

$$\omega^p \cdot k_{\text{hi}} + \Psi(G^{(X)}, \rho|_{Q_{\text{lo}}^{(X)}}; \rho_\tau).$$

All residual exponents, including ρ_τ , are below p , so this tail is strictly below ω^p . Maximize over X to prove the theorem. $\square$

Corollary 48.2 (Highest coefficient; 04). *If the largest lower exponent p exceeds ρ_τ , the coefficient of ω^p in Ψ is exactly $v_{\text{ur}}(G_{\text{hi}})$. If $p = \rho_\tau$, its coefficient is $v_{\text{ur}}(G_{\text{hi}}) + 1$. If $p < \rho_\tau$, the only surviving coefficient is the terminal coefficient 1.*

This gives an exact answer to the leading-coefficient part of the complexity item of Section 37 (07's question whether the leading coefficient exceeds a threshold). The answer is graph-theoretic and independent of the magnitudes of the ordinal exponents, but it remains computationally hard in general (Corollary 46.3).

48.2 Why a maximum high matching is not enough

Example 48.3 (A tie requiring lower-priority lookahead; 04). Let

$$\mathcal{N}(h) = \{u, v\}, \quad \mathcal{N}(\ell) = \{u\}, \quad \rho_h = 3, \quad \rho_\ell = 2, \quad \rho_\top = 1.$$

The high graph has maximum uniquely restricted matching size 1 and two possible upper endpoint sets. Choosing $X = \{u\}$ also retires the low row, so its residual value is ω and the resulting value is $\omega^3 + \omega$. Choosing $X = \{v\}$ leaves the low row alive, so the residual is a single edge of weight ω^2 and the result is

$$\Psi = \omega^3 + \omega^2 + \omega.$$

Both high matchings are optimal at the leading scale. Only one is globally optimal. Thus arbitrary tie-breaking in a maximum high matching is incorrect.

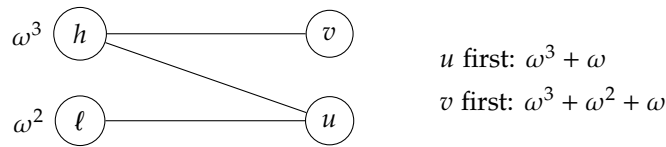


Figure 6: Example 48.3 (04). The residual graph, not just the cardinality of the high matching, determines the lower coefficients.

48.3 What the recurrence does and does not simplify

The recursion decreases the number of distinct lower priorities until the terminal scale absorbs everything left. It identifies exactly which residual information controls the next coefficient: the low rows whose neighbourhoods avoid the selected high columns, together with their remaining incidence data. Different admissible orders of the *same* high endpoint set need not be retained at this stage; their lower terms occur before the final high term and are erased. Distinct endpoint sets, however, need not be equivalent.

This is a structural reduction, not a polynomial-time algorithm. Enumerating all maximum uniquely restricted matchings can be exponential, and even finding their maximum cardinality is NP-hard. The recurrence is nevertheless useful for proofs, exact branch-and-bound designs, and testing a different implementation against the subset dynamic program. It also suggests that equivalence classes of residual graphs, rather than arbitrary matching witnesses, are the right objects for further compression.

49 Exact dynamic programming and finite certificates

49.1 A subset recurrence

Let $r = |Q_{lo}|$ and $s = |Q_{up}|$. For a processed subset $X \subseteq Q_{up}$ and a new column $u \notin X$ put

$$b(X, u) = \max\{\omega^{\rho_\ell} : u \in \mathcal{N}(\ell), \mathcal{N}(\ell) \cap X = \emptyset\}, \quad (49.1)$$

with value 0 when the set is empty. This depends on the processed *set*, not its order. Let $\mathfrak{f}(X)$ be the greatest ordinary sum of retirement monomials over orderings of X , without a terminal term.

Theorem 49.1 (Bellman recurrence; 04). *The exact values satisfy*

$$\mathfrak{f}(\emptyset) = 0, \quad \mathfrak{f}(X) = \max_{u \in X} \left(\mathfrak{f}(X \setminus \{u\}) + b(X \setminus \{u\}, u) \right), \quad \Psi = \mathfrak{f}(Q_{\text{up}}) + \omega^{\rho^\tau}. \quad (49.2)$$

Proof. Partition the orderings of X according to their last column u . The preceding retirement history has processed exactly $X \setminus \{u\}$, so the last summand is (49.1), regardless of its earlier order. Ordinary addition by this fixed right summand is weakly increasing in the prefix value. The largest preceding value is attained because the set of orderings is finite. Thus its addition gives the maximum for this last column. Maximize over u . The final terminal addition is handled by the same monotonicity and attainment argument. $\square$

Strict monotonicity in the left argument is not required and would be false for ordinary addition. Weak monotonicity suffices. This is exactly the distinction that permits a scalar state value $\mathfrak{f}(X)$ despite ordinal absorption. On a height-two skeleton this is the specialization of Part III's support-ideal program (78) (09) to the ideals $Q_{\text{lo}} \cup Q_{\text{iso}} \cup X$, whose value differs only by the placement of the terminal weight; the proof above does not use Part III.

49.2 Representation and complexity

A finite result is stored as a list

$$[(e_1, c_1), \dots, (e_d, c_d)], \quad e_1 > \dots > e_d > 0,$$

representing $\omega^{e_1}c_1 + \dots + \omega^{e_d}c_d$. Appending ω^e deletes the trailing entries of exponent below e , then increments the last coefficient if its exponent is e , or appends $(e, 1)$ otherwise. The empty list represents 0. Lexicographic comparison of these lists gives ordinal comparison.

Computing a step weight directly scans the r lower rows. With at most d distinct priorities, a straightforward implementation takes

$$O(2^s s(r + d)) \quad (49.3)$$

elementary operations and $O(2^s d)$ stored ordinal entries, apart from the incidence input. This expression treats a bit-mask intersection or exponent comparison as an elementary operation. In a bit-cost model, multiply by the appropriate polynomial factor in mask length, exponent bit length and $\log(r + 1)$ for coefficients. In particular the problem is fixed-parameter tractable in s with ordinary binary integer input.

The program's default guard is $s \leq 22$. It is a protection against accidental state-space explosion, not a theorem restriction or a promise that every 22-column instance is inexpensive. The implementation stores full certificates; this can dominate memory long before arithmetic becomes difficult.

49.3 A dual dynamic program on covered rows

A second state representation is useful when the lower side is smaller. For $C \subseteq Q_{\text{lo}}$ retain the maximum prefix value among productive sequences covering exactly C ; unreachable states have no value. Start with $\mathfrak{c}(\emptyset) = 0$ and, for every column u with $\mathcal{N}(u) \setminus C \neq \emptyset$, relax the transition

$$C \longrightarrow C \cup \mathcal{N}(u), \quad \mathfrak{c}(C) + \max_{\ell \in \mathcal{N}(u) \setminus C} \omega^{\rho^\ell}. \quad (49.4)$$

The target state's value is the maximum of its incoming candidates.

Proposition 49.2 (Covered-row state sufficiency; 04). *The recurrence (49.4) has exact final value $\mathfrak{c}(Q_{\text{lo}}) + \omega^{\rho^\tau} = \Psi$. It takes $O(2^r s(r + d))$ elementary operations in a straightforward implementation.*

Proof. A previously used column has its whole neighbourhood inside the current covered set. It can never again make a productive transition. Thus every productive transition allowed by the covered set automatically uses a previously unused column, without needing to store the used-column set. Conversely every productive sequence traces these transitions. Delete all zero steps from a full schedule, or extend a productive sequence covering Q_{10} by the remaining zero steps, to pass between the two representations without changing the value. The transition strictly increases $|C|$, so induction and weak monotonicity of ordinal addition prove the maximization recurrence. There are 2^r states and at most s transitions per state, with at most r rows scanned per weight. $\square$

The delivered function `solve_by_covered_rows` implements this value-only variant; it does not emit the upper-subset certificate. The verification suite compares it with the production optimizer. Choosing the smaller side gives an exponential dependence on $\min(r, s)$, with polynomial factors in the full input length, before any twin compression.

49.4 A certificate of optimality, not just a good schedule

A schedule certifies a lower bound but generally does not certify optimality. The program therefore emits, for every $X \subseteq Q_{\text{up}}$, a proposed value $v(X)$ and, for nonempty X , one selected last column $p(X) \in X$.

Theorem 49.3 (Bellman certificate soundness; 04). *Suppose a finite table satisfies $v(\emptyset) = 0$, and for every nonempty X and every $u \in X$,*

$$v(X \setminus \{u\}) + b(X \setminus \{u\}, u) \leq v(X), \quad (49.5)$$

$$v(X \setminus \{p(X)\}) + b(X \setminus \{p(X)\}, p(X)) = v(X). \quad (49.6)$$

Then $v(X) = \mathfrak{f}(X)$ for all X , and $v(Q_{\text{up}}) + \omega^{\rho_{\text{T}}} = \Psi$.

Proof. Induct on $|X|$. The already verified predecessor values are the true maxima. The inequalities bound every candidate in (49.2) above, while the selected equality exhibits one candidate equal to the proposed value. Hence the proposed value is the maximum. The base case and final terminal addition complete the proof. $\square$

The supplied checker does not call the optimizer. It validates the instance, the entire state table, every Bellman inequality, each selected equality, the terminal value, and an explicit attaining schedule with its claimed retirement groups. It does share the program's exact ordinal arithmetic and input representation. It is an independent checking path, not an independent trusted formal kernel. These certificates can be exponentially large; their existence does not put the threshold problem in a stronger complexity class. Part III's support certificates (09) have the same shape over ideals of Q .

49.5 Running the delivered implementation

The package uses only the Python standard library and requires Python 3.10 or later. Its solver is shipped as `code/06-ur-matchings-ordinal_matchings.py` and needs no sibling module. A complete input is, for example,

```
{
  "upper_count": 2,
  "neighbours": [[0, 1], [0]],
  "priorities": [3, 2],
  "terminal": 1
}
```

Rows are indexed by their position, and upper columns by $0, \dots, \text{upper_count} - 1$. Every row must have a nonempty list of neighbours. Duplicate indices, out-of-range indices, zero priorities, and Boolean values used as integer priorities are rejected. The terminal priority is explicit, even for an empty residual graph. Exponents and terminal priorities must be positive integers; zero means an empty group internally, not ω^0 . Empty lower and upper lists with a terminal represent a nonempty terminal antichain, not the empty skeleton. From this report's directory, with the output written outside the repository:

```
py code/06-ur-matchings-ordinal_matchings.py solve \
  data/06-ur-matchings-tie_requires_lookahead.json --output /scratch/certificate.json
py code/06-ur-matchings-ordinal_matchings.py check \
  data/06-ur-matchings-tie_requires_lookahead.json /scratch/certificate.json
```

The first command reports $\omega^3 + \omega^2 + \omega$. The second checks the finite optimum. Neither command checks a transfinite height theorem. The solver returns the exact Cantor normal form as decreasing `[exponent, coefficient]` pairs (an empty list is zero), a maximizing upper order, its retirement groups and a full Bellman table. It is a pure-positive-priority solver, not an implementation for finite or successor fibres, and it keeps the terminal priority fixed when residual graphs or twin quotients are formed. The function `compress_twins` applies equality-only weighted compression; the command line does not silently transform inputs. The test script `code/06-ur-matchings-verify.py` imports its sibling under the delivered name `ordinal_matchings` and rewrites its examples and ledgers; it must be run on a copy that recreates the delivered layout (see the README).

50 Safe compression and structural parameters

Problem 37.2 asks for algorithms avoiding the full frontier expansion when neighbourhood data have few types. The following weighted compression rules provide one answer for height-two skeletons.

Theorem 50.1 (Weighted twin compression; 04). *The value of Ψ is preserved by the following operations, with the terminal exponent kept unchanged.*

- (i) *Replace all lower rows with the same upper-neighbour set by one row whose exponent is the maximum exponent of the class.*
- (ii) *Replace all upper columns with the same lower-neighbour set by one column.*
- (iii) *Delete columns with no lower neighbours.*

Proof. Rows with equal neighbourhoods retire at exactly the same first column in every schedule. Their only contribution to that group's maximum is the largest of their exponents, proving (i).

For columns with equal neighbourhoods, only the first scheduled member of the class can retire any rows. Every later member has an empty group. Delete those later members from a schedule; the first member represents the class and all nonzero groups are unchanged. Conversely, a schedule on the quotient expands by inserting all other members of each class after its representative, where they contribute zero. This proves (ii). A column with no neighbours contributes zero in every position, proving (iii). $\square$

In the transfinite interpretation, removing an upper or isolated fibre must not silently lower the terminal maximum. It is enough to retain that maximum as the separate terminal parameter. Alternatively one can keep a virtual isolated fibre of height ω^{ρ_τ} in the represented skeleton.

Corollary 50.2 (Few distinct upper neighbourhoods; 04). *Let t be the number of distinct nonempty neighbourhoods of upper vertices, after safe row compression if desired. The weighted height-two objective is computable in $f(t)N^{O(1)}$ bit operations, for example by the subset dynamic program on the t retained columns. The number and binary sizes of the priorities are part of the input length N , not additional fixed parameters.*

Proof. Hashing or sorting incidence rows and columns computes the quotient in polynomial time. Theorem 50.1 preserves the answer and Theorem 49.1 uses 2^t states; the remaining work and bit costs are polynomial in the input length. $\square$

Corollary 50.3 (Few distinct upper-neighbour sets of lower rows; 04). *Let a be the number of distinct sets $N(\ell) \subseteq Q_{\text{up}}$ as ℓ ranges over the lower rows. The weighted objective is computable in $2^a N^{O(1)}$ bit operations.*

Proof. Merge equal lower neighbourhoods, retaining their largest priority. This leaves a rows without changing the objective. Apply the covered-row dynamic program of Proposition 49.2; all other costs are polynomial in the encoded input. $\square$

This distinguishes the two possible orientations of “neighbourhood types”. Both are tractable parameters: distinct neighbourhoods of upper vertices use the upper-subset program, and distinct upper-neighbour sets of lower rows use the covered-row program. The latter is the direct height-two answer to the upper-neighbour-set parameter of Problem 37.2.

Corollary 50.4 (Vertex-cover parameter; 04). *If the incidence graph has a vertex cover of size k , the weighted objective is fixed-parameter tractable in k .*

Proof. Let the cover be $C = C_{\text{lo}} \sqcup C_{\text{up}}$. Every upper vertex outside C_{up} has all its neighbours in C_{lo} , so there are at most $2^{|C_{\text{lo}}|} - 1$ nonempty neighbourhood types outside the cover and at most $|C_{\text{up}}|$ additional types. Consequently

$$t \leq |C_{\text{up}}| + 2^{|C_{\text{lo}}|} - 1 \leq k + 2^k.$$

Apply Corollary 50.2. If a cover is not supplied, the standard edge-branching search chooses one endpoint of an uncovered edge at each step and finds a cover of size at most k , if one exists, in $2^k N^{O(1)}$ time. Thus no unmentioned cover oracle is needed. The resulting double-exponential parameter dependence is a simple bound, not an optimality claim. $\square$

These positive results concern height-two skeletons. They do not answer Problem 37.2 for arbitrary finite skeletons. They also do not contradict Corollary 46.3: a bound on incidence types controls combinatorial structure, whereas a bound on priorities alone does not.

Example 50.5 (Containment is not a safe deletion rule; 04). Return to the path-shaped graph $N(a) = \{c\}$, $N(b) = \{c, d\}$ of Example 47.3 (the N -poset), but give every fibre weight ω . It has a two-edge uniquely restricted matching and uniform height $\omega \cdot 3$. Deleting a merely because $N(a) \subseteq N(b)$ leaves a one-row graph with matching number 1 and height $\omega \cdot 2$. Thus neighbourhood *equality* is safe, but replacing it with containment in the compression proof is false, already on a four-vertex path.

51 Exact classes and algebraic upper certificates

51.1 Components, forests and degree two

Proposition 51.1 (04). *For disjoint graph components $G_1, \dots, G_m$,*

$$v_{\text{ur}}(G_1 \sqcup \dots \sqcup G_m) = \sum_{i=1}^m v_{\text{ur}}(G_i).$$

If G is a forest, then $v_{\text{ur}}(G)$ is its ordinary maximum matching number.

Proof. Every alternating cycle lies inside one component. Thus a matching is uniquely restricted exactly when its restrictions to the components are, and component optima may be united. A forest has no cycle at all, so every matching is uniquely restricted. $\square$

Corollary 51.2 (04). *For a path P_n on n vertices and a bipartite cycle C_{2m} ,*

$$v_{\text{ur}}(P_n) = \lfloor n/2 \rfloor, \quad v_{\text{ur}}(C_{2m}) = m - 1 \quad (m \geq 2).$$

Hence for a nonempty skeleton whose incidence graph has maximum degree at most two, the uniform height is

$$\alpha \cdot \left(1 + \sum_{P_n \text{ component}} \lfloor n/2 \rfloor + \sum_{C_{2m} \text{ component}} (m - 1) \right). \quad (51.1)$$

Proof. A path is a forest and has a matching of size $\lfloor n/2 \rfloor$, which is also the trivial vertex-count upper bound. A cycle has two perfect matchings, so a matching of size m is not uniquely restricted. Removing one edge from one perfect matching gives a matching of size $m - 1$. Its saturated induced subgraph is a proper induced subgraph of the cycle, hence a forest, so this matching is uniquely restricted. Apply component additivity and Theorem 45.1. $\square$

For a general forest, an ordinary maximum matching can also be found directly by repeatedly matching a leaf to its neighbour and deleting their endpoints. An exchange argument proves that some maximum matching contains that edge: if its neighbour is matched elsewhere, exchange the incident matching edge; if both are unmatched, add the leaf edge. This gives a simple polynomial algorithm and makes the forest consequence independent of a matching-algorithm citation.

Beyond these elementary classes, published algorithms transfer without a new ordinal argument. For example, Francis, Jacob and Jana give a linear-time algorithm for maximum uniquely restricted matching in bipartite permutation graphs [11]; it therefore computes the uniform height coefficient for that incidence class. Such a transfer concerns the uniform objective only. An algorithm maximizing matching cardinality is not automatically an algorithm for the ordered weighted objective of Theorem 47.1.

51.2 Field-rank bounds

Let A_G be the 0–1 matrix with lower rows and upper columns, interpreted over any chosen field F .

Proposition 51.3 (04). *For every field F ,*

$$v_{\text{ur}}(G) \leq \text{rank}_F(A_G).$$

More generally the same bound holds for any matrix over F with exactly the same zero–nonzero pattern as G .

Proof. An admissibly ordered matching of size k selects a $k \times k$ submatrix whose diagonal entries are nonzero and whose entries below the diagonal vanish, by (43.1). Its determinant is the product of its nonzero diagonal entries, so it is nonsingular. Therefore the full matrix has rank at least k . Maximize over the matching. $\square$

This is the matrix-rank connection motivating the established matching notion [12], restated as an upper certificate for an ordinal height. A triangular matching of size k , together with a rank- k factorization of a matrix with the required zero-nonzero pattern, certifies equality $\nu_{\text{ur}}(G) = k$ using only finite algebra and incidence checks. It need not exist for an arbitrary graph at its optimum; no general equality with matrix rank is claimed.

Example 51.4 (A characteristic-dependent certificate; 04). For the six-cycle incidence matrix

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix},$$

the first two rows are independent over $\mathbb{F}_2$ and the third is their sum. Hence $\text{rank}_{\mathbb{F}_2}(A) = 2$. A two-edge triangular matching gives the opposite bound, so the corresponding uniform height is $\alpha \cdot 3$. Over $\mathbb{Q}$ the determinant is 2, and the same matrix has rank 3, which gives a weaker upper certificate. Choice of field can therefore matter.

52 Worked weighted examples

The example inputs and their complete Bellman tables are shipped (Table 13). The following values can also be checked by hand.

52.1 A weighted six-cycle

Take lower rows 0, 1, 2 and upper columns 0, 1, 2 with

$$\mathcal{N}(0) = \{0, 1\}, \quad \mathcal{N}(1) = \{1, 2\}, \quad \mathcal{N}(2) = \{0, 2\}, \quad (\rho_0, \rho_1, \rho_2) = (3, 2, 1), \quad \rho_{\top} = 1.$$

The order (0, 2, 1) retires rows $\{0, 2\}$, then $\{1\}$, then nothing, producing $\omega^3 + \omega^2 + \omega$. To see optimality from the layers, the unique high row can be matched to column 0 or 1. The former leaves the priority-2 row alive; the latter retires it immediately. The best residual is therefore $\omega^2 + \omega$. Unlike the uniform cycle example, the proof must consider which high endpoint is selected, not only the high matching size. This is the example following Theorem 35.1 (09's data/03-wpo-height_two_overlap.json), with the upper vertices 3, 4, 5 there renamed 0, 1, 2; both manuscripts obtain the same value by different certificates.

52.2 Independent edges and terminal absorption

For three disjoint edges with lower priorities 3, 1, 2 and terminal priority 1, schedule the priorities in decreasing order to obtain

$$\omega^3 + \omega^2 + \omega \cdot 2.$$

Any order with a smaller term before a larger one erases that smaller term, so this is optimal. More generally, for disjoint edges the optimal prefix is the natural sum of their monomials, followed by the ordinary addition of the terminal monomial (compare (80)). This statement is special to unconstrained independent edges; Example 47.3 explains why it fails for general incidence.

With two disjoint edges of priorities 1, 2 but terminal priority 3, the answer is simply ω^3 . The entire lower schedule is absorbed. An isolated maximal fibre can thus change which lower information survives, even though it has no effect on the unweighted matching number.

52.3 Summary of shipped examples

Input (shipped as data/06-ur-matchings-name.json)	Exact value
antichain	ω^3
single_edge	$\omega \cdot 2$
six_cycle	$\omega \cdot 3$
weighted_cycle	$\omega^3 + \omega^2 + \omega$
forced_absorption	$\omega^2 + \omega$
tie_requires_lookahead	$\omega^3 + \omega^2 + \omega$
terminal_absorption	ω^3
weighted_twins	$\omega^4 + \omega^3 + \omega$
parallel_edges	$\omega^3 + \omega^2 + \omega \cdot 2$

Table 13: 04's nine example inputs and their exact values; each input ships with its full certificate ..._certificate.json (delivered in examples/).

The antichain input has no lower or upper vertices in the finite objective and an explicit terminal parameter. It represents a nonempty antichain of pure fibres through their maximum height, not the empty skeleton.

53 Verification, reproducibility and limits

53.1 Executed checks

The recorded run used Python 3.13.5 with pseudorandom seed 20260929. It performed the following exhaustive families. For every $0 \leq r, s \leq 3$ it enumerated every $r \times s$ binary incidence matrix with no zero lower row, every assignment of lower priorities from $\{1, 2, 3\}$, and each terminal priority in $\{1, 2, 3\}$. This yields 31,755 weighted instances. Zero columns are permitted. For each one the subset program was compared against all permutations, the covered-row program, all admissibly ordered matchings, and the priority-layer recurrence, and its certificate and twin compression were checked.

The run also enumerated all $15^4 = 50,625$ nonzero-row 4×4 incidence matrices with every priority equal to 1, comparing the program against all upper permutations. This is an exhaustive incidence-matrix test, not an enumeration of unlabelled graphs and not a check of every weighted 4×4 case. An additional 120 seeded weighted instances used up to five vertices per side, with lower priorities selected from 1, 2, 5, 9, 10^{12} and terminal priorities from 1, 3, 8. The large-priority tests check that no expansion proportional to the numerical exponent occurs.

The counts 31,884 include the 31,755 exhaustive weighted instances, the 120 seeded instances and the nine named examples. The exact ledger is data/06-ur-matchings-verification.json; the console record data/06-ur-matchings-verification_console.txt prints two count lines followed by the same ledger. Not every category is run on every family.

53.2 Independence and what was not checked

The brute schedule evaluator locates each row's first neighbour by a position map, rather than by the production program's incremental group rule. Its ordinal normalizer scans the

Recorded assertion category	Count
Subset program versus all upper permutations, weighted suite	31,884
Subset program versus covered-row program	31,884
Subset program versus all admissibly ordered matchings	31,884
Subset program versus the priority-layer recurrence	31,884
Full Bellman certificate verification	31,884
Twin-compression invariance	31,884
Maximum high endpoints cover the high rows	33,245
Closure-system height versus matching number	445
Uniform 4×4 program versus all permutations	50,625
Tampered terminal-value rejection	9
Malformed mathematical input rejection	5
Total recorded assertions	275,633

Table 14: 04’s recorded assertion counts, from `data/06-ur-matchings-verification.json`.

monomial word from right to left, rather than using the production left-to-right ordinal adder. The ordered-matching evaluator checks all forbidden earlier cross-edges directly. The priority recurrence independently enumerates maximum high matchings and residuals. These comparisons reduce implementation-specific error risk; they are not a substitute for the proofs.

Some components remain shared: the instance representation and the basic matching enumeration are reused by validation paths, and the certificate checker shares ordinal arithmetic with the optimizer. The counts are numbers of assertions, not measures of proof strength. No finite test establishes a transfinite theorem, exhausts arbitrary graphs, verifies an arbitrary ordinal notation system, or establishes historical novelty. No Lean or Rocq source was compiled as part of this work. The finite graph results hold for arbitrary allowed priorities; the tests do not establish that extension.

53.3 Formalization roadmap (04)

A small formal development can be organized without importing the report’s entire transfinite theory. First formalize finite bipartite incidence, triangular matchings and the productive-witness equivalence. Next formalize ordinal words as finite priority-coefficient lists, prove their normalization semantics, and verify the Bellman checker. Then prove the priority recurrence over the finite set of occurring priorities, using the coverage lemma and the ordered-matching principle as separate dependencies.

For the transfinite endpoint, formalize the support-profile description and the strict map (45.5); the attaining-chain construction then proves the uniform theorem directly. This separates the graph-theoretic layer, the finite ordinal-algorithm layer and the set-sized rank layer. The nonuniform height application should be enabled only after Theorem 35.1 is itself proved or explicitly assumed. A future formalization should report its axioms and assumptions rather than labelling Python execution as kernel checking.

53.4 A compact proof-dependency map (04)

Endpoint	Essential dependencies
Productive count = ν_{ur}	Fresh witnesses; triangular order; alternating-cycle characterization, reproved.
Uniform transfinite height	Finite progress function $\text{prog}(X)$; exact downset profiles; natural-sum rank map; explicit α -blocks.
Uniform NP-completeness	Uniform height theorem; published bipartite uniquely restricted matching NP-completeness; explicit polynomial reduction.
Weighted matching formula	Fresh maximum-priority witnesses; weak monotonicity of ordinary addition.
Priority-layer recurrence	Weighted formula; high-subgraph matching bound; coverage lemma; residual low-row characterization.
Weighted height conclusions	Above finite theorems plus Theorem 35.1.
Bellman certificate soundness	Finite subset induction; all candidate inequalities; one attaining predecessor.
Structural FPT bounds	Equal-neighbourhood compression; subset program; elementary vertex-cover type count.

The part contains conventional mathematical proofs for these arrows, not formal proof terms. It is suitable for external review and for the staged formalization described above.

54 Research questions of Part IV

54.1 Questions of Parts II–III answered or partly answered here

- *Complexity* (07 Question 4; 09 §14.4; 12 Problem 10.3; the item of Section 37 and the closing sentence of Section 35): **answered** for the NP-hardness alternative. The threshold problem is NP-complete already for height-two skeletons with every fibre ω , one exponent scale (Theorem 46.2); with Corollary 20.4 the threshold problem for the explicit pure-block encoding is NP-complete (Remark 46.5). 09's uniform case (largest number of nonempty retirement groups) is the maximum uniquely restricted matching number and the Grundy covering number of the neighbourhood family (Theorem 43.3, Section 43.1); 09's mixed case is the ordered-matching objective of Theorem 47.1, which 04 does not identify with a previously named problem. 07's leading-coefficient threshold is NP-complete at one scale (Corollary 46.3), and the leading coefficient is a uniquely restricted matching number (Corollary 48.2). Input-comparison cost is separated from combinatorial cost as 09 asked: exponents are binary labels copied, never expanded (Proposition 46.4). The "separate reduction" that item called for is Theorem 46.2, from the published NP-completeness of bipartite uniquely restricted matching [12]. Nothing here gives a polynomial algorithm, which would imply $P = NP$.
- *Structural parameters* (Problem 37.2): **partly answered** for height-two skeletons. The number of distinct upper-neighbour sets of lower rows gives a $2^a N^{O(1)}$ algorithm (Corollary 50.3); the number of distinct neighbourhoods of upper vertices and the vertex-cover number are also FPT parameters (Corollaries 50.2 and 50.4). The number of distinct exponent priorities is *not* an FPT or even an XP parameter unless $P = NP$ (Corollary 46.3). Width parameters, decomposition widths and all skeletons of height three or more remain open.

54.2 Further questions (04)

The complexity question and the height-two neighbourhood-type parameter have been addressed above. The following are further targets relative to these results; 04 does not assert that all of them are previously unstudied in the literature.

Problem 54.1 (Weighted forests; 04). Determine the complexity of the ordered ordinal objective on incidence forests. Can the priority-layer recurrence be implemented in polynomial time without enumerating all maximum high matchings? A successful tree dynamic program must remember enough boundary information to distinguish the two residual outcomes in Example 48.3.

The unweighted forest problem is elementary because every matching is uniquely restricted. That fact does not resolve the weighted precedence and residual selection problem. A first milestone is a complete algorithm for paths with arbitrary binary priority labels and an explicit proof that local state merging preserves all surviving lower coefficients.

Problem 54.2 (Weighted decomposition-width algorithms; 04). Find exact algorithms parameterized by an incidence-graph decomposition width, with priorities part of the input. Seek substantially better dependence than the double-exponential vertex-cover bound of Corollary 50.4. Specify whether the algorithm handles the whole ordinal or only its leading coefficient.

An algorithm for maximum-cardinality uniquely restricted matching alone is insufficient: one needs either ordered boundary data or a compositional form of the residual priority theorem. The number of priorities by itself is excluded as a tractable parameter by Corollary 46.3. This is the height-two case of the decomposition-width clause of Problem 37.2.

Problem 54.3 (Compression of maximum high endpoint sets; 04). For fixed Q_{hi} , characterize when two sets in $\mathcal{S}_{hi}$ produce equivalent residual objectives. Can one enumerate only a small representative family, perhaps parameterized by the number of low neighbourhood types or by a separator?

The recurrence proves that the internal ordering of a chosen high matching is irrelevant after that layer, but its upper endpoint set may be indispensable. A useful certificate would describe equivalence of residual objectives without solving each residual instance independently.

Problem 54.4 (Succinct Cantor coefficients; 04). Extend the algorithms to ordinal-chain capacities specified by binary Cantor coefficients, without replacing a coefficient c by c separate pure blocks. Handle finite tails and successor corrections explicitly.

This is Problem 37.1, which stays open. The present binary priority representation does not answer it: storing the exponent 10^{12} as a label is different from processing a fibre $\omega^p \cdot 10^{12}$ without expansion. A fixed skeleton with two repeated interacting chains is an appropriate first test, as 12 also suggested.

Problem 54.5 (Beyond height two; 04). Identify a finite combinatorial object playing the role of a triangular matching for general finite skeletons. It must record vertices that can be both newly activated and later retired, not merely a single lower-to-upper incidence.

The height-two reduction depends on every nonmaximal vertex retiring at a first maximal neighbour. Intermediate levels create multiple lifetimes and precedence constraints. Any proposed generalization should reduce to the present matching formula and preserve the retirement-path formulas already known in this report (Theorems 21.1 and 27.1).

Problem 54.6 (Nonpure non-chain components; 04). For lexicographic sums of well-partial-orders whose finitary powersets have nonpure heights, find the local rank information needed in place of a single pure monomial per fibre.

The current finite calculus assumes one pure contribution at each retirement. The report's ordinal-chain block expansion does not automatically transfer to arbitrary non-chain components. A small counterexample to scalar-height substitution, or a provably sufficient finite profile invariant, would be a substantive next step rather than another application of the pure formula. This is the nonpure clause of the non-chain item of Section 37 (09 §14.2), which stays open.

Problem 54.7 (Short upper certificates; 04). Determine useful graph classes on which a matching lower witness and a low-rank matrix with the same zero–nonzero pattern always certify the optimum. More generally, find polynomial-size upper certificates for structured priority-layer residuals that avoid a full 2^s Bellman table.

The field can matter, as Example 51.4 shows. Matrix-rank bounds are sound but not asserted to be complete. Any general certificate claim should be reconciled with the NP-completeness result, rather than inferred from successful small examples.

Problem 54.8 (Counting and stability; 04). Count optimal upper schedules or optimal high endpoint sets in tractable incidence classes, and classify how these sets change when two adjacent priority levels exchange order.

Strictly increasing relabelling preserves all maximizers, but changing the weak-order pattern can erase or restore lower terms. A chamber-by-chamber analysis could expose combinatorial structure hidden by the single optimum coefficient vector. Compare the chamber questions of Problem 37.6 (09).

Problem 54.9 (Certified integration; 04). Formalize the finite matching and Bellman layers, then connect the checker to the direct uniform ordinal rank theorem. Extend the trust boundary to the nonuniform case only after its transfinite scheduling theorem is checked.

This is a concrete route to a machine-checked answer to the original complexity question, and a height-two instance of Problem 37.7. It avoids making the finite computation depend on an unnecessarily large library while preserving a precise theorem-level connection to the actual finitely generated downsets.

55 What Part IV does not claim, and choices made in writing it

55.1 Non-claims

Every limitation stated by 04, in its article, delivery README and proof audit, is kept. The manuscript is AI-assisted and unrefereed; it is not a Lean or Rocq formalization and no proof-assistant source was compiled. No historical priority is claimed; the unweighted matching and covering-sequence notions are classical and credited, and the targeted literature search does not establish that every formulation is unprecedented. The NP-hardness is imported from Golumbic–Hirst–Lewenstein; 04 does not reproduce or claim a new proof of the original matching hardness reduction. The nonuniform transfinite readings import Theorem 35.1, which 04 neither re-proves nor formally checks; only the uniform height theorem, and the finite weighted theorems, are proved independently. There is no polynomial-time general solver, no approximation ratio for an ordinal objective, and no proof that matrix-rank bounds are

always tight. Arbitrary higher-skeleton scheduling, nonpure WPO substitution and binary Cantor-coefficient compression are not solved. The solver is a pure-positive-priority solver, not an implementation for finite or successor fibres; its guard of 22 upper vertices is not a performance guarantee, and the certificate checker certifies the finite optimum, not the transfinite interpretation. The finite tests are implementation checks, not proofs of transfinite statements and not evidence of historical novelty. 04 did not exhaustively compare the report archives that had newly arrived at its pin, and states that it resolves an identified repository question, not an unspecified famous open problem in matching theory.

55.2 Choices made in writing Part IV

1. *Placement.* 04 continues Part III's height-two theorem and Section 37, so it is added to this report as a fourth part, after the closing part of Parts II–III and before Part I's appendices.
2. *Numbering.* Equations of Part IV are numbered within sections, and the equation counter is restored after this part, so that no section, theorem or equation number of Parts I–III or of the appendices moves; in particular (83), which already moved once in batch 36, keeps its number.
3. *Notation.* 04's symbols were renamed as in Table 12: its L , U , Z , $N(\ell)$ follow Part III's renaming of the same sets; S , τ , H , K_H , d , R , F , $D(C)$, V , p and cl were renamed because each already names something else in this report. No normalization changed: every sum is ordinary unless written $\#$, and empty maxima are 0, as in Part III.
4. *Repository input.* 04's "Theorem (repository input)" is printed as Remark 42.2, a restatement of Theorem 35.1, not as a new theorem; 04's citations of "Theorem 35.1", "Section 37" and the pinned article became cross-references.
5. *Editorial additions.* Remarks 44.2 and 46.5, the identification of 04's examples with this report's N -poset and with 09's height-two example, the relations to Part III's support-ideal program, and the placement-time check of Section 41.6 are the merge's own.
6. *Stale sentences.* Sentences of Parts II–III that said the complexity was not classified, that a hardness claim needed a separate reduction, or that NP membership is not NP-completeness, were kept as provenance of the pinned state and given dated pointers to this part. The abstract on the title page gained one dated sentence, and the title page's top spacing was reduced by 12 mm so that it still fits on one page.
7. *Bibliography.* 04's entries for Golumbic–Hirst–Lewenstein, Brešar–Henning–Rall and Francis–Jacob–Jana were added to the report's list; its entry for this report itself became internal cross-references.
8. *Files.* 04's code, data, examples, proof audit and source record are shipped byte-identical under the prefix 06-ur-matchings-; its examples, delivered in examples/, are in data/. The shipped text still uses the delivered names; the README maps them and gives a rerun recipe that works on a copy.

56 Conclusion of Part IV

For height-two skeletons, uniform finitary-powerset height is governed by a classical graph invariant:

$$\begin{aligned} \text{height coefficient} - 1 &= \text{maximum productive schedule length} \\ &= \text{maximum uniquely restricted matching size.} \end{aligned}$$

The equivalence supplies an NP-completeness proof with only the fibre ω , and the explicit ordinal ranking establishes that conclusion independently of the inherited weighted height theorem. It also identifies the finite maximal frontier height and gives immediate exact formulas for forests and cycles.

For unequal priorities, the correct object is an admissibly ordered matching. The leading coefficient comes from the highest-priority incidence subgraph; all lower coefficients are optimized in residual graphs selected by maximum high matchings. This separates what is determined by matching cardinality from what depends on preserving low-priority rows. Safe twin compression, structural parameter bounds and finite Bellman certificates turn the theory into an exact computational continuation, while leaving the genuinely new transfinite and succinct-encoding questions visible.

A Finite nonuniform lexicographic sums

This appendix gives a useful consequence of the same absorption theorem. It concerns the lexicographic sum itself, not its finitary powerset.

For a nonzero ordinal

$$\eta = \omega^{\delta_1} a_1 + \cdots + \omega^{\delta_s} a_s, \quad \delta_1 > \cdots > \delta_s,$$

define $\deg(\eta) = \delta_1$ and

$$\mathrm{tr}_{\geq \tau}(\eta) = \sum_{\{j: \delta_j \geq \tau\}} \omega^{\delta_j} a_j.$$

The latter is the initial Cantor-normal-form part whose exponents are at least τ .

Lemma A.1 (Replacing a fiber by a maximal linearization). *For a finite poset Q and WPOs P_q , let $\eta_q = o(P_q)$. Then*

$$o\left(\sum_{q \in Q}^{\mathrm{lex}} P_q\right) = o\left(\sum_{q \in Q}^{\mathrm{lex}} \eta_q\right).$$

Empty fibers can be deleted from the index poset, retaining its induced order on the remaining indices.

Proof. Choose maximal linearizations of the fibers. Replacing their orders by these linearizations adds comparisons to the lexicographic sum, so the right-hand side is at most the left-hand side.

Conversely, take an arbitrary linearization L of the original sum. Its restriction to P_q has some type $\zeta_q \leq \eta_q$. After identifying each restricted fiber with its ordinal type, L is a linearization of $\sum_q^{\mathrm{lex}} \zeta_q$. This sum embeds as an induced subposet in $\sum_q^{\mathrm{lex}} \eta_q$, by initial-segment embeddings within each fiber. Consequently

$$\mathrm{type}(L) \leq o\left(\sum_q^{\mathrm{lex}} \zeta_q\right) \leq o\left(\sum_q^{\mathrm{lex}} \eta_q\right).$$

Taking the maximum over L proves the reverse inequality. $\square$

Theorem A.2 (Successor-degree truncation). *Let Q be finite and let P_q be WPOs with $\eta_q = o(P_q)$. Ignore indices whose fibers are empty. For every remaining q , put*

$$\tau(q) = \max(\{0\} \cup \{\deg(\eta_r) : r > q, \eta_r > 0\}).$$

Then

$$o\left(\sum_{q \in Q}^{\mathrm{lex}} P_q\right) = \#_{\{q: \eta_q > 0\}} \mathrm{tr}_{\geq \tau(q)}(\eta_q). \quad (83)$$

Proof. By Lemma A.1, it suffices to use the ordinal fibers η_q . Expand each of their finite Cantor normal forms into a finite sequence of pure blocks, repeating a block according to its coefficient. Within a fiber these blocks appear in nonincreasing exponent order. Between fibers, order every block of q below every block of r whenever $q < r$. This produces a finite pure-block stratification with full comparisons between comparable blocks, hence in particular a finite cofinal stratification.

A block of exponent δ is not absorbed by a later block of its own fiber, since internal exponents never increase. It is absorbed by a block of a strict successor fiber exactly when some such fiber has leading exponent greater than δ . Therefore it survives exactly when $\delta \geq \tau(q)$. Theorem 4.2 now says to take the natural sum of all surviving blocks, which is (83). $\square$

A.1 Recovery of the finite uniform formula

Suppose all n fibers have the same nonzero type $\eta = \omega^\delta c + \sigma$, where $\sigma < \omega^\delta$, and let k be the number of maximal elements of Q . Every nonmaximal fiber loses its tail σ , while every maximal fiber retains it. Thus

$$o\left(\sum_{q \in Q}^{\text{lex}} P\right) = \omega^\delta \cdot (cn) + (\sigma \otimes k).$$

This is the finite-index specialization of the uniform lexicographic product result treated in [4]. For finite η , the same expression simply says ηn . Its appearance here is a consistency check and a consequence of the more general finite-block argument, not an assertion that the uniform problem remained open.

A.2 A small tail example

For two incomparable fibers of types $\omega + 1$ and $\omega + 2$, no term is truncated, giving $\omega \cdot 2 + 3$. If the first fiber is below the second, its constant term is deleted, giving $\omega \cdot 2 + 2$. This is why ordinary concatenation of whole impure fibers is not a sufficient lower-bound construction for arbitrary partial index orders: maximal linearizations may separate their Cantor-normal-form blocks. The pure-stratum theorem avoids that difficulty by starting at the appropriate block granularity.

B A compact proof-dependency audit

The proof graph has three standard inputs: maximal-linearization attainment, the disjoint-union/natural-sum formula, and the Cartesian-product/natural-product formula. Induced-subposet monotonicity and elementary Cantor-normal-form arithmetic are used throughout.

From these, Lemma 3.1 proves the crucial upper bound for a cofinal pure component. Theorem 4.2 adds finite grouping and an explicit block-order lower bound. Lemmas 5.2, 6.1 and 6.2 establish the finite frontier model without any ordinal estimate. The product sandwich then feeds Theorem 4.2, yielding Theorem 7.3.

The height proof is separate. Its upper bound is an explicit ordinal rank map, and its lower bound is an explicit chain. It does not infer height from maximal order type or from a finite numerical test. The counting and pure-output results are finite corollaries of the frontier formula. Finally, Appendix A uses maximal linearizations only after proving that fiber replacement preserves the quantity being calculated.

No step depends on a claim that cofinality preserves arbitrary maximal order types, that frontier strata are fully comparable, that finite cutoffs compute infinite ranks, or that all ordinal widths equal maximal order types. The counterexamples and limitations in Section 13 are included to make these potential failure modes explicit.

References

- [1] Sergio Abriola, Simon Halfon, Aliaume Lopez, Sylvain Schmitz, Philippe Schnoebelen and Isa Vialard. *Measuring well-quasi-ordered finitary powersets*. arXiv:2312.14587v2, 17 July 2024. Open extension question: Section 6, p. 23. <https://arxiv.org/abs/2312.14587>.
- [2] D. H. J. de Jongh and Rohit Parikh. Well-partial orderings and hierarchies. *Indagationes Mathematicae*, 39(3):195–207, 1977.

- [3] Mirna Džamonja, Sylvain Schmitz and Philippe Schnoebelen. On ordinal invariants in well quasi orders and finite antichain orders. In *Well Quasi-Orders in Computation, Logic, Language and Reasoning*, Trends in Logic 53, pp. 29–54. Springer, 2020. Updated preprint: arXiv:1711.00428v5, 28 January 2025. <https://arxiv.org/abs/1711.00428>.
- [4] Isa Vialard. On maximal order type of the lexicographic product. *Logic Journal of the IGPL*, 33(4):jzaf051, 2025. Published 17 July 2025. <https://doi.org/10.1093/jigpal/jzaf051>.
- [5] Jacob Hilton. The topological pigeonhole principle for ordinals. arXiv:1410.2520v4, 12 July 2016. Section 2.5, Theorem 2.20 (finite Milner–Rado formula). <https://arxiv.org/abs/1410.2520>.
- [6] Alberto Marcone, Antonio Montalbán and Richard A. Shore. Computing maximal chains. arXiv:1201.4408, 2012. Theorem 1.2 states Wolk’s maximal-chain theorem. <https://arxiv.org/abs/1201.4408>.
- [7] D. Schmidt. The relation between the height of a well-founded partial ordering and the order types of its chains and antichains. *Journal of Combinatorial Theory, Series B*, 31(2):183–189, 1981.
- [8] Isa Vialard. On the width of the Cartesian product of ordinals. *Order*, 42:37–58, 2025 (published online 2024). <https://doi.org/10.1007/s11083-024-09668-8>.
- [9] E. S. Wolk. Partially well ordered sets and partial ordinals. *Fundamenta Mathematicae*, 60(2):175–186, 1967. Theorem 9 (maximal chains). <https://doi.org/10.4064/fm-60-2-175-186>.
- [10] Boštjan Brešar, Michael A. Henning and Douglas F. Rall. Total dominating sequences in graphs. *Discrete Mathematics*, 339:1665–1676, 2016. <https://doi.org/10.1016/j.disc.2016.01.017>. Author manuscript: <https://arxiv.org/abs/1601.07525>.
- [11] Mathew C. Francis, Dalu Jacob and Satyabrata Jana. Uniquely restricted matchings in interval graphs. *SIAM Journal on Discrete Mathematics*, 32(1):148–172, 2018. <https://doi.org/10.1137/16M1074631>. Author manuscript: <https://arxiv.org/abs/1604.07016>.
- [12] Martin C. Golumbic, Tirza Hirst and Moshe Lewenstein. Uniquely restricted matchings. *Algorithmica*, 31:139–154, 2001. <https://doi.org/10.1007/s00453-001-0004-z>.